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Microlocal Neural Operators: Sparse Canonical Packet Matrices for Oscillatory Operator Learning

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Abstract

Neural operators for oscillatory PDEs face a representation problem that is not resolved by point-space correlation or global frequency filtering alone. After wavepacket analysis, a short-time hyperbolic propagator is represented by a sparse matrix indexed by phase-space packets, with significant entries concentrated near a canonical relation. We introduce the *Microlocal Neural Operator* (MiNO), a neural-operator family built around this sparse canonical packet-matrix primitive.

MiNO tokenizes fields into localized packets, updates packet metadata by a learned near-canonical map, applies a local packet-kernel symbol, and reconstructs through carrier-bound transported synthesis plus a low-frequency residual branch. The theorem-facing model, MiNO-Core, admits an explicit-rate short-time half-wave approximation theorem in ℓ^2 packet-matrix and L^2 field norms. Its error decomposes into transport, local-kernel symbol, residual, packet-tail, frame/asymptotic/covering, and executable synthesis budgets. The continuous input is classical microlocal analysis: off-canonical-tube decay for half-wave propagators gives packet sparsity, and bounded canonical stencils plus a Schur row/column transfer convert that sparsity into an operator-level packet-matrix law. The finite packet landing layer, including bounded-stencil propagation, shell truncation, and cross-resolution transfer, is machine-checked in Lean 4.

Empirically, the evidence is organized around identifiability of the packet primitive. On controlled wave probes, corrupting transported metadata produces a consistent field-level degradation: mean relative ℓ^2 increases from 0.380 to 0.714 on `wave_bicharacteristic_control`, from 0.316 to 0.839 on `wave_chirp_propagation`, and from 0.204 to 0.777 on `wave_two_packet_no_interaction`. Learned finite landing narrows the absolute gap to same-refinement UNO/WNO controls while preserving a transport penalty. On cached diffusion rows, MiNO-Core/MiNO-Plus are competitive with imported references, whereas high-k Helmholtz remains a stress regime that exposes the need for anisotropic, resolvent-aware packet structure. The contribution is a bounded, executable, and falsifiable sparse canonical packet-matrix primitive for oscillatory operator learning.

Keywords: neural operators, microlocal analysis, wavepacket frames, sparse packet matrices, canonical propagation, machine-checked proof, phase-space learning

1 Introduction

Oscillatory solution operators have a representation that is largely invisible in point space and only partially visible in a global Fourier basis. After wavepacket analysis, a short-time hyperbolic propagator becomes a sparse matrix whose rows and columns are indexed by

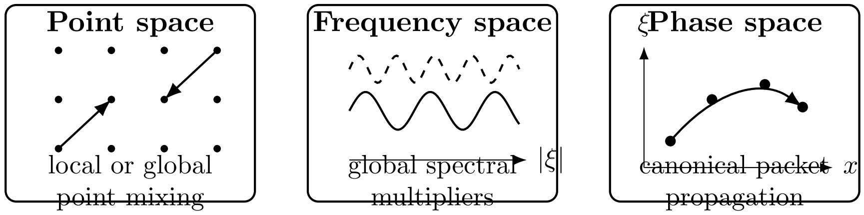

Figure 1: Three operator-learning primitives. Point-space and frequency-space primitives can be effective in smooth or stationary regimes. MiNO targets the oscillatory regime where localized packets move along a canonical relation in phase space.

phase-space packets, and whose significant entries lie near canonical relations. Most neural operators instead build their primitive interactions from point correlations, global spectral multipliers, or generic attention maps (Kovachki et al., 2023; Li et al., 2021; Lu et al., 2021). Those primitives are effective in smooth, stationary, or translation-invariant regimes, but they do not expose the canonical packet geometry that controls high-frequency propagation.

The primitive in this paper is a *sparse canonical packet matrix*. The underlying microlocal fact is classical: wave packets propagate along canonical relations and acquire local symbol factors. The contribution is to make that fact an operator-learning object. In MiNO, packet coefficients and metadata form the network state, canonical packet motion is represented by an explicit learned branch, symbol modulation acts locally on transported packets, and the finite landing layer is bounded and machine-checkable.

The resulting discrete question is not "which pixels attend to which pixels?" or "which global Fourier modes are amplified?" but rather: which packets with state (x, ξ, σ) interact after canonical motion, with what local amplitude, and with what smoothing or dissipative remainder? The Microlocal Neural Operator (MiNO) is built around this packet-matrix question.

This distinction matters outside wave propagation. Hyperbolic operators occupy the nontrivial-canonical-graph regime of the calculus. Elliptic parametrices occupy the identity-canonical pseudodifferential regime, where packets should mostly stay near their phase-space locations and the local symbol branch carries the leading action. Parabolic semigroups occupy a dissipative-symbol regime, where the main microlocal effect is high-frequency damping rather than packet relocation. Thus MiNO's identity is not that packets must always move far. Its identity is that the operator is exposed as a structured phase-space packet matrix: canonical transport when the PDE transports singularities, identity-canonical symbols when the PDE is elliptic, and dissipative symbols when the PDE smooths.

The paper keeps one central theorem in focus. Standard pseudodifferential and Fourier-integral-operator estimates supply the continuous wave/FIO structure; the operator layer supplies an ℓ^2 sparse packet-matrix bound; and the finite layer supplies the packet landing lemma used by the architecture. The finite landing lemma is checkable and implementable:

- an analytic discrete wavepacket frame defines packet coefficients and packet metadata;
- a reference operator acts on those coefficients through packet transport and local symbol modulation;

- an approximate MINO layer perturbs that transport, perturbs that symbol, and adds a residual truncation term;

- the resulting coefficient error is controlled by transport, symbol, and truncation budgets.

This finite law is the *finite transported-symbol perturbation bound*. It is the machine-checked algebraic landing layer of the manuscript and fixes the packet-level error budget used by the continuous and empirical sections. The main theorem then upgrades those local budgets to an $\ell^2 \to \ell^2$ sparse packet-matrix estimate through a Schur row/column transfer.

The finite/discrete viewpoint supports two additional layers. The first is a *continuous microlocal extension*: for hyperbolic pseudodifferential propagators with real principal symbol, standard Hörmander calculus and packet almost-diagonalization imply that the continuous propagator reduces to a transported-symbol packet law plus a controllable residual. The second is *semi-discrete PDE transplantation*: once a discretized PDE solver, written in a packet basis, is close to that transported-symbol form, the finite perturbation bound applies immediately. These layers turn the point-space/frequency-space dichotomy into a phase-space packet calculus.

Two further pieces complete the theorem spine: a *quantitative cross-resolution transfer theorem* and the main *Microlocal Propagation Theorem for* MINO-*Core*. Resolution transfer becomes a scale-indexed theorem. The PDE layer becomes an explicit-rate approximation theorem combining bounded-stencil packet-matrix control, packet sparsity, wave off-graph decay, and compact-window neural approximation rates for the canonical map and local packet-kernel symbol.

The theorem constrains the architecture without uniquely determining every implementation choice. Its hypotheses identify the objects that must be approximated—a packet analysis/synthesis layer, a canonical transport map, a local symbol, and a residual/tail term. The carrier-bound MINO implementation is one explicit realization of that template:

1. an analytic local Fourier / wavepacket tokenizer;

2. a canonical propagation layer that updates packet states through a near-canonical transport map with local kernel interaction;

3. a symbol-modulation branch that acts like a learned pseudodifferential multiplier on transported packets;

4. a transported synthesis and high-frequency carrier path that makes moved metadata visible in the reconstructed field;

5. an identity-canonical / low-frequency residual branch that preserves competitiveness on smooth regimes while still being interpretable as the pseudodifferential limit of the same packet calculus.

Because the finite theorem is stated over packet coefficients, the mathematical object and implementation share the same coordinates. High-frequency coefficients interact through transported metadata, local symbol weights, and a bounded retained stencil. The comparison theorem is a representation-separation statement for restricted primitives: when a propagated packet leaves the identity tube and remains outside a fixed Fourier-limited residual span, explicit packet transport avoids an error floor that identity-local or Fourier-limited primitives cannot avoid within that span. The matching experiment asks whether corrupting transported metadata damages the learned field map after low-frequency refinement is restricted and baselines receive the same local refinement advantage. The controlled carrier-bound rows answer this question affirmatively.

The contributions are best read in three layers.

Conceptual contribution. The central design hypothesis is that phase-space packet structure, not point correlation or global frequency filtering alone, is the right primitive to test for oscillatory and heterogeneous operator learning. In the hyperbolic setting this structure becomes canonical propagation. In elliptic and parabolic settings it becomes identity-canonical or dissipative symbol action. MiNO implements this hypothesis by making the packet the token, phase-space transport the primary update when transport is present, symbol modulation the amplitude and damping mechanism, and a residual branch the controlled place for smoothing and off-stencil effects.

Mathematical contribution. The operator-level theorem is a sparse packet-matrix theorem in the natural energy norm. A Schur row/column transfer converts bounded canonical stencils and local entrywise transport/symbol defects into an $\ell^2 \to \ell^2$ packet-matrix bound, and synthesis gives the corresponding L^2 field estimate. The finite transported-symbol perturbation bound remains in the paper as the Lean-checked landing lemma: it decomposes a single landed packet into transport, symbol, and residual budgets, and its bounded-stencil and cross-resolution extensions are machine-checked in Lean 4. The continuous part then uses classical hyperbolic/FIO theory to prove off-canonical-tube decay for the short-time half-wave family, converts that decay into packet sparsity, and combines the pieces with compact-window neural approximation rates. A Hamiltonian metadata lemma separates cheap first-order Hamiltonian bias from a separable Störmer–Verlet branch that is symplectic by construction in packet metadata variables under a separable or splitting-based metadata model. The result is the Microlocal Propagation Theorem for MiNO-Core: an explicit-rate field-facing theorem whose error depends on packet scale, stencil budget, covering radius, residual size, transported-synthesis/carrier defect, and the parameter budgets of the metadata and local packet-kernel symbol branches. Elliptic and parabolic examples are interpreted through the same microlocal lens as identity-canonical pseudodifferential and dissipative-symbol regimes rather than as departures from the theory.

Empirical and artifact contribution. The source repository contains the PyTorch backbone, staged benchmark runner, imported TCNO/DMNO reference ingestion, symbolic certificates, interval-style checks, Lean companion, and manuscript source. The empirical evidence is organized around branch identifiability rather than a single leaderboard. The carrier-bound protocol measures symplectic defect, high-frequency transport gating, packet-wavefront transport, Sobolev error, symbol order, core-only error, refinement energy, and same-refinement baselines. In controlled wave probes, removing transport or randomizing metadata consistently degrades field error. The learned finite-landing controls then show that improving the executable synthesis layer narrows the absolute-accuracy gap while preserving a transport penalty.

Scope. The primary theorem is short-time hyperbolic: the Microlocal Propagation Theorem treats the half-wave family in a no-caustic window. Helmholtz appears through a local-window corollary and as an oscillatory stress test; Navier–Stokes appears only through a one-step transport–diffusion reading. The continuous Hörmander/FIO layer supplies the classical propagation input, and the Lean companion checks the finite packet landing layer. The empirical section evaluates whether the resulting carrier-bound phase-space packet backbone uses transported metadata as a field-level mechanism.

2 Related Work

The most direct background is the neural-operator literature itself. The survey of Kovachki et al. (2023) frames the field as learning maps between function spaces and places FNO, DeepONet, geometric variants, and physics-informed variants within a shared operator-learning view. FNO (Li et al., 2021) established the modern spectral-convolution template; DeepONet (Lu et al., 2021) emphasized branch–trunk decomposition; Geo-FNO (Li et al., 2023) and related boundary/geometry-aware models such as BENO (Wang et al., 2024) show that geometry handling is essential when regular grids are no longer sufficient. CANO (Calvello et al., 2025) makes attention itself a function-space operator and proves a universal approximation theorem for transformer neural operators; DDO (Lim et al., 2023) gives a parallel example of lifting a finite-dimensional generative primitive to function space. Multiwavelet and wavelet neural operators (Gupta et al., 2021; Tripura and Chakraborty, 2022) already suggest that localization beyond the global Fourier basis matters. The Windowed Fourier Propagator of Cai et al. (2026) is especially close in spirit for wave equations: it exploits frequency locality and compact windowed propagation in heterogeneous media. MiNO differs by making the packet matrix explicitly phase-space-valued and by attaching the learned interaction to a canonical transport/symbol calculus rather than only to frequency-window locality.

High-frequency Helmholtz is a separate comparison tier. Specialized neural solvers such as MGCFNN (Xie et al., 2025) and probabilistic high-frequency Helmholtz operators (Zou et al., 2026) target the same phase-sensitive regime more directly than generic operator-learning baselines. Classical numerical analysis also gives the correct scaling language: pollution-effect discussions and *hp*-FEM results compare degrees of freedom against the natural k^d high-frequency resolution scale (Spence, 2023). For this reason, the Helmholtz rows in this paper are treated as a stress path unless the experiment reports frequency scaling, phase/radiation residuals, and specialized Helmholtz references.

The comparison with spectral, U-shaped, continuous-attention, function-space diffusion, and discretization-aware operator models is therefore one of primitives. FNO and UNO-style models may learn transported laws implicitly through global frequency filters, multiscale decoding, or refinement. CANO-style models emphasize continuum attention and discretization stability, while DDO-style models lift score-based diffusion to function space by perturbing functions with Gaussian-process noise and learning function-valued scores. MiNO targets the complementary oscillatory regime in which the sparse canonical packet matrix is exposed as network state. Proposition 39 formalizes the single-packet primitive advantage: a global Fourier-limited residual or identity-local multiscale decoder cannot land a high-frequency packet that has moved along a nontrivial canonical graph unless the relevant transported subspace is represented indirectly. The many-packet rank-growth theorem in Appendix C gives the corresponding representation-complexity reading: simultaneously landing many almost-orthogonal transported packets forces a fixed spectral/residual primitive to grow its effective rank, while a canonical packet matrix keeps bounded local degree.

The signal-processing side of the story comes from wavelet, wavepacket, and curvelet representations (Mallat, 2009; Candès and Donoho, 2004; Candès and Demanet, 2005). These constructions provide localized atoms that simultaneously carry spatial and frequency information, and the curvelet representation of wave propagators is a direct mathematical antecedent of the packet sparsity story used here. Classical microlocal analysis interprets oscillatory propagation in terms of phase-space motion and canonical relations (Alazard, 2021; Hörmander, 1985a; Zworski, 2012). FIO-inspired learning for wave-based imaging has also used this geometry as an architectural prior; FIONet (Kothari et al., 2020), for example, learns geometry associated with wave propagation in imaging inverse problems. On the pseudodifferential side, ΨDONet (Bubba et al., 2021) gives an explicit precedent for neural architectures motivated by pseudodifferential operators and wavelet structure. MiNO combines these two microlocal directions in a forward-operator setting: packet metadata become network state, canonical tubes become sparse stencil constraints, symbol amplitudes become learned local branches, and the finite truncation layer becomes machine-checkable.

There is also a methodological contrast with architectures whose primary claim is scaling or generic attention. Those models can be strong empirical baselines, but their primitive remains indirect for the microlocal regime targeted here. MiNO uses the microlocal viewpoint as the design principle: packet metadata form the state, canonical transport defines the update geometry, the ℓ^2 packet-matrix theorem supplies the operator-level error language, and the finite landing lemma supplies the machine-checked algebraic core.

The closest internal precedent is TCNO. Its learned local phase transport can be read as a low-order, local-Fourier special case of the present construction. MiNO generalizes that idea from local phase ramps to packet metadata, bounded canonical stencils, symbol modulation, and a wave/FIO theorem that explains why sparse canonical packet matrices should be the right object in short-time hyperbolic regimes.

The manuscript integrates five components that are usually separated: a concrete backbone, a rigorous discrete error law, a continuous microlocal reduction, a Lean finite-bridge companion, and an executable benchmark artifact. The resulting evidence standard is mechanism-first: canonical transport is credited only when transport corruption, metadata randomization, or identity-only transport measurably degrades field or packet diagnostics under the residual-limited protocol. The carrier-bound MiNO rows satisfy this standard on controlled wave probes, which is the empirical mechanism claim made here.

3 Discrete Phase-Space Formulation

This section turns the phase-space primitive into finite objects. A packet carries a coefficient and metadata; a reference operator moves that packet through a transport map and multiplies it by a local symbol; an approximate MiNO layer perturbs those two operations

Table 1: Positioning of MiNO relative to representative neural-operator primitives. The table is schematic: it records the primitive emphasized by each family, not every implementation detail.

Method family	Main primitive	Phase-aware	Theorem role	Machine check
FNO	global Fourier multiplier	no	operator approximation theory	no
WNO	multiscale wavelet filtering	partial scale locality	localized approximation	no
PDNO	differential / local operator structure	indirect	PDE-informed operator bias	no
TCNO	local phase transport	partial	finite transport decomposition	no
CANO	continuous attention	no	continuum stability	no
Grid-robust NO	discretization robustness	no	cross-grid stability	no
DDO-style diffusion	function-valued score dynamics	no	function-space generative modeling	no
MiNO	sparse canonical packet matrix	yes	explicit-rate wave/FIO bridge	finite bridge

and leaves a residual. These are the objects formalized in Lean and the objects used later to connect the architecture to wave/FIO theory.

3.1 Analytic packet states

Let $I = \{1, \ldots, n\}$ index a finite packet family. Each packet carries a coefficient and metadata:

$$\mathsf{p}_i = \big(a_i, x_i, \xi_i, \sigma_i\big),$$

where $a_i \in \mathbb{C}$ is the coefficient, $x_i \in \mathbb{R}^d$ is a normalized position, $\xi_i \in \mathbb{R}^d$ is a normalized local wavevector, and $\sigma_i \in \mathbb{R}_+$ is a scale parameter. In the implementation, the packet family is obtained by patchifying the input field, windowing each patch, taking a local real-FFT, and retaining a controlled set of local modes. This construction appears in `mino/models/wavepacket.py`.

For the mathematical statements below, packet coefficients and symbols are complex-valued: $a_i, s_i^\star, \hat{s}_i \in \mathbb{C}$. This is the natural coefficient type for Gabor packets, local Fourier modes, FIO phases, and half-wave propagation. The Lean companion formalizes the real finite landing layer; the manuscript uses the complex version because the same add–subtract proof only uses the complex modulus and the triangle inequality.

The analytic tokenizer is deliberately *not* fully learned. The point of the theorem spine is that packet identities and packet metadata should remain mathematically interpretable. Learning enters later, in the transport and symbol-modulation components. One may regard this as the phase-space analogue of fixing the Fourier basis in FNO while learning the multiplier; the difference is that the present basis is local and metadata-aware.

3.2 Reference transported-symbol operators

The simplest discrete operator class we need is the following.

Definition 1 (Reference transported-symbol operator) *Let $a \in \mathbb{C}^n$ be the vector of packet coefficients. A* reference transported-symbol operator *is an operator $\mathcal{T}^\star : \mathbb{C}^n \to \mathbb{C}^n$ of the form*

$$(\mathcal{T}^\star a)_i = s_i^\star \, a_{\pi^\star(i)},$$

where $\pi^\star : I \to I$ is a reference transport map and $s_i^\star \in \mathbb{C}$ is a reference symbol value attached to packet i.

The map $\pi^\star$ is the discrete stand-in for canonical propagation, while $s_i^\star$ is the stand-in for a local amplitude factor. This class is intentionally minimal. It already captures the decisive microlocal split between *where packets go* and *how packets are modulated.*

In the continuous language one would write an oscillatory operator as a Fourier integral operator plus lower-order corrections. Definition 1 is the finite packet counterpart of that decomposition and already explains the architecture.

3.3 Approximate MiNO layers

The implemented model uses a more flexible layer. Besides perturbing the transport map and the symbol, it allows a residual term that collects truncation, neglected interactions, low-order effects, and finite-frame leakage.

Definition 2 (Approximate MiNO layer) *An* approximate MiNO layer *is a map $\widehat{\mathcal{T}} : \mathbb{C}^n \to \mathbb{C}^n$ of the form*

$$(\widehat{\mathcal{T}} a)_i = \hat{s}_i \, a_{\hat{\pi}(i)} + \rho_i(a),$$

where $\hat{\pi} : I \to I$ is the approximate transport map, $\hat{s}_i \in \mathbb{C}$ is the approximate symbol, and $\rho_i(a)$ is a residual term.

The residual term is where the finite theorem makes room for practicality. In the implementation it absorbs the mismatch between the finite local packet frame and the idealized packet class, neglected off-stencil interactions, and any low-frequency remainder not represented by the transported-symbol path.

3.4 Theorem-grade MiNO subclass

The executable code separates the theorem-facing Core from the empirical Plus model, but both share the same carrier-bound packet primitive. For the mathematics in this paper we isolate the narrower subclass, denoted MiNO-*Core*, and then treat the broader engineering model MiNO-*Plus* as an empirical enlargement of the same transported-packet architecture.

Definition 3 (Theorem-grade MiNO block) *Fix a packet system at scale h with packet centers $z_i = (x_i, \xi_i)$, raw phase-space distance d, normalized packet distance $d_h = h^{-1/2} d$, normalized covering defect δ_h, canonical-tube radius parameter $\rho > 0$, and stencil budget*

$K \in \mathbb{N}$. *Thus the raw covering radius is* $h^{1/2}\delta_h$. *A* theorem-grade MiNO block *is an approximate layer of the form*

$$(\widehat{\mathcal{T}}_h a)_i = \sum_{j \in \widehat{\Pi}_h(i)} \hat{s}_{h,ij}\, a_j + \rho_{h,i}(a)$$

with the following additional restrictions:

1. ***bounded canonical stencil:*** *each output packet uses at most K source packets,*
$$\#\widehat{\Pi}_h(i) \le K;$$
2. ***canonical-tube localization:*** *every retained source packet lies inside a canonical tube of normalized radius $\rho + \delta_h$ around a learned near-canonical image,*
$$d_h\big(z_i, \widehat{\kappa}_h(z_j)\big) \le \rho + \delta_h \qquad \textit{for every } j \in \widehat{\Pi}_h(i);$$
3. ***local packet-kernel law:*** *each coefficient $\hat{s}_{h,ij}$ is produced by a metadata-local branch and may depend on the source packet and the normalized tube coordinate $\nu_{ij} = h^{-1/2}(z_i - \widehat{\kappa}_h(z_j))$; source-only multipliers are the special case with no ν dependence;*
4. ***transported field realization:*** *the inverse packet synthesis uses the learned final packet centers, and the high-frequency carrier is synthesized from transported input packet coefficients rather than from an unrestricted physical-space skip;*
5. ***small residual:*** *all nonlocal leakage, unresolved multibranch effects, and low-frequency mismatch are placed in $\rho_{h,i}(a)$ and measured separately.*

Definition 3 is the mathematical class justified by the continuous theory. The repository implements its finite analogue as MiNO-Core: a practical packet tokenizer, a top-K truncated propagation kernel, a local symbol branch, transported synthesis, a transported high-frequency carrier, and an explicit low-frequency residual. The ideal-frame versus executable-tokenizer gap is measured by packetization defects and tokenizer diagnostics. The broader empirical model MiNO-Plus keeps the same tokenizer and transported-symbol decomposition, but enlarges the propagation block with confidence-gated multi-stencil transport and adds a low-pass local field-space refinement head after synthesis.

This theorem-grade class is the bounded-stencil, tube-truncated core of the broader implementation. The special case $K = 1$ recovers the exact single-destination finite landing lemma formalized in Lean; the case $K > 1$ is the natural target predicted by packet sparsity in the continuous theory. The theorem statements attach to MiNO-Core. MiNO-Plus is an empirical superset containing MiNO-Core as the special case with unit confidence gates, no local post-synthesis refinement, and the same bounded canonical stencil.

3.5 From definitions to architecture

The architecture in `mino/models/mino.py` follows the finite definitions almost literally.

Tokenizer. The local Fourier tokenizer produces the finite packet family and metadata. The executable frame menu has five levels: `local_fft` for the fast legacy path, `gabor_gaussian` for the theorem-aligned Gaussian packet path, `multiscale_gabor` for scale stress tests, `anisotropic_gabor` for rectangular packet windows in two principal orientations, and `directional_gabor` for a richer axis-aligned directional packet family with short–long and long–short windows. The last two options give elongated-packet probes and an explicit E_{frame} hook; full curvelet/shearlet scaling would require an additional tight directional frame construction.

Canonical propagation layer. The propagation block updates packet metadata and then applies local kernel mixing in the updated metadata geometry. The repository now exposes three transport parameterizations. The backward-compatible option, `mlp_displacement`, predicts a direct displacement in (x, ξ). The intermediate option, `hamiltonian_euler`, learns a scalar Hamiltonian and applies one explicit Hamiltonian-vector-field step; it is useful as a cheap canonical-bias diagnostic, but it is not symplectic by construction. The structured option, `hamiltonian_verlet`, learns a separable or splitting-biased Hamiltonian

$$H_\theta(q, p; \tau) = V_\theta(q; \tau) + T_\theta(p; \tau), \qquad q = (y, x), \quad p = (k_y, k_x),$$

with packet features τ held fixed during the metadata update, and applies the Störmer–Verlet composition

$$p_{1/2} = p_0 - \frac{\Delta}{2}\nabla_q V_\theta(q_0; \tau), \qquad q_1 = q_0 + \Delta \nabla_p T_\theta(p_{1/2}; \tau), \qquad p_1 = p_{1/2} - \frac{\Delta}{2}\nabla_q V_\theta(q_1; \tau).$$

For fixed packet features this is a symplectic map in the metadata variables. This is an architectural bias for near-canonical metadata motion, not a general representation theorem for the nonseparable half-wave Hamiltonian $c(x)|\xi|$. The main packet-matrix theorem only assumes a learned near-canonical map with the displayed transport budget; Verlet is one clean parameterization of that budget.

In MiNO-Core the resulting mixing is explicitly top-K truncated in metadata space, so the retained interaction pattern matches the bounded-stencil hypothesis of Definition 3. The backward-compatible code path computes a dense pairwise distance matrix before masking to top-K; the theorem-aligned profile keeps the same dense distance search but performs sparse top-K aggregation after neighbor selection. Thus the mathematical stencil is bounded in both paths, while replacing dense distance evaluation by a genuine sparse-neighbor search remains a speed and memory optimization. In MiNO-Plus the same top-K primitive is retained, but the propagated message is multiplied by a learned confidence gate before local refinement. The mechanism-identifiability profile adds a wavefront-aware confidence bias: packets with larger normalized $|\xi|$ receive higher transport-gate logits, making high-frequency routing visible to the transport diagnostics. For finite-branch profiles the router can also use a frequency-sector warm start, `metadata_frequency_softmax`: learned metadata logits are added to a soft angular prior over packet wavevector sectors, giving a geometric initialization for the routing budget E_{route}.

Symbol modulation. After transport, a separate branch modulates coefficients using a metadata-conditioned MLP. This is the analogue of a local pseudodifferential symbol.

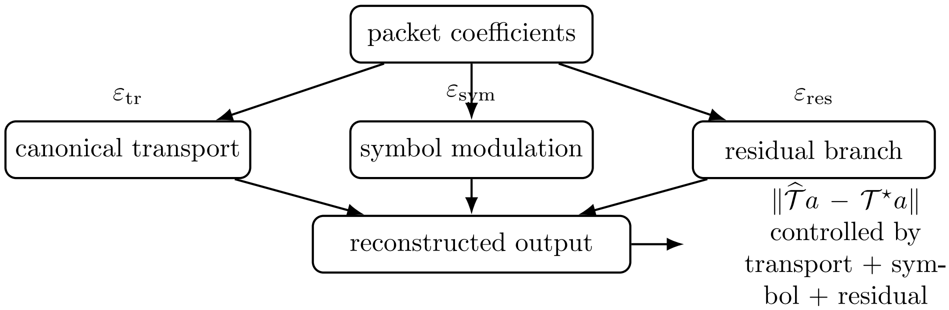

Figure 2: Theorem–architecture correspondence. The finite propagation theorem assigns one error budget to each architectural branch, making the microlocal design directly falsifiable.

Low-frequency residual. The spectral residual branch handles the identity-canonical and dissipative-symbol faces of the same calculus. A translation-invariant Fourier multiplier is the special case in which the symbol depends only on ξ; MiNO's packet symbol branch allows the more general local packet-kernel form $m_\theta(x, \xi, \nu; h)$, with the residual path absorbing smoothing remainders, finite-frame leakage, and unresolved low-frequency components. In MiNO-Plus this branch is complemented by a small local refinement head after inverse synthesis; this refinement is tracked separately as an empirical enlargement outside the theorem-facing Core class.

3.6 Why the architecture is split

The transport/symbol split is the central design claim. A generic token mixer can, in principle, represent both transport and amplitude modulation. The problem is that it does not state which of those is meant to be the primary primitive. A microlocal design should state this explicitly:

1. transport is the primary motion law;
2. symbol modulation is a lower-order amplitude law;
3. residual correction accounts for finite-frame and low-order mismatch.

The theorem in the next section makes this split quantitative.

3.7 Two useful finite baselines

Before stating the theorem, it is helpful to identify two reduced regimes.

Proposition 4 (Identity-canonical pseudodifferential reduction) *Suppose that $\hat{\pi}(i) = i$ for every packet index and that the approximate symbol is metadata-local. Then the approximate* MiNO *layer is an identity-canonical pseudodifferential action in packet coordinates:*

$$(\widehat{\mathcal{T}}a)_i = \hat{s}_i a_i + \rho_i(a).$$

If the symbol depends only on frequency metadata and the packet frame collapses to the global Fourier frame, this further reduces to a standard spectral-multiplier block.

Proof If $\hat{\pi}(i) = i$, then the displayed identity follows directly from Definition 2. Ignoring the residual for the purpose of the reduction claim, this is diagonal in the packet basis and therefore represents a packetized pseudodifferential multiplier with identity canonical relation. When the packet frame is the global Fourier frame and the symbol is x-independent, diagonal action in packet coordinates is exactly global spectral multiplication. The residual branch then becomes an ordinary smoothing or spectral remainder. ■

In smoothing regimes, MiNO specializes the same packet calculus to the identity canonical relation. Hyperbolic operators stress the transport branch, elliptic parametrices stress the local symbol branch, and parabolic updates stress dissipative symbol structure such as $m_\theta(x, \xi, \Delta t) \approx \exp[-\Delta t\, q_\theta(x, \xi)]$.

The opposite reduced regime is a global-scalar interior law, which is useful precisely because it fails in the microlocal setting.

Proposition 5 (Global-scalar obstruction) *Let two packets i, j have target outputs $y_i^\star = s_i^\star a_{\pi^\star(i)}$ and $y_j^\star = s_j^\star a_{\pi^\star(j)}$. For any scalar $c \in \mathbb{C}$,*

$$\max\{|ca_{\pi^\star(i)} - y_i^\star|,\ |ca_{\pi^\star(j)} - y_j^\star|\} \geq \frac{1}{2}\left|y_i^\star - y_j^\star\right|$$

whenever $a_{\pi^\star(i)} = a_{\pi^\star(j)} \neq 0$.

Proof Write $b = a_{\pi^\star(i)} = a_{\pi^\star(j)} \neq 0$. Then the two target outputs are $y_i^\star = s_i^\star b$ and $y_j^\star = s_j^\star b$. If both errors were strictly smaller than $\frac{1}{2}|y_i^\star - y_j^\star|$, the triangle inequality would give

$$|y_i^\star - y_j^\star| \leq |cb - y_i^\star| + |cb - y_j^\star| < \frac{1}{2}|y_i^\star - y_j^\star| + \frac{1}{2}|y_i^\star - y_j^\star|,$$

a contradiction. Therefore at least one of the two errors is at least the stated lower bound. ■

Proposition 5 is elementary, but it captures a real obstruction. Once two packets with the same incoming coefficient require different local symbol values, a single global scalar cannot match both. The point is not that scalar baselines are always bad, but that they are structurally incapable of representing packetwise amplitude separation once local wavevector variation matters.

4 The MiNO Architecture

We summarize the executable model in a form aligned with the theory.

Input and output. The default code path assumes a real-valued field on a regular grid. The tokenizer converts the input into patch-local Fourier coefficients and packet metadata. The inverse synthesis reconstructs a field from the learned coefficients. When a benchmark supplies genuine real/imaginary output channels, the training loop can additionally use

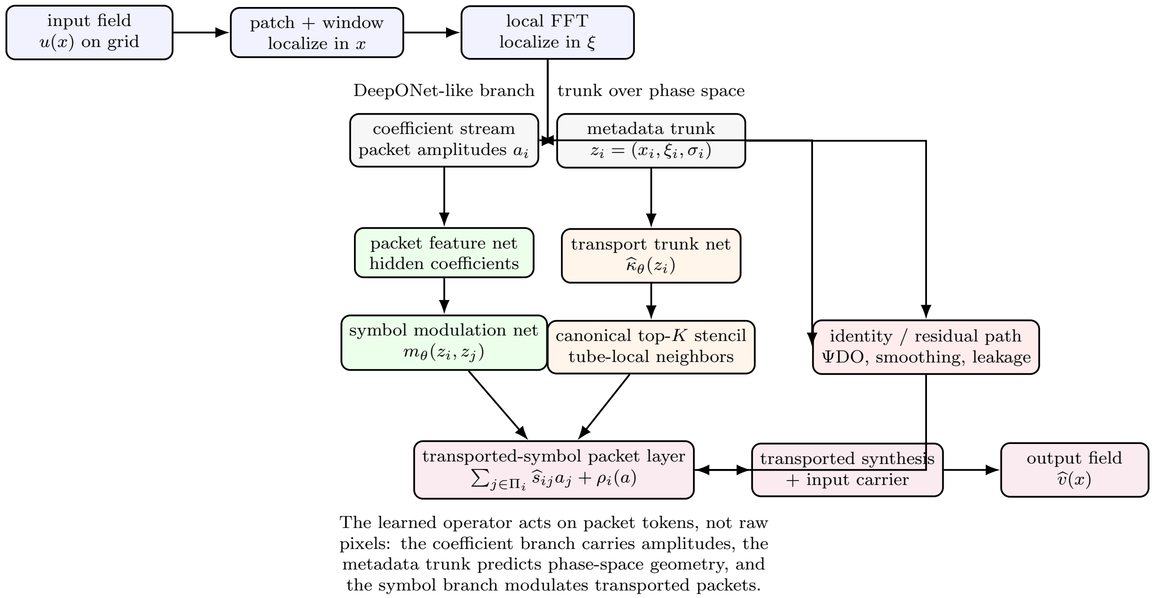

Figure 3: Neural-network view of carrier-bound MiNO. The input field is first converted into packet coefficients and packet metadata. Analogously to a branch–trunk split, the coefficient stream carries amplitudes while the metadata trunk carries phase-space state. The central layer learns a bounded canonical stencil and local symbol modulation, and the reconstructed field uses transported synthesis plus a transported high-frequency input carrier.

explicit complex-pair relative and phase/coherence losses. These losses are inert for scalar real-valued PDE rows; the artifact does not manufacture an imaginary channel from a real scalar field.

Propagation. The propagation layer can use either a metadata-conditioned displacement MLP or the theorem-aligned separable Hamiltonian Verlet update described above. The latter is the default in the mechanism-identifiability profile because it makes the transport branch symplectic by construction in packet metadata variables. In MiNO-Plus, the transport confidence gate may also receive a normalized high-frequency packet bias, so wavefront-bearing packets are pushed toward the transport branch rather than being absorbed entirely by post-synthesis refinement.

Symbol branch. The generic symbol branch takes packet features and updated metadata as input and returns scale and shift terms that modulate the propagated packet features; the shift is treated as an empirical local source/residual term, not as the theorem-facing symbol law. The theorem-grade object is a local packet kernel $s_\theta(z, \nu; h)$, where ν is the packet-radius normalized displacement inside the retained canonical tube. The implementation exposes a symbol-order parameter: the generic hyperbolic symbol branch defaults to order 0, the identity-ΨDO branch uses the configured order m, and the dissipative branch constrains the multiplier to $(0, 1]$. The stricter `spectral_order` parameterization is a source-only finite symbol proxy. It is multiplier-only: it removes feature-conditioned symbol gates and additive shifts, predicts packet multipliers from phase-space metadata, log radial frequency, and normalized frequency direction, then applies the fixed order weight $\langle\xi\rangle^m$. Equivalently, it is a finite shell-local approximation to the source-only special case

$$\hat{s}_\theta(x,\xi) = \langle\xi\rangle^m\, b_\theta(x,\hat{\xi},\log\langle\xi\rangle), \qquad \hat{\xi} = \xi/|\xi|,$$

on the compact phase window retained by the packet frame. The local-kernel theorem allows the richer dependence on ν and therefore uses the larger approximation dimension $m_\Omega + q_{\mathrm{loc}}$; the deployed `spectral_order` profile corresponds to $q_{\mathrm{loc}} = 0$. This is a finite S^m-proxy: the order is explicit, log-frequency scaling is measurable, and the theorem-aligned runner can penalize deviation from a configured target order. The code also exposes a finite first-seminorm proxy and an identity regularizer on the local tube and retained-edge symbol corrections, used as executable smoothness and attribution diagnostics. The high-frequency runner therefore treats `source_only_symbol`, `no_edge_symbol`, and `no_resolvent_phase` as separate ablations rather than assuming that every local-kernel refinement is a field-error improvement. Full $S^m_{1,0}$ enforcement would require higher symbol-seminorm control; the present branch supplies the finite symbol-calculus layer used by the mechanism profile.

Calculus branches. The current MiNO-Plus code now exposes two optional branches that make the restricted Hörmander-style interpretation explicit. The identity-ΨDO branch leaves packet metadata fixed and learns an order-weighted local symbol, representing elliptic parametrices and the x-dependent generalization of spectral multipliers. The dissipative-ΨDO branch applies a multiplier of the form

$$m_\theta(x,\xi,\Delta t) = \exp[-\Delta t\ \mathrm{softplus}(q_\theta(x,\xi))],$$

representing parabolic damping. These branches are disabled by default in legacy benchmark configurations and enabled in the calculus-identifiability campaign, so the empirical comparison can ask whether each microlocal regime is actually being used.

For nonlinear or quasilinear transport–diffusion problems, the intended calculus reading is an operator-splitting profile rather than a new unrelated architecture: an advective FIO branch transports high-frequency packets, a dissipative ΨDO branch damps packet amplitudes according to a local symbol, and the residual path carries pressure, forcing, and unresolved closure effects. This gives Navier–Stokes-style experiments a specific microlocal hook in the extension ledger, with branch ablations and coupling diagnostics deciding when the hook becomes an empirical claim.

Branch-factored oscillatory mode. Some oscillatory operators are better represented by a finite canonical relation than by a single canonical graph on the retained window. The general extension is a finite FIO-sum mode, rather than a Helmholtz-only module: multiple canonical branches predict candidate packet transports, a tokenwise router assigns packet mass across branches, and the transported synthesis/carrier/landing path is applied branchwise before summation. The router may be fully learned from metadata or initialized by the frequency-sector prior described above. The default MiNO profile remains the single-branch case $L = 1$. The optional branch-factored profile is activated when the PDE model or diagnostic campaign asserts a finite canonical-relation decomposition, and its theorem budget contains separate routing, branch cross-talk, per-branch transport/symbol, phase-factor, and finite-landing terms. A high-k Helmholtz branch probe therefore measures whether those finite-branch budgets are small at the tested resolution and training budget.

Transported synthesis and residual branch. The final MiNO field realization has two transported paths. Learned packet coefficients are synthesized at the final packet metadata, and a high-frequency carrier is synthesized from the input packet coefficients transported by the same metadata. The implementation also exposes an optional transport-conditioned finite landing decoder. Its inputs are the transported coefficient synthesis, transported input carrier, and low-pass lifted field; its convolutions are local and bias-free; and its output is multiplied by a transported-energy gate so that the learned landing correction vanishes when the transported paths vanish. Its role is to reduce the executable landing defect tracked as E_{land} (or, in the nonlearned path, E_{tsyn} and E_{car}). The low-frequency branch is a compact spectral residual block, and the post-synthesis refinement head is projected through a low-pass filter in mechanism tests. This makes the refinement path a smoothing residual: it repairs low-order field mismatch while leaving high-frequency wavefront content to packet transport. The thesis of the paper is that each operator family should expose the microlocal structure it has: canonical motion for waves, identity-canonical local symbols for elliptic parametrices, dissipative symbols for parabolic smoothing, and explicit residuals for effects outside the retained packet law.

Theorem-aligned training mode. The code exposes a stricter mechanism-identification mode in which the core is optimized before the local refinement path is released. During the warmup epochs the primary field loss is applied to the core prediction, the post-synthesis refinement parameters can be frozen, and the released refinement is low-pass filtered. This training path forces the packet transport and symbol branches to carry the high-frequency part before the empirical residual head corrects smooth mismatch.

Complexity. If n is the number of packets and w is the hidden width, the raw distance computation is quadratic in n, while the interaction kernel used by MiNO-Core and MiNO-Plus is explicitly truncated to a bounded stencil of size K. The executable propagation law realizes the logical top-K packet interaction assumed by the finite theorem. The `sparse_topk` option avoids dense aggregation after the top-K set has been found; the nearest-neighbor search remains dense.

Artifact status. The repository under `MiNO/` contains working carrier-bound MiNO-Core and MiNO-Plus PyTorch models, tests, a Lean companion, a unified scenario registry over synthetic and cached PDE families, official baseline wrappers borrowed from the TCNO benchmark code, and a staged benchmark runner. The Lean component verifies the finite landing layer; the completed branch-identifiability run shows that the final carrier-bound architecture makes transport corruption field-causal on controlled wave probes.

5 Sparse Packet-Matrix Error Decomposition

The previous section introduced the transported-symbol class as the finite version of canonical propagation. The main mathematical object is now the sparse packet matrix itself. The flagship norm is therefore the energy norm $\ell^2 \to \ell^2$ on packet coefficients, and the finite ℓ^1 identity below is used as a machine-checked landing lemma rather than as the main operator theorem. The section first records the Schur packet-matrix transfer that turns local row/column budgets into an ℓ^2 operator bound. It then gives the Lean-facing finite transported-symbol lemma, whose proof is intentionally elementary.

The finite theorem is the $K = 1$ machine-checked core of Definition 3: one dominant transported packet, one local symbol value, and one residual term. This is the exact statement formalized in Lean up to naming differences.

This finite lemma is the landing layer for the main MiNO-Core approximation theorem. The short-time wave-family bound assembled later has three additional ingredients:

1. the finite bookkeeping in the present section;
2. the continuous wave/FIO packet sparsity theorem of Section 6;
3. the PDE-family transplantation step of Section 7.

The present section therefore establishes two bridges. The Schur bridge is the operator-level packet-matrix statement used by the main theorem. The finite algebraic bridge is the proof-assistant-checked landing layer that keeps the implementation and theorem budgets aligned.

5.1 Continuum MiNO primitive

The finite architecture is best understood as a discretization of a function-space packet operator, not merely as a neural network on a fixed grid. Fix a semiclassical packet analysis/synthesis pair

$$W_h : L^2(\mathbb{R}^d) \to \ell^2(\Gamma_h), \qquad W_h^* : \ell^2(\Gamma_h) \to L^2(\mathbb{R}^d),$$

where Γ_h is a compact phase-space packet cloud with packet radius comparable to $h^{1/2}$. We use d for raw phase-space distance and $d_h = h^{-1/2}d$ for packet-radius normalized distance. The symbol δ_h denotes the normalized covering defect, so the corresponding raw covering radius is $h^{1/2}\delta_h$. The continuum MiNO-Core primitive at scale h is

$$\mathcal{M}_{\theta,h}u = W_h^* \left[\mathcal{T}_{\widehat{\kappa}_\theta,\widehat{s}_\theta}(W_h u) + R_{\theta,h}(W_h u)\right].$$

Here $\widehat{\kappa}_\theta$ is the learned near-canonical packet map and $\widehat{s}_\theta$ is the learned local packet kernel. On a retained canonical tube it is better to write this symbol as

$$\widehat{s}_\theta(z,\nu;h), \qquad \nu = h^{-1/2}\big(z_\gamma - \widehat{\kappa}_\theta(z_\eta)\big),$$

where z_η is the source packet, z_γ is the output packet, and ν is the packet-radius normalized local tube coordinate. A source-only multiplier is the special case in which the dependence on ν is suppressed. The discrete implementation replaces W_h by the executable packet tokenizer, replaces the continuum stencil by top-K packet neighbors in transported metadata geometry, and approximates $\widehat{\kappa}_\theta$ and $\widehat{s}_\theta$ by finite neural branches.

This definition is the function-space primitive that corresponds to the Neural Operator/CANO style of formulation. Theorem 6 is the packet-matrix bridge for this primitive, while Theorem 8 is only the finite landing lemma. The main Microlocal Propagation Theorem, stated later as Theorem 36, is the field-level statement that a short-time half-wave propagator lies near such a continuum packet primitive and that the finite MiNO-Core realization inherits an explicit error law.

5.2 Operator-level packet-matrix bridge

Theorem 6 (Schur packet-matrix transfer) *Let $A, \widehat{A} : \ell^2(\Gamma_h) \to \ell^2(\Gamma_h)$ be two packet matrices on the active compact packet window. Assume their difference $D = \widehat{A} - A$ has at most K_{row} significant entries per row and at most K_{col} significant entries per column after discarding a tail matrix D_{tail}. Suppose the retained entries satisfy the uniform envelope*

$$|D_{\gamma\eta}| \le E_{\text{entry}} := C_{\text{tr}}\big(\epsilon_{\kappa,h} + \delta_h\big) + C_{\text{sym}}\epsilon_{\text{sym}} + C_{\text{res}}\epsilon_{\text{res}},$$

and the discarded part satisfies

$$\|D_{\text{tail}}\|_{\ell^2\to\ell^2} \le E_{\text{tail}} + E_{\text{frame}} + E_{\text{asymp}} + C_{\text{cov}}\delta_h.$$

Then

$$\|\widehat{A} - A\|_{\ell^2\to\ell^2} \le \sqrt{K_{\text{row}}K_{\text{col}}}\, E_{\text{entry}} + E_{\text{tail}} + E_{\text{frame}} + E_{\text{asymp}} + C_{\text{cov}}\delta_h.$$

In the bounded-degree canonical stencil regime $K_{\text{row}}, K_{\text{col}} \le C_{\text{deg}}K$, this gives

$$\|\widehat{A}-A\|_{\ell^2\to\ell^2} \le CK\Big[C_{\text{tr}}\big(\epsilon_{\kappa,h}+\delta_h\big)+C_{\text{sym}}\epsilon_{\text{sym}}+C_{\text{res}}\epsilon_{\text{res}}\Big]+E_{\text{tail}}+E_{\text{frame}}+E_{\text{asymp}}+C_{\text{cov}}\delta_h.$$

Proof Schur's test gives

$$\|D_{\text{ret}}\|_{\ell^2\to\ell^2} \le \Big(\sup_{\gamma}\sum_{\eta}|D_{\gamma\eta}|\Big)^{1/2}\Big(\sup_{\eta}\sum_{\gamma}|D_{\gamma\eta}|\Big)^{1/2}.$$

The row sum is at most $K_{\text{row}}E_{\text{entry}}$ and the column sum is at most $K_{\text{col}}E_{\text{entry}}$. Adding the tail, frame, asymptotic, and covering terms by the triangle inequality gives the first estimate. The second follows from the bounded-degree hypothesis. ■

Theorem 6 is the norm upgrade used by the main PDE theorem. It is still finite, but it is an operator theorem rather than a coefficient-counting theorem. The old ℓ^1 bound below remains useful because it is the exact algebra checked in Lean and because it identifies the same transport, symbol, and residual budgets at the level of individual landed packets.

5.3 Assumptions

Let $a \in \mathbb{C}^n$ be an input coefficient vector. Let $\mathcal{T}^\star$ be the reference transported-symbol operator from Definition 1, and let $\widehat{\mathcal{T}}$ be the approximate MINO layer from Definition 2. We assume the following coefficient-level budgets for some nonnegative constants

$$\varepsilon_{\text{tr}}, \qquad \varepsilon_{\text{sym}}, \qquad \varepsilon_{\text{res}}, \qquad B_a, \qquad B_s.$$

Assumption 7 (Finite transport, symbol, and residual budgets) *For every packet index $i \in I$,*

$$\big|a_{\hat\pi(i)} - a_{\pi^\star(i)}\big| \le \varepsilon_{\text{tr}}, \tag{1}$$

$$\big|\hat s_i - s_i^\star\big| \le \varepsilon_{\text{sym}}, \tag{2}$$

$$\big|\rho_i(a)\big| \le \varepsilon_{\text{res}}, \tag{3}$$

$$\big|a_{\hat\pi(i)}\big| \le B_a, \tag{4}$$

$$\big|s_i^\star\big| \le B_s. \tag{5}$$

Assumption 7 is exactly the form one expects from a finite packet approximation. Equation (1) measures transport mismatch after a finite matching has already been chosen, (2) measures symbol mismatch, and (3) measures residual truncation. Bounds (4) and (5) are envelope assumptions used to convert those mismatches into coefficient errors. All absolute values in this section are complex moduli. The proof is therefore the same triangle-inequality proof as in the real finite Lean specialization.

It is important not to confuse this finite coefficient budget with the geometric transport budget used in the main Microlocal Propagation Theorem. For the continuous theorem, the natural quantity is a canonical-map or packet-kernel mismatch, for example

$$\widetilde{\varepsilon}_\kappa := \sup_{z\in\Omega_T} d\big(\widehat{\kappa}_\theta(z), \kappa_t(z)\big), \qquad \varepsilon_{\kappa,h} := h^{-1/2}\widetilde{\varepsilon}_\kappa, \qquad \varepsilon_{\mathrm{ker}} := \sup_\eta \sum_\gamma \big|\widehat{K}^{\mathrm{tr}}_{\gamma\eta} - K^{\star,\mathrm{tr}}_{\gamma\eta}\big|.$$

Here $\widetilde{\varepsilon}_\kappa$ is measured in raw phase-space units, while $\varepsilon_{\kappa,h}$ is the corresponding packet-radius normalized error. $K^{\star,\mathrm{tr}}$ and $\widehat{K}^{\mathrm{tr}}$ are the reference and learned transported packet kernels before the residual is added. Under the standard packet-overlap Lipschitz estimate on the compact phase window, the classical bridge has the form

$$\varepsilon_{\mathrm{ker}} \le C_{\mathrm{ov}}\big(\varepsilon_{\kappa,h} + \delta_h\big) = C_{\mathrm{ov}}\big(h^{-1/2}\widetilde{\varepsilon}_\kappa + \delta_h\big).$$

Thus the main theorem uses geometric or kernel transport error. The coefficient-level Assumption 7 is only the finite landing lemma consumed after this continuous packet estimate has already produced a retained finite stencil.

5.4 Finite algebraic core

Theorem 8 (Finite transported-symbol perturbation bound) *Under Assumption 7, the pointwise coefficient error satisfies*

$$\big|(\widehat{\mathcal{T}}a)_i - (\mathcal{T}^\star a)_i\big| \le B_s\,\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}\,B_a + \varepsilon_{\mathrm{res}} \qquad \textit{for every } i \in I.$$

Consequently, the induced ℓ^1 coefficient error satisfies

$$\|\widehat{\mathcal{T}}a - \mathcal{T}^\star a\|_{\ell^1} \le n\Big(B_s\,\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}\,B_a + \varepsilon_{\mathrm{res}}\Big).$$

Proof Fix $i \in I$. By Definitions 1 and 2,

$$(\mathcal{T}^\star a)_i = s_i^\star a_{\pi^\star(i)}, \qquad (\widehat{\mathcal{T}}a)_i = \hat{s}_i a_{\hat{\pi}(i)} + \rho_i(a).$$

Subtracting and adding the term $s_i^\star a_{\hat{\pi}(i)}$ gives

$$\begin{aligned}(\widehat{\mathcal{T}}a)_i - (\mathcal{T}^\star a)_i &= \hat{s}_i a_{\hat{\pi}(i)} + \rho_i(a) - s_i^\star a_{\pi^\star(i)} \\ &= s_i^\star\big(a_{\hat{\pi}(i)} - a_{\pi^\star(i)}\big) + (\hat{s}_i - s_i^\star)a_{\hat{\pi}(i)} + \rho_i(a).\end{aligned}$$

Taking absolute values and applying the triangle inequality,

$$\big|(\widehat{\mathcal{T}}a)_i - (\mathcal{T}^\star a)_i\big| \le \big|s_i^\star\big|\,\big|a_{\hat{\pi}(i)} - a_{\pi^\star(i)}\big| + \big|\hat{s}_i - s_i^\star\big|\,\big|a_{\hat{\pi}(i)}\big| + \big|\rho_i(a)\big|.$$

Assumption 7 then yields

$$\left|(\widehat{\mathcal{T}}a)_i - (\mathcal{T}^\star a)_i\right| \le B_s\,\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}\,B_a + \varepsilon_{\mathrm{res}}.$$

Summing over all $i \in I$ gives

$$\begin{aligned}\|\widehat{\mathcal{T}}a - \mathcal{T}^\star a\|_{\ell^1} &= \sum_{i=1}^{n}\left|(\widehat{\mathcal{T}}a)_i - (\mathcal{T}^\star a)_i\right| \\ &\le \sum_{i=1}^{n}\Big(B_s\,\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}\,B_a + \varepsilon_{\mathrm{res}}\Big) \\ &= n\Big(B_s\,\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}\,B_a + \varepsilon_{\mathrm{res}}\Big).\end{aligned}$$

This proves both statements. ■

In plain language: if the packet goes to approximately the right phase-space location, arrives with approximately the right amplitude, and leaves only a small residual, then the layer approximates the reference transported-symbol law. The proof decomposition is complete: every error falls into one of the three architectural categories:

1. the packet goes to the wrong place or wrong local wavevector state;
2. the packet arrives with the wrong symbol value;
3. the neglected residual is too large.

This is the algebraic meaning of the microlocal primitive. The PDE approximation theorem is obtained by adding the continuous packet theorem that supplies a sparse transported-symbol reference law and the learned-network estimates that make the three budgets small.

Example. Take $n = 4$ packets with $B_a = B_s = 1$. Suppose the approximate transport has budget $\varepsilon_{\mathrm{tr}} = 0.10$, the symbol branch has budget $\varepsilon_{\mathrm{sym}} = 0.05$, and the residual branch has budget $\varepsilon_{\mathrm{res}} = 0.02$. The pointwise coefficient envelope is

$$E_\infty = B_s\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}B_a + \varepsilon_{\mathrm{res}} = 0.17,$$

and the conservative ℓ^1 bound is $4E_\infty = 0.68$. More importantly, the decomposition identifies the branch-level error that would dominate if the finite budgets were realized by training: here the transport budget contributes 0.10 of the 0.17 pointwise envelope. The theorem is therefore useful as an error language for diagnostics and architecture design, while the empirical ablations determine which budgets are actually controlled by the trained model.

5.5 Approximation corollary

Corollary 9 (Finite budget corollary) *Under Assumption 7, if*

$$n\Big(B_s\,\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}\,B_a + \varepsilon_{\mathrm{res}}\Big) \le \varepsilon,$$

then

$$\|\widehat{\mathcal{T}}a - \mathcal{T}^\star a\|_{\ell^1} \le \varepsilon.$$

Proof This is immediate from Theorem 8. ■

Corollary 9 is simple but important. It turns the theorem into an explicit training target. If the learned transport map, learned symbol branch, and residual branch can drive their respective budgets below the stated threshold, the finite operator error falls below ε. The conditional language is essential: this corollary does not prove that a particular neural network optimizer will learn the budgets. It states the exact finite consequence once transport, symbol, and residual errors have been controlled.

Budget assumptions versus network approximation. The learned-budget hypotheses in Assumption 7 are where approximation theory and empirical diagnostics enter. Section 6.6 supplies the rate theorem for the flagship short-time wave family: under compact-window C^r assumptions on the canonical map κ_t and transported symbol s_t, the metadata and symbol branches approximate those objects with explicit neural-network rates, which then feed into the finite bound here. The theorem establishes realizability of the PDE operator class by the theorem-grade MiNO-Core architecture; the proxy-loss and ablation protocol measure whether training drives the relevant budgets down.

5.6 Bounded canonical stencil extension

The previous theorem is the exact $K = 1$ core. The next theorem is the finite bounded-stencil extension that matches the revised theorem-grade MiNO subclass more closely.

Theorem 10 (Bounded canonical stencil propagation) *Fix a stencil budget $K \in \mathbb{N}$. Let*

$$(\mathcal{T}^{\star(K)}a)_i = \sum_{m=1}^{K} s^{\star}_{i,m}\, a_{\pi^{\star}_m(i)}, \qquad (\widehat{\mathcal{T}}^{(K)}a)_i = \sum_{m=1}^{K} \hat{s}_{i,m}\, a_{\hat{\pi}_m(i)} + \rho_i(a).$$

Assume that for every packet index i and slot m,

$$\begin{aligned} \left|a_{\hat{\pi}_m(i)} - a_{\pi^{\star}_m(i)}\right| &\le \varepsilon_{\mathrm{tr}}, \\ \left|\hat{s}_{i,m} - s^{\star}_{i,m}\right| &\le \varepsilon_{\mathrm{sym}}, \\ \left|a_{\hat{\pi}_m(i)}\right| &\le B_a, \\ \left|s^{\star}_{i,m}\right| &\le B_s, \end{aligned}$$

and that $|\rho_i(a)| \le \varepsilon_{\mathrm{res}}$ for every i. Then

$$\left|(\widehat{\mathcal{T}}^{(K)}a)_i - (\mathcal{T}^{\star(K)}a)_i\right| \le K\big(B_s\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}B_a\big) + \varepsilon_{\mathrm{res}}$$

for every i, and consequently

$$\|\widehat{\mathcal{T}}^{(K)}a - \mathcal{T}^{\star(K)}a\|_{\ell^1} \le n\Big(K\big(B_s\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}B_a\big) + \varepsilon_{\mathrm{res}}\Big).$$

Proof Subtract the two stencil actions and add-subtract the mixed terms $s^{\star}_{i,m}a_{\hat{\pi}_m(i)}$ slot-wise. For each fixed output packet i, this gives

$$(\widehat{\mathcal{T}}^{(K)}a)_i - (\mathcal{T}^{\star(K)}a)_i = \sum_{m=1}^{K}\Big[s^{\star}_{i,m}\big(a_{\hat{\pi}_m(i)} - a_{\pi^{\star}_m(i)}\big) + (\hat{s}_{i,m} - s^{\star}_{i,m})a_{\hat{\pi}_m(i)}\Big] + \rho_i(a).$$

Taking absolute values, applying the triangle inequality, and using the uniform per-slot envelopes yields

$$\big|(\widehat{\mathcal{T}}^{(K)}a)_i - (\mathcal{T}^{\star(K)}a)_i\big| \le \sum_{m=1}^{K}\big(B_s\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}B_a\big) + \varepsilon_{\mathrm{res}},$$

which is the first claim. Summing over i gives the ℓ^1 bound. ■

Theorem 10 is the exact finite bridge between the machine-checked core and the continuous bounded-stencil theorem of Section 6. It says that once a canonical tube has reduced the true packet matrix to at most K significant interactions per output packet, the same transport/symbol/residual bookkeeping survives with only a factor of K.

5.7 Quantitative cross-resolution transfer

The theorem above controls one fixed packet system. For neural operators, however, a central scientific question is whether a packetized model trained or validated on one resolution transfers in a controlled way to another. The next theorem gives an explicit finite rate statement for the simplest such protocol.

Let coarse packet coefficients live in $\mathbb{C}^{n_c}$ and fine packet coefficients in $\mathbb{C}^{n_f}$. Let

$$R_{f\to c} : \mathbb{C}^{n_f} \to \mathbb{C}^{n_c}, \qquad P_{c\to f} : \mathbb{C}^{n_c} \to \mathbb{C}^{n_f}$$

denote restriction and prolongation maps. Let $\mathcal{T}_c^\star$ be a coarse reference transported-symbol operator, $\mathcal{T}_f^\star$ a fine reference operator, and $\widehat{\mathcal{T}}_c$ an approximate coarse MiNO layer. The transferred fine predictor is

$$P_{c\to f}\widehat{\mathcal{T}}_c R_{f\to c}.$$

Theorem 11 (Quantitative cross-resolution transfer) *Fix a fine input coefficient vector $a_f \in \mathbb{C}^{n_f}$ and suppose that:*

1. *the prolongation map is ℓ^1-stable with constant $\kappa \ge 0$, meaning*

$$\|P_{c\to f}x - P_{c\to f}y\|_{\ell^1} \le \kappa\|x-y\|_{\ell^1} \qquad \text{for all } x, y \in \mathbb{C}^{n_c};$$

2. *the coarse approximation error obeys*

$$\|\widehat{\mathcal{T}}_c R_{f\to c}a_f - \mathcal{T}_c^\star R_{f\to c}a_f\|_{\ell^1} \le C_c h^\beta;$$

3. *the reference transfer defect obeys*

$$\|P_{c\to f}\mathcal{T}_c^\star R_{f\to c}a_f - \mathcal{T}_f^\star a_f\|_{\ell^1} \le C_t h^\alpha.$$

Then the transferred coarse predictor satisfies

$$\|P_{c\to f}\widehat{\mathcal{T}}_c R_{f\to c}a_f - \mathcal{T}_f^\star a_f\|_{\ell^1} \le \kappa C_c h^\beta + C_t h^\alpha.$$

Proof Write

$P_{c\to f}\widehat{\mathcal{T}}_c R_{f\to c}a_f - \mathcal{T}_f^\star a_f = \big(P_{c\to f}\widehat{\mathcal{T}}_c R_{f\to c}a_f - P_{c\to f}\mathcal{T}_c^\star R_{f\to c}a_f\big) + \big(P_{c\to f}\mathcal{T}_c^\star R_{f\to c}a_f - \mathcal{T}_f^\star a_f\big).$

Set

$$\Delta_{\mathrm{model}} := \|P_{c\to f}\widehat{\mathcal{T}}_c R_{f\to c}a_f - P_{c\to f}\mathcal{T}_c^\star R_{f\to c}a_f\|_{\ell^1}, \qquad \Delta_{\mathrm{ref}} := \|P_{c\to f}\mathcal{T}_c^\star R_{f\to c}a_f - \mathcal{T}_f^\star a_f\|_{\ell^1}.$$

Applying the triangle inequality and then the ℓ^1 stability of $P_{c\to f}$ gives

$$\begin{aligned}\|P_{c\to f}\widehat{\mathcal{T}}_c R_{f\to c}a_f - \mathcal{T}_f^\star a_f\|_{\ell^1} &\le \Delta_{\mathrm{model}} + \Delta_{\mathrm{ref}} \\ &\le \kappa\|\widehat{\mathcal{T}}_c R_{f\to c}a_f - \mathcal{T}_c^\star R_{f\to c}a_f\|_{\ell^1} + \Delta_{\mathrm{ref}}.\end{aligned}$$

The two assumed rate bounds then yield the claim. ■

The scale-indexed theorem is materially stronger than the single-resolution bound: it introduces a scale parameter, separates *model error on the coarse packet system* from *reference inconsistency across packet resolutions*, and yields an explicit transfer rate. In the PDE sections below, microlocal analysis justifies decay of the reference defect term for suitably nested packet systems and short-time wave-type propagators.

Corollary 12 (Machine-checked bounded-stencil transfer bridge) *Fix a stencil budget $K \in \mathbb{N}$ and let the coarse operator be a theorem-grade* MiNO *block*

$$(\widehat{\mathcal{T}}_c^{(K)}a)_i = \sum_{m=1}^{K} \hat{s}_{i,m}\, a_{\hat{\pi}_m(i)} + \rho_i(a)$$

with reference coarse operator

$$(\mathcal{T}_c^{\star(K)}a)_i = \sum_{m=1}^{K} s^\star_{i,m}\, a_{\pi^\star_m(i)}.$$

Assume the slotwise transport, symbol, residual, coefficient, and reference-symbol envelopes of Theorem 10. Assume also that the prolongation map is ℓ^1-stable with constant $\kappa \ge 0$ and that the lifted coarse reference differs from the fine reference by a tail-plus-cover defect:

$$\big\|P_{c\to f}\mathcal{T}_c^{\star(K)}R_{f\to c}a_f - \mathcal{T}_f^\star a_f\big\|_{\ell^1} \le E_{\mathrm{tail}} + E_{\mathrm{cover}}.$$

Then

$$\big\|P_{c\to f}\widehat{\mathcal{T}}_c^{(K)}R_{f\to c}a_f - \mathcal{T}_f^\star a_f\big\|_{\ell^1} \le \kappa\, n_c\Big(K(B_s\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}B_a) + \varepsilon_{\mathrm{res}}\Big) + E_{\mathrm{tail}} + E_{\mathrm{cover}}.$$

If in addition

$$n_c\Big(K(B_s\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}B_a) + \varepsilon_{\mathrm{res}}\Big) \le C_c h^\beta, \qquad E_{\mathrm{tail}} + E_{\mathrm{cover}} \le C_t h^\alpha,$$

then

$$\big\|P_{c\to f}\widehat{\mathcal{T}}_c^{(K)}R_{f\to c}a_f - \mathcal{T}_f^\star a_f\big\|_{\ell^1} \le \kappa C_c h^\beta + C_t h^\alpha.$$

Proof Combine Theorem 10 with Theorem 11. The first theorem converts the coarse theorem-grade MiNO budgets into an explicit coarse ℓ^1 error. The second theorem lifts that error across resolutions and adds the reference tail-plus-cover defect. The scale-indexed statement is the same composition with the stated rate envelopes. ■

5.8 Interpretation

The theorem separates *representation* from *estimation*. It does not say how the approximate transport and symbol should be learned; it says that once they are learned to a certain budget, the resulting operator error is controlled. This is the finite analogue of why one studies Fourier integral operators and pseudodifferential operators separately from numerical schemes that approximate them.

The theorem also clarifies why the transported-symbol split is more than an aesthetic preference. If one uses a generic token mixer, the transport and symbol terms are entangled. The present theorem shows that there is value in disentangling them, because the resulting budgets have different meanings and different PDE interpretations:

- transport error corresponds to phase-space drift;
- symbol error corresponds to amplitude or lower-order modulation error;
- residual error corresponds to finite-frame leakage and neglected interactions.

The cross-resolution theorem adds a second distinction that is crucial for neural operators: even when the coarse model error is small, the lifted predictor can still fail if the packet system itself does not transfer coherently across resolutions. This is why the manuscript treats packet resolution and packet transfer as mathematical objects rather than only as engineering details.

5.9 Empirical packet-budget bridge

The finite theorem uses abstract budgets $(\varepsilon_{\mathrm{tr}}, \varepsilon_{\mathrm{sym}}, \varepsilon_{\mathrm{res}})$. For benchmarks without a known reference canonical graph or symbol, the experiments estimate the three terms through packet-level proxy measurements in the same transport–symbol–residual language.

For an input field u, target field v, and model prediction $\widehat{v}$, the code constructs packet encodings $W_h u$, $W_h v$, and the diagnostic packet output produced by MiNO. It then records three proxies:

$$\begin{aligned}
\widehat{\varepsilon}_{\mathrm{tr}} &= \text{energy-weighted packet transport mismatch},\\
\widehat{\varepsilon}_{\mathrm{sym}} &= \text{post-alignment packet amplitude mismatch},\\
\widehat{\varepsilon}_{\mathrm{res}} &= \|\widehat{v} - v\|_2 / \|v\|_2.
\end{aligned}$$

These proxies are deliberately weaker than the theorem hypotheses. They do not certify that Assumption 7 holds. They do, however, give the falsifiable diagnostic that the theorem demands: if canonical propagation is the right primitive, improvements in field error on transport-sensitive regimes should usually be accompanied by lower transport or symbol proxies, not only by a larger physical-space refinement head.

Identifiability constraint. The theorem also explains why the empirical architecture must separate the microlocal branches from the residual branch. If an unrestricted physical-space refinement head is allowed to solve the whole task, the measured field error can improve while the transport and symbol budgets remain unidentified. The current runner therefore includes an identifiable variant of MiNO-Plus in which the final refinement is treated as a small residual:

$$\widehat{u} = \widehat{u}_{\mathrm{core}} + r_\theta(x)\,\eta_\theta(x), \qquad \|r_\theta \eta_\theta\|_2^2 / \|u\|_2^2 \text{ penalized.}$$

The training objective adds

$$\lambda_{\mathrm{core}}\|\widehat{u}_{\mathrm{core}}-u\|_2^2+\lambda_{\mathrm{res}}\frac{\|r_\theta\eta_\theta\|_2^2}{\|u\|_2^2}+\lambda_{\mathrm{route}}\|r_\theta\|_1+\lambda_{\mathrm{ord}}\big(\widehat{m}_\theta-m_{\mathrm{target}}\big)^2$$

to the packet transport and symbol proxies. Here $\widehat{m}_\theta$ is the executable log-frequency scaling proxy of the symbol branch. It is a finite counterpart of the full $S^m_{1,0}$ seminorm target and makes the symbol-order claim measurable in the runner. This is the empirical counterpart of the residual and symbol budgets in Theorem 8. It makes the ablation test meaningful: if the field error remains unchanged after removing transport or symbol under this residual-limited, order-aware protocol, then the corresponding branch is not identified by that experiment.

The stricter runner also uses a core-first schedule. For the first warmup epochs the field loss is applied to $\widehat{u}_{\mathrm{core}}$ rather than the full MiNO-Plus output, and the post-synthesis refinement parameters may be frozen. After warmup, the refinement head is released as a low-pass residual. This schedule is not part of the theorem; it is an optimization constraint designed to make the theorem's sufficient mechanism reachable by training.

5.10 From coefficient budgets to field-level interpretation

The finite ℓ^1 corollaries above state a conservative worst-case bound. The factor n appears because the theorem sums a uniform pointwise coefficient envelope over all packet indices. The empirical regime is better described by the packet-local law

$$\big|(\widehat{\mathcal{T}}a)_i-(\mathcal{T}^\star a)_i\big|\le E_\infty:=B_s\varepsilon_{\mathrm{tr}}+\varepsilon_{\mathrm{sym}}B_a+\varepsilon_{\mathrm{res}},$$

and this estimate is independent of the ambient packet count.

To translate coefficient errors back to fields, let W_h^* denote the packet synthesis map and suppose the frame has an upper synthesis bound

$$\|W_h^*e\|_{L^2}\le B_W\|e\|_{\ell^2}.$$

If the error is supported on an active packet set of size s after canonical truncation, then

$$\|W_h^*e\|_{L^2}\le B_W\sqrt{s}\,E_\infty.$$

For the bounded-stencil theorem, the relevant support size is controlled by active input mass and the retained canonical stencil, not by the full ambient discretization. The continuous packet sparsity theorem in Section 6 is precisely the mechanism that makes this distinction meaningful: off-graph decay turns the dense packet matrix into a sparse canonical stencil plus a tail. Thus the ℓ^1 bounds are the safest machine-checked finite statements, while the field-level interpretation depends on frame bounds, active packet support, and truncation rates.

5.11 Machine-checked status

The finite landing spine is formalized in Lean 4 across the following modules:

- `FinitePacket.lean`, `Propagation.lean`, and `StencilPropagation.lean`;

- `Approximation.lean`, `Reduction.lean`, and `Obstruction.lean`;
- `Transplantation.lean`;
- `ResolutionTransfer.lean`;
- `StencilWaveTransfer.lean`.

The Lean theorem names are written against finite coefficient functions over `Fin n`. The manuscript statement above matches those files in substance:

- `pointwise_propagation_bound` corresponds to the pointwise estimate in Theorem 8;
- `l1_propagation_bound` corresponds to the ℓ^1 estimate in Theorem 8;
- `pointwise_stencil_propagation_bound` and `l1_stencil_propagation_bound` correspond to Theorem 10;
- `finite_approximation_corollary` corresponds to Corollary 9.
- `cross_resolution_transfer_bound` and `cross_resolution_transfer_rate` correspond to Theorem 11.
- `bounded_stencil_transfer_bound` and `bounded_stencil_wave_transfer_rate` correspond to Corollary 12.

The machine-checked spine establishes an important precedent: a neural-operator architecture can be organized around a proof-assistant-verified finite core. For short-time hyperbolic propagators, classical FIO theory produces exactly the transported-symbol and sparse canonical-stencil structure that the finite theorem assumes.

6 Continuous Microlocal Extension

Why should a learned packet layer follow a transported-symbol law at all? The answer comes from classical microlocal analysis. Short-time hyperbolic propagators are Fourier integral operators associated with canonical graphs, and their packet matrices are concentrated near those graphs. This section turns that classical fact into the continuous bridge used by the paper.

The finite landing spine is machine-checked. The continuous layer explains why sparse canonical packet matrices are natural for a concrete PDE family. The short-time wave propagator furnishes the first main continuous theorem, which in turn feeds an abstract canonical-graph packet sparsity theorem. The result is a dual spine:

1. a *short-time half-wave off-graph decay theorem*;
2. an *abstract canonical-graph packet sparsity theorem*.

Together they explain why the theorem-grade MiNO class should use a bounded canonical stencil rather than either a dense interaction map or a purely single-destination law.

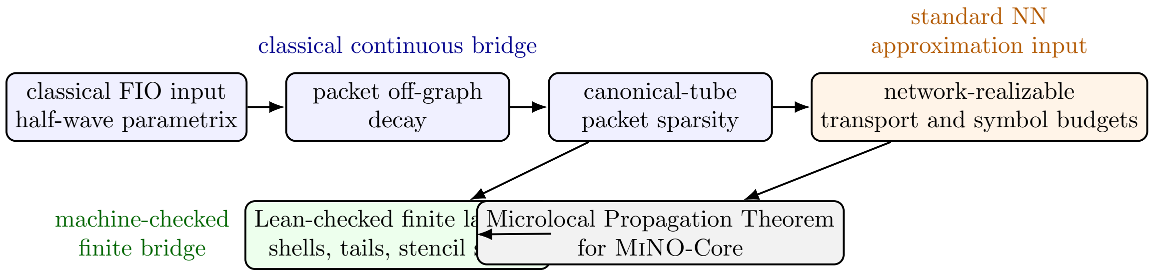

Figure 4: Proof-boundary map for the continuous extension. The oscillatory-integral and packet almost-diagonalization inputs are classical; the shell/tail/stencil landing layer is the machine-checked part; the learned transport and symbol terms enter through compact-window neural approximation budgets.

6.1 Classical ingredients

Write $S^m = S^m_{1,0}$ for the standard Hörmander symbol class on $\mathbb{R}^d_x \times \mathbb{R}^d_\xi$, and $\mathrm{Op}(a)$ for the Kohn–Nirenberg quantization of $a \in S^m$. We use the following standard ingredients:

1. Calderón–Vaillancourt continuity for $\mathrm{Op}(a)$ with $a \in S^0$;
2. symbolic calculus for compositions of pseudodifferential operators;
3. short-time Fourier-integral parametrices for hyperbolic propagators generated by real principal symbols;
4. packet almost-diagonalization for canonical-graph Fourier integral operators;
5. non-stationary phase estimates obtained by repeated integration by parts.

The relevant references are Hörmander (1985a,b); Zworski (2012); Alazard (2021); Candès and Demanet (2005). The contribution of the present section is not to replace those results. It is to organize them around the exact packet objects used by MiNO and to expose the proof chain in a form that can be mirrored by a Lean-friendly finite abstraction.

6.2 Packet frames, shells, and canonical tubes

Fix a semiclassical scale $h \in (0, h_0]$ and a wavepacket frame $\{\varphi^h_\gamma\}_{\gamma\in\Gamma_h}$ with packet centers

$$z_\gamma = (x_\gamma, \xi_\gamma) \in T^*\mathbb{R}^d,$$

analysis map $W_h u = (\langle u, \varphi^h_\gamma\rangle)_\gamma$, synthesis map W_h^*, and bounded overlap. Let d denote raw phase-space distance and set $d_h = h^{-1/2}d$. Throughout this section δ_h is the *normalized* covering defect, so the raw covering radius is $h^{1/2}\delta_h$. For a canonical map κ and a source packet η, we measure output packets relative to the propagated center $\kappa(z_\eta)$ using the shell index

$$\ell_\kappa(\gamma, \eta) := \Big\lfloor h^{-1/2}\,\mathrm{dist}\big(z_\gamma, \kappa(z_\eta)\big)\Big\rfloor.$$

The corresponding canonical tube of radius parameter ρ is the set of packets with shell index at most ρ. The finite Lean files introduced in this revision formalize exactly this shell-and-tube bookkeeping layer, independently of the continuous oscillatory-integral input.

Assumption 13 (Quantitative packet-frame constants) *Throughout the continuous bridge, the packet family is restricted to a compact phase window $\Omega \subset T^*\mathbb{R}^d$ and satisfies the following scale-uniform bounds:*

1. Frame stability: *there are constants $0 < A_W \le B_W < \infty$, independent of h, such that*

$$A_W \|u\|_{L^2}^2 \le \sum_{\gamma \in \Gamma_h} |\langle u, \varphi_\gamma^h \rangle|^2 \le B_W \|u\|_{L^2}^2$$

   for all u microlocalized in Ω.

2. Covering: *the packet centers cover Ω at raw radius $h^{1/2}\delta_h$, with $\delta_h \le C_{\mathrm{cov}}$.*

3. Counting: *canonical balls of radius $Rh^{1/2}$ contain at most $C_{\mathrm{cnt}} R^q$ packet centers.*

4. Uniform localization: *packet envelopes and their derivatives up to the order used in stationary and non-stationary phase are bounded by constants collected in $C_{\mathrm{loc},N}$ after the standard $h^{1/2}$ rescaling.*

The constants in the continuous theorems below are allowed to depend on

$$\mathfrak{C}_{\mathrm{frame},N} := (A_W^{-1}, B_W, C_{\mathrm{cov}}, C_{\mathrm{cnt}}, C_{\mathrm{loc},N})$$

and on the compact-window symbol and flow seminorms. They are not allowed to depend on h, the number of packets, the retained stencil size K, or the branch parameter budgets. This convention is the quantitative version of the packetization defect tracked in the experiments: changing from local FFT packets to Gaussian/Gabor or multiscale Gabor packets changes $\mathfrak{C}_{\mathrm{frame},N}$ and δ_h, hence it changes the theorem constants rather than the algebraic finite bridge.

Lemma 14 (Constant ledger for the propagation theorem) *In Theorem 22 and the Microlocal Propagation Theorem below, the constants may be chosen with the following dependency classes:*

$$C_{\mathrm{tail}} = C(C_{\mathrm{cnt}}, q, N, C_{T,N}), \qquad C_{\mathrm{disc}} = C(T, M, \mathfrak{C}_{\mathrm{frame},M}, \|a_t\|_{S^0,M}),$$

$$C_{\mathrm{tr}} = C(K_{\kappa,T}, \mathfrak{C}_{\mathrm{frame},2}, \|s_t\|_{C^1}, C_{\mathrm{deg}}), \qquad C_{\mathrm{sym}} = C(B_W, C_{\mathrm{deg}}, \|s_t\|_{L^\infty}),$$

and

$$C_{\mathrm{field}} = C(A_W^{-1}, B_W, E_{\mathrm{proj}}^{\mathrm{in}}, E_{\mathrm{proj}}^{\mathrm{out}}).$$

Here $K_{\kappa,T}$ denotes compact-window Lipschitz and inverse-Lipschitz bounds for the no-caustic canonical graph, $\|a_t\|_{S^0,M}$ collects the amplitude seminorms used by stationary phase, and C_{deg} is the row-column stencil conversion constant from Lemma 18. These constants do not depend on h, the packet count, K, P_{tr}, or P_{sym} once those quantities are kept in the explicit error terms.

Proof The tail constant is the dyadic shell-sum constant from Theorem 17. The discretization constant is the stationary-phase and packet-covering constant from Theorem 19. The transport constant is the Lipschitz conversion from canonical-map error to packet-kernel mismatch on the retained tube, followed by the bounded-degree row-column transfer. The symbol constant is the retained-stencil multiplier bound. The field constant is the frame synthesis and projection constant used to pass from packet coefficients back to L^2 fields. Each step uses only compact-window seminorms and the frame constants listed in Assumption 13. ∎

6.3 Spine A: short-time half-wave off-graph decay

We now specialize to the cleanest PDE family in the paper. Let $c \in C^\infty(\mathbb{R}^d)$ satisfy

$$0 < c_{\min} \le c(x) \le c_{\max} < \infty,$$

and define the half-wave principal symbol

$$p(x, \xi) = c(x)|\xi|.$$

Let $U_+(t)$ denote the positive half-wave propagator generated microlocally by p, and let κ_t be the Hamiltonian flow of p. We work on a time interval $|t| \le T$ chosen so that the flow remains a smooth canonical graph on the frequency band seen by the packet frame, with no caustic formation or branch crossing.

Theorem 15 (Short-time half-wave off-graph decay) *Fix $T > 0$ inside a no-caustic window for the half-wave flow generated by $p(x, \xi) = c(x)|\xi|$. Let $\{\varphi_\gamma^h\}_{\gamma \in \Gamma_h}$ be an analytic Gaussian/Gabor-type packet frame with normalized covering defect δ_h. Then for every integer $N \ge 1$ there is a constant $C_{T,N}$ such that*

$$\big|\langle U_+(t)\varphi_\eta^h, \varphi_\gamma^h\rangle\big| \le C_{T,N}\Big(1 + h^{-1/2}\,\mathrm{dist}\big(z_\gamma, \kappa_t(z_\eta)\big)\Big)^{-N} \tag{6}$$

uniformly for $|t| \le T$, $h \in (0, h_0]$, and $\gamma, \eta \in \Gamma_h$.

Proof [Proof sketch] The proof has four classical steps.

Step 1: parametrix. For $|t| \le T$, the short-time half-wave propagator is microlocally represented by an oscillatory integral

$$U_+(t)u(x) \sim (2\pi h)^{-d} \iint e^{\frac{i}{h}(\Phi_t(x,\xi) - y\cdot\xi)} a_t(x, \xi; h)\, u(y)\, dy\, d\xi,$$

where Φ_t generates the canonical graph of κ_t and a_t is a classical amplitude. The specific input is the usual real-principal-type short-time parametrix: on a compact frequency band before caustic formation, the half-wave group is a semiclassical Fourier integral operator associated with the canonical graph of the Hamiltonian flow. This is the part supplied by the classical Hörmander calculus and its semiclassical formulations (Hörmander, 1985a,b; Zworski, 2012; Alazard, 2021). The rest of the proof only uses this parametrix through the generated phase, amplitude symbol bounds, and the absence of competing stationary branches.

Step 2: test against packets. Insert φ_η^h and φ_γ^h into the kernel representation. The resulting phase is the half-wave phase plus the packet phases centered at z_η and z_γ.

Step 3: phase-gradient lower bound outside the canonical tube. If z_γ lies outside a fixed canonical tube around $\kappa_t(z_\eta)$, then the packet-localized total phase has gradient bounded from below by

$$|\nabla\Psi_{t,\gamma,\eta}| \geq c_T h^{-1/2} \operatorname{dist}\big(z_\gamma, \kappa_t(z_\eta)\big) - C_T\delta_h$$

on the support of the localized amplitude, for constants depending on the compact phase-space window and the time horizon T. After enlarging the canonical tube by a fixed multiple of the normalized covering defect δ_h, the last term is absorbed and the lower bound becomes

$$|\nabla\Psi_{t,\gamma,\eta}| \geq c_T' h^{-1/2} \operatorname{dist}\big(z_\gamma, \kappa_t(z_\eta)\big).$$

This is exactly where the no-caustic hypothesis is used: the canonical graph is smooth, locally single-valued, and the phase has no competing nearby stationary point away from the propagated packet center.

Step 4: repeated integration by parts. Applying the non-stationary phase operator

$$L = \frac{\nabla\Psi_{t,\gamma,\eta}\cdot\nabla}{i|\nabla\Psi_{t,\gamma,\eta}|^2}$$

repeatedly transfers derivatives to the localized amplitude. Each application gains one factor of

$$\Big(1 + h^{-1/2}\operatorname{dist}\big(z_\gamma, \kappa_t(z_\eta)\big)\Big)^{-1},$$

and the analytic packet envelope keeps the differentiated amplitude uniformly summable. After N iterations one obtains (6). ∎

Theorem 15 is a classical manuscript-level theorem and the first continuous spine of the paper.

Proposition 16 (Restricted non-stationary phase bridge) *Fix a packet-localized oscillatory representation of a canonical-graph propagator on a short time window. Assume only the following restricted inputs:*

1. *a single-graph oscillatory parametrix on the chosen frequency band;*
2. *uniform derivative bounds for the localized amplitude after packet testing;*
3. *a phase-gradient lower bound outside a raw canonical tube of radius $O(h^{1/2}(1+\delta_h))$.*

Then repeated integration by parts yields the off-graph decay law (6). In particular, the manuscript does not need the full Hörmander calculus at the point where the packet theorem starts. It needs only this restricted bridge from canonical-graph geometry to packet-matrix decay.

Proof The proposition is the rescaled packet form of Steps 3–4 in the proof of Theorem 15. A full statement and proof are given in Proposition 46. In brief, once the phase-gradient lower bound is available on the packet-localized support, the operator

$$L = \frac{\nabla\Psi \cdot \nabla}{i|\nabla\Psi|^2}$$

can be iterated without encountering a stationary point. After rescaling packet variables by $h^{1/2}$, the large parameter is

$$D_{\gamma\eta} = 1 + h^{-1/2}\mathrm{dist}(z_\gamma, \kappa(z_\eta)).$$

The differentiated coefficients of the adjoint vector field are $O(D_{\gamma\eta}^{-1})$, while the rescaled packet amplitude has uniformly bounded C^N-L^1 seminorms. Iterating N times therefore gives a factor $D_{\gamma\eta}^{-N}$, which is exactly (6). ■

6.4 Spine B: abstract canonical-graph packet sparsity

The second spine abstracts away from the wave equation and isolates the exact packet argument that converts off-graph decay into bounded-stencil sparsity.

Theorem 17 (Abstract canonical-graph packet sparsity) *Let F_h be a packet matrix associated with a canonical graph κ, and suppose that for every $N \in \mathbb{N}$ there is $C_N > 0$ such that*

$$\big|\langle F_h\varphi_\eta^h, \varphi_\gamma^h\rangle\big| \le C_N\Big(1 + h^{-1/2}\,\mathrm{dist}\big(z_\gamma, \kappa(z_\eta)\big)\Big)^{-N}.$$

Assume in addition that there exist constants $q > 0$ and $C_{\mathrm{cnt}} > 0$ such that for every source packet index η and every shell radius $R \ge 1$,

$$\#\Big\{\gamma \in \Gamma_h : \mathrm{dist}\big(z_\gamma, \kappa(z_\eta)\big) \le Rh^{1/2}\Big\} \le C_{\mathrm{cnt}}R^q. \tag{7}$$

Let $F_h^{(K)}$ be the packet matrix obtained by keeping, for each source packet, only the K output packets nearest to $\kappa(z_\eta)$. Assume the same shell-counting estimate for the inverse graph, equivalently the retained matrix and its adjoint have the same tail envelope on the compact no-caustic window. Then for every $N > q + 1$ there is a constant $C_{N,q}$ such that

$$\|F_h - F_h^{(K)}\|_{\ell^2\to\ell^2} \le C_{N,q}\,K^{1-N/q}.$$

Proof [Proof sketch] Decompose the packet indices into dyadic shells around the propagated center $\kappa(z_\eta)$. On shell ℓ, the counting hypothesis gives $O(2^{q\ell})$ packets and the off-graph decay gives $O(2^{-N\ell})$ per entry. Hence the column shell mass is $O(2^{(q-N)\ell})$. The inverse-graph hypothesis gives the same row shell mass. Summing over shells above level L gives row and column tails of size $O(2^{(q-N)L})$. Since the number of packets retained up to shell L is $O(2^{qL})$, one has $K \asymp 2^{qL}$; Schur's test gives the displayed ℓ^2 bound. ■

Theorem 17 is the manuscript's abstract FIO theorem. It is independent of the wave equation, and Theorem 15 shows that short-time half-wave propagation satisfies its hardest hypothesis. The corresponding Lean files formalize the shell/tube/truncation layer that starts once the off-graph decay and shell counting laws are available.

Lemma 18 (Row-column bounded-stencil transfer) *Assume, in addition to the shell-counting hypothesis above, that κ is bi-Lipschitz on the compact packet window and that the packet cover has bounded overlap at scale $h^{1/2}$. If a source-wise truncation keeps at most K output packets in a canonical tube around $\kappa(z_\eta)$, then there is a constant $C_{\deg}$, depending only on the bi-Lipschitz and overlap constants, such that every output packet receives contributions from at most $C_{\deg}K$ retained source packets. Conversely, an output-wise canonical tube stencil has at most $C_{\deg}K$ retained outputs per source.*

Proof [Proof sketch] Fix an output packet γ. If γ is retained by a source packet η, then z_γ lies in a tube of radius $R_K h^{1/2}$ around $\kappa(z_\eta)$, where R_K is the shell radius corresponding to K retained packets. Since κ^{-1} is Lipschitz on the compact window, z_η lies in a tube of radius $CR_K h^{1/2}$ around $\kappa^{-1}(z_\gamma)$. The bounded-overlap shell-counting law bounds the number of such source packets by $C_{\deg} R_K^q$. Because the retained source-wise tube has $K \asymp R_K^q$ packets, this is $O(K)$. The reverse implication is identical with κ and κ^{-1} interchanged. ■

Lemma 18 resolves the apparent direction mismatch between the abstract sparsity theorem and the executable layer. The sparsity proof is most naturally columnwise, retaining output packets for each source packet. The MiNO implementation is outputwise, aggregating source packets for each output packet. On compact no-caustic windows these two descriptions differ only by the bounded degree constant above, which is absorbed into the stencil budget and constants below.

6.5 Continuous transported-symbol reduction with bounded canonical stencil

We now return to the actual operator class used by MiNO. The previous theorem says that a canonical-graph packet matrix can be truncated to a bounded stencil with explicit error. The next result combines that with stationary phase near the graph.

Theorem 19 (Continuous transported-symbol reduction) *Under the assumptions of Theorem 15, fix an integer $M \geq 1$ and a stencil budget $K \in \mathbb{N}$. Then the packet-space realization*

$$K_h(t) := W_h U_+(t) W_h^*$$

admits a decomposition

$$K_h(t) = \mathcal{T}_h^{\star(K)}(t) + R_h^{(K,M)}(t),$$

where

$$(\mathcal{T}_h^{\star(K)}(t)a)_\gamma = \sum_{\eta \in \Pi^\star_{h,K}(t,\gamma)} s^\star_{h,\gamma\eta}(t)\, a_\eta$$

has an output-wise canonical stencil of size at most $C_{\deg}K$ after the row-column transfer of Lemma 18 (renamed as K below), with each retained packet center lying in a raw tube of radius $O(h^{1/2}(1+\delta_h))$ around the canonical preimage, and where the residual satisfies

$$\|R_h^{(K,M)}(t)\|_{\ell^2\to\ell^2} \le C_{T,N,q}K^{1-N/q} + E_{\mathrm{frame}}(h) + E_{\mathrm{asymp}}(h,M) + C_{\mathrm{cov}}\delta_h$$

for every $N > q+1$.

Proof [Proof sketch] Apply Theorem 17 to discard all but the K nearest packets to the canonical graph in the source-to-output direction. Lemma 18 converts this into the output-to-source stencil used by MiNO, changing only constants and the effective value of K. This contributes the $K^{1-N/q}$ tail. On the retained tube, stationary phase around the propagated packet center identifies the leading amplitude samples

$$s^\star_{h,\gamma\eta}(t) = a_t(z_\eta, z_\gamma) + O(h^{1/2}(1+\delta_h)),$$

while bounded overlap of the packet frame and Schur's test convert the local coefficient error into an ℓ^2 operator remainder. We separate this finite-to-asymptotic term into the frame defect $E_{\mathrm{frame}}(h)$, the retained stationary-phase defect $E_{\mathrm{asymp}}(h,M)$, and the normalized covering defect $C_{\mathrm{cov}}\delta_h$. Summing the truncation and local approximation terms gives the stated bound. ■

Theorem 19 is the continuous theorem that the paper now wants to stand behind. It is stronger than the old transported-symbol statement because it identifies the correct *local rank* of the reference operator in the packet energy norm: not dense, not necessarily one, but bounded by a packet sparsity budget K.

Theorem 19 is also the paper's restricted packet almost-diagonalization bridge. The off-graph part comes from Proposition 16 plus shell counting, and the on-graph part comes from a local stationary-phase sample of the retained amplitude. This is the exact point where the continuous argument becomes a theorem-grade packet reduction rather than a broad appeal to "standard FIO theory." The remaining classical input is therefore narrow and inspectable.

Corollary 20 (Continuous-to-discrete bounded-stencil MiNO corollary) *Under the assumptions of Theorem 19, let $\widehat{\mathcal{T}}_h^{(K)}(t)$ be a theorem-grade* MiNO *block with the same stencil budget K. This corollary is the finite landing statement: it assumes coefficient-level budgets after a retained stencil has already been matched. The main wave theorem below replaces this finite transport hypothesis by a geometric canonical-map error and the induced packet-kernel mismatch. Suppose that for every retained pair (γ,η) one has transport and symbol budgets*

$$|a_{\hat\pi_{\gamma\eta}} - a_{\pi^\star_{\gamma\eta}}| \le \varepsilon_{\mathrm{tr}}, \qquad |\hat s_{h,\gamma\eta} - s^\star_{h,\gamma\eta}| \le \varepsilon_{\mathrm{sym}},$$

and residual envelope $|\rho_{h,\gamma}(a)| \le \varepsilon_{\mathrm{res}}$, together with coefficient/symbol envelopes $|a_\eta| \le B_a$ and $|s^\star_{h,\gamma\eta}| \le B_s$. Then

$$\|\widehat{\mathcal{T}}_h^{(K)}(t)a - K_h(t)a\|_{\ell^1} \le nK(B_s\varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}}B_a) + n\varepsilon_{\mathrm{res}} + \Big(C_{T,N,q}K^{1-N/q} + E_{\mathrm{frame}}(h) + E_{\mathrm{asymp}}(h,M) + C_{\mathrm{cov}}\delta_h\Big)\|a\|_{\ell^1}.$$

Proof The retained stencil part is bounded by applying the finite propagation bookkeeping packetwise across at most K retained interactions per output packet. The discarded shells and packet-covering defect are then absorbed into the residual term from Theorem 19. ■

6.6 Network-realizable short-time wave approximation

Corollary 20 assumes small learned transport and symbol budgets. The next theorem closes that gap at the level of approximation theory: under the same no-caustic wave assumptions, a theorem-grade MINO-Core block with sufficiently large metadata and symbol networks realizes the transported-symbol approximation with an explicit rate.

Let $\Omega_T \subset T^*\mathbb{R}^d$ be the compact packet window swept out by the no-caustic half-wave flow for $|t| \le T$, and let $m_\Omega = \dim(\Omega_T)$. We use the following standard compact-window neural approximation input: if $f \in C^r(\Omega_T; \mathbb{R}^p)$ with $\|f\|_{C^r} \le A_r$, then the branch network class with P trainable parameters contains a map $\hat{f}_P$ such that

$$\|f - \hat{f}_P\|_{L^\infty(\Omega_T)} \le C_{\mathrm{nn}} A_r P^{-r/m_\Omega}, \tag{8}$$

up to the usual architecture-dependent logarithmic factors. For ReLU networks this is the standard compact smooth-function approximation rate (Yarotsky, 2017); other smooth activations may be substituted without changing the structure of the theorem.

Proposition 21 (Hamiltonian metadata branch consistency) *Let $H \in C^{r+2}(\Omega_T)$ generate the canonical flow κ_t by*

$$\dot{z} = X_H(z) = J\nabla H(z), \qquad z = (x, \xi),$$

on the compact no-caustic window Ω_T, and suppose $\|H\|_{C^{r+2}} \le A_{r+2}$. Assume a scalar Hamiltonian branch realizes H_P with

$$\|H - H_P\|_{C^2(\Omega_T)} \le C_H A_{r+2} P_H^{-(r-2)/m_\Omega}, \qquad r > 2.$$

Let $\widehat{\kappa}_{t,P,\Delta}$ be the m-step explicit Hamiltonian-vector-field update

$$z_{j+1} = z_j + \Delta\, J\nabla H_P(z_j), \qquad \Delta = t/m,$$

projected back to Ω_T when needed. Then, for $|t| \le T$,

$$\sup_{z\in\Omega_T} d\big(\widehat{\kappa}_{t,P,\Delta}(z), \kappa_t(z)\big) \le C_T \Big(P_H^{-(r-2)/m_\Omega} + \Delta\Big).$$

Moreover one explicit step has canonical-defect bound

$$\sup_{z\in\Omega_T} \big\|D\Psi_{P,\Delta}(z)^\top \Omega_0 D\Psi_{P,\Delta}(z) - \Omega_0\big\| \le C_T \Delta^2 \|H_P\|^2_{C^2(\Omega_T)},$$

where $\Psi_{P,\Delta}(z) = z + \Delta J\nabla H_P(z)$ and Ω_0 is the standard symplectic matrix.

If, in addition, H_P is represented in separable form

$$H_P(q,p) = V_P(q) + T_P(p),$$

and the metadata branch uses the Störmer–Verlet step

$$p_{1/2} = p_0 - \frac{\Delta}{2}\nabla_q V_P(q_0), \qquad q_1 = q_0 + \Delta\nabla_p T_P(p_{1/2}), \qquad p_1 = p_{1/2} - \frac{\Delta}{2}\nabla_q V_P(q_1),$$

then each step is symplectic in the metadata variables. The canonical-defect term vanishes at the map level, up to floating-point and autograd implementation error, and the discretization term in the flow error is the standard second-order $O(\Delta^2)$ term for smooth separable Hamiltonians.

Proof The vector-field error is

$$\|X_H - X_{H_P}\|_{C^1} \leq C\|H - H_P\|_{C^2}.$$

Standard Gronwall stability for ODE flows on the compact window gives an $O(\|X_H - X_{H_P}\|_{C^0})$ flow perturbation over $|t| \leq T$. The explicit Euler discretization for the C^1 vector field X_{H_P} contributes the usual first-order global error $O(\Delta)$, with a constant depending on the C^1 and Lipschitz bounds of X_{H_P}. Combining the two estimates gives the displayed flow error. For the canonical defect, write

$$D\Psi_{P,\Delta} = I + \Delta\, DX_{H_P}.$$

Since $DX_{H_P} = J\nabla^2 H_P$ is an infinitesimal Hamiltonian matrix, the first-order term in

$$(I + \Delta DX_{H_P})^\top \Omega_0 (I + \Delta DX_{H_P}) - \Omega_0$$

cancels. The remaining term is quadratic in Δ and bounded by $C\Delta^2\|DX_{H_P}\|^2$, hence by $C_T\Delta^2\|H_P\|_{C^2}^2$ on the compact window.

For the separable branch, the Störmer–Verlet map is a composition of three exact symplectic shears: a potential half-step, a kinetic full-step, and another potential half-step. A composition of symplectic maps is symplectic, so the discrete canonical defect is zero at the exact map level. The trajectory approximation is the standard second-order consistency estimate for smooth separable Hamiltonian flows on a compact window. ■

Proposition 21 records the scope of the Hamiltonian metadata branch. It justifies `hamiltonian_euler` as a cheap Hamiltonian vector-field bias and `hamiltonian_verlet` as the theorem-facing separable-Hamiltonian transport branch. The latter is symplectic by construction in the packet metadata variables under the stated separability and fixed-feature assumptions. The wave theorem below uses the broader compact-window transport approximation hypothesis; the Verlet branch is a clean structural specialization of that hypothesis.

Theorem 22 (Network-realizable short-time wave MiNO-Core approximation) *Assume the hypotheses of Theorems 15, 17, and 19. Assume that the canonical map $\kappa_t : \Omega_T \to \Omega_T$ has C^r norm at most A_r on the compact packet window, with $r \geq 1$. Assume also that the retained local packet kernel has the form*

$$s_t = s_t(z, \nu; h), \qquad \nu = h^{-1/2}\big(z_\gamma - \kappa_t(z_\eta)\big),$$

with C^r norm at most A_r on a compact tube-coordinate window of dimension $m_\Omega + q_{\mathrm{loc}}$. The source-only multiplier case is recovered by taking $q_{\mathrm{loc}} = 0$. Let the metadata and symbol branches have budgets P_{tr} and P_{sym}, both satisfying (8) on their respective input dimensions. Then there exists a theorem-grade bounded-stencil MiNO*-Core block $\mathcal{M}^{\mathrm{core}}_{h,K,P}(t)$ such that, for every $N > q+1$ and every retained asymptotic order $M \geq 1$,*

$$\begin{aligned}\|W_h U_+(t) W_h^* - \mathcal{M}^{\mathrm{core}}_{h,K,P}(t)\|_{\ell^2\to\ell^2} \leq{}& C_{\mathrm{tail}} K^{1-N/q} + E_{\mathrm{frame}}(h) + E_{\mathrm{asymp}}(h,M) + C_{\mathrm{cov}}\delta_h \\ &+ C_{\mathrm{tr}} K\big(h^{-1/2} P_{\mathrm{tr}}^{-r/m_\Omega} + \delta_h\big) + C_{\mathrm{sym}} K P_{\mathrm{sym}}^{-r/(m_\Omega+q_{\mathrm{loc}})} \\ &+ C_{\mathrm{res}}\varepsilon_{\mathrm{res}}. \end{aligned} \tag{9}$$

The constants depend on the time window, packet frame bounds, shell-counting geometry, compact-window C^r norms, local tube dimension, and the branch architecture class, but not on h, K, P_{tr}, or P_{sym}. The normalized covering defect δ_h is an explicit packetization term: it vanishes only for an oversampled packet cloud with normalized covering radius tending to zero. For a critically sampled frame, δ_h should be read as a fixed finite-frame defect rather than as an asymptotically vanishing quantity.

Proof [Proof sketch] The proof combines five ingredients.

Step 1: reduce the PDE operator to a sparse packet matrix. Theorem 15 gives off-canonical-tube decay for the half-wave propagator. Theorem 17 turns that decay into the K-term packet tail $C_{\mathrm{tail}} K^{1-N/q}$. Theorem 19 then identifies the retained part with a bounded-stencil transported-symbol packet law, up to the frame, asymptotic, and covering defects $E_{\mathrm{frame}}(h) + E_{\mathrm{asymp}}(h,M) + C_{\mathrm{cov}}\delta_h$.

Step 2: approximate the canonical map. Apply (8) to κ_t. The metadata branch can therefore realize a map $\widehat{\kappa}_t$ with

$$\|\widehat{\kappa}_t - \kappa_t\|_{L^\infty(\Omega_T)} \leq C_{\mathrm{nn}} A_r P_{\mathrm{tr}}^{-r/m_\Omega}.$$

Because packet tubes have natural radius $h^{1/2}$ in the semiclassical scaling, this geometric error enters the packet coefficient law as

$$\varepsilon_{\mathrm{tr}} \leq C_\kappa\big(h^{-1/2} P_{\mathrm{tr}}^{-r/m_\Omega} + \delta_h\big).$$

The $h^{-1/2}$ factor is the cost of measuring canonical-map error in packet-radius units.

Step 3: approximate the local packet kernel. Apply (8) to the retained local packet kernel $s_t(z,\nu;h)$ on the compact source–tube coordinate window. The symbol branch can realize $\widehat{s}_t$ with

$$\varepsilon_{\mathrm{sym}} \leq C_s P_{\mathrm{sym}}^{-r/(m_\Omega+q_{\mathrm{loc}})}.$$

Step 4: insert the operator-level bounded-stencil bridge. The retained packet law has at most K interactions per output packet after Lemma 18. The geometric canonical-map error first gives a packet-kernel transport mismatch of size $O(h^{-1/2} P_{\mathrm{tr}}^{-r/m_\Omega} + \delta_h)$ on the retained tube. Theorem 10 then converts that kernel mismatch, together with the symbol approximation error, into the learned-budget contribution

$$C_{\mathrm{tr}} K \varepsilon_{\mathrm{tr}} + C_{\mathrm{sym}} K \varepsilon_{\mathrm{sym}} + C_{\mathrm{res}}\varepsilon_{\mathrm{res}}.$$

Equivalently, Theorem 6 gives the same bounded-stencil contribution directly in $\ell^2 \to \ell^2$ operator norm after row and column stencil bounds are imposed.

Step 5: combine defects. Summing the tail, frame, asymptotic, covering, transport, local-kernel, and residual terms gives (9). ■

Theorem 22 is the rate-realization input for the main Microlocal Propagation Theorem stated in Section 7. Its scope is compact-window, no-caustic, short-time hyperbolic propagation, with classical off-graph and stationary-phase inputs and a finite machine-checked landing layer. Within that scope, the theorem proves that the MiNO-Core primitive approximates a concrete PDE operator family in packet energy norm, with an explicit rate in packet scale, stencil budget, covering radius, local-kernel dimension, and neural-network size.

Corollary 23 (Sobolev field-level MiNO-Core approximation) *Assume the hypotheses of Theorem 22. In addition, assume that the packet analysis and synthesis maps are Sobolev-stable on the compact phase window: for every s in a fixed compact interval $I \subset \mathbb{R}$ there are constants $A_{W,s}, B_{W,s}$, independent of h and the packet count, such that the weighted packet norms induced by the window satisfy*

$$\|W_h v\|_{\ell^2_s} \le A_{W,s}\|v\|_{H^s}, \qquad \|W_h^* b\|_{H^s} \le B_{W,s}\|b\|_{\ell^2_s},$$

and the retained short-time canonical graph changes the corresponding weights by at most a constant $C_{\kappa,s}$ on Ω_T. Then, for microlocalized inputs and for every $s \in I$,

$$\|W_h^* \mathcal{M}^{\mathrm{core}}_{h,K,P}(t) W_h - U_+(t)\|_{H^s \to H^s} \le C_s\big(E_{h,K,P} + E^{\mathrm{in}}_{\mathrm{proj},s} + E^{\mathrm{out}}_{\mathrm{proj},s}\big),$$

where $E_{h,K,P}$ denotes the right-hand side of Theorem 22, and

$$E^{\mathrm{in}}_{\mathrm{proj},s} = \|U_+(t)(I - \mathcal{P}_h)\|_{H^s \to H^s}, \qquad E^{\mathrm{out}}_{\mathrm{proj},s} = \|(I - \mathcal{P}_h)U_+(t)\mathcal{P}_h\|_{H^s \to H^s}.$$

The same statement with the executable transported-synthesis, carrier, and admissible landing defects added holds for the carrier-bound field realization.

Proof Apply Theorem 22 in the weighted packet norm ℓ^2_s. The compact-window canonical graph preserves Sobolev weights up to $C_{\kappa,s}$, and the retained symbol is order zero for the half-wave branch, so the same packet error envelope $E_{h,K,P}$ applies with a Sobolev-dependent constant. Sobolev-stable synthesis converts the packet error to H^s, while analysis stability converts the input H^s norm to the weighted packet norm. The two displayed projection defects account for replacing the true field by its packet projection before and after propagation. The executable transported-synthesis, carrier, and finite-landing terms enter by the same triangle inequality as in the L^2 field-level statement. ■

6.7 Branch-factored oscillatory extension

The preceding theorem treats a single canonical graph. This is the cleanest short-time wave setting, but it should not be confused with the full range of oscillatory PDE operators. Local outgoing Helmholtz windows, reflected waves, and scattering parametrices may be better modeled by a finite sum of canonical-graph FIOs, while elliptic and parabolic regimes remain identity-canonical or smoothing components. The architectural extension is therefore not Helmholtz-specific: it is a finite-branch packet realization of a local FIO sum.

Let an oscillatory target on a compact packet window admit the microlocal decomposition

$$T_h = \sum_{\ell=1}^{L} T_{\ell,h} + A_h + S_h,$$

where each $T_{\ell,h}$ is a canonical-graph FIO with graph κ_ℓ, retained symbol s_ℓ, and local packet tube, A_h is an identity-canonical pseudodifferential component, and S_h is smoothing or outside the retained window. A branch-factored MINO realizes

$$\mathcal{M}^{(L)}_{\theta,h} u = W_h^* \left[\sum_{\ell=1}^{L} G_{\theta,\ell}(z)\, \mathcal{T}_{\hat{\kappa}_{\theta,\ell}, \hat{s}_{\theta,\ell}, \hat{\phi}_{\theta,\ell}}(W_h u) + \mathcal{A}_\theta(W_h u) + R_\theta(W_h u) \right],$$

with token-wise router $\sum_\ell G_{\theta,\ell} = 1$. The optional phase factor $\hat{\phi}_{\theta,\ell}$ is included only in regimes where the packet landing error is dominated by unresolved high-frequency phase. It does not change the single-branch wave theorem; it records the sharper implementation requirement that phase errors scale like h^{-1} in oscillatory output.

Lemma 24 (Phase-factor landing defect) *On a local output window, suppose a branch is represented as*

$$T_{\ell,h} u = e^{i\phi_\ell/h} B_{\ell,h} u, \qquad \widehat{T}_{\ell,h} u = e^{i\hat{\phi}_\ell/h} \widehat{B}_{\ell,h} u,$$

with $\|B_{\ell,h}\|_{L^2\to L^2} \le C_B$ *and* $\|u\|_{L^2} \le 1$. *Then*

$$\|(T_{\ell,h} - \widehat{T}_{\ell,h})u\|_{L^2} \le \|(B_{\ell,h} - \widehat{B}_{\ell,h})u\|_{L^2} + C_B h^{-1} \|\phi_\ell - \hat{\phi}_\ell\|_{L^\infty}.$$

The same estimate holds in packet norm after multiplying the right-hand side by the packet frame synthesis constant.

Proof Use $|e^{ia/h} - e^{ib/h}| \le h^{-1}|a-b|$ and the triangle inequality. The packet statement follows by applying the same synthesis bound used in Corollary 23. ■

Theorem 25 (Branch-factored oscillatory MiNO envelope) *Assume that* T_h *admits the finite decomposition above, that the canonical tubes of the retained FIO branches have finite overlap, and that the ideal router* $G^\star_\ell$ *assigns retained packets to the corresponding branch up to router defect* E_{route}. *For branch* ℓ, *let* $E_{\mathrm{tr},\ell}$, $E_{\mathrm{sym},\ell}$, *and* $E_{\mathrm{phase},\ell}$ *denote its transport, symbol, and optional phase-factor defects; let* $E_{\mathrm{tsyn},\ell}$, $E_{\mathrm{car},\ell}$, *and* $E_{\mathrm{land},\ell}$ *denote the executable transported-synthesis, carrier, and finite-landing defects; and let* E_{cross} *denote* *packet cross-talk between retained branch tubes. Then the branch-factored carrier-bound realization satisfies*

$$\|\mathcal{M}^{(L)}_{\theta,h} - T_h\|_{\ell^2\to\ell^2} \le \sum_{\ell=1}^{L}\big(E_{\mathrm{tr},\ell} + E_{\mathrm{sym},\ell} + E_{\mathrm{phase},\ell} + E_{\mathrm{tsyn},\ell} + E_{\mathrm{car},\ell} + E_{\mathrm{land},\ell}\big)$$
$$+ E_{\mathrm{route}} + E_{\mathrm{cross}} + E_{\Psi} + E_{\mathrm{res}} + E_{\mathrm{tail}} + E_{\mathrm{frame}} + E_{\mathrm{asymp}} + C_{\mathrm{cov}}\delta_h.$$

Here E_Ψ is the identity-canonical pseudodifferential approximation defect for A_h. The corresponding field-level L^2 and Sobolev estimates add the same projection constants as Corollary 23.

Proof Apply the single-branch packet reduction to each retained canonical tube and sum the per-branch transport, symbol, phase, and executable landing defects. Replacing the ideal branch assignment by the learned router gives E_{route}. Finite overlap and almost-orthogonality of the retained tubes control the non-diagonal interactions by E_{cross}. The identity-canonical component is bounded by the symbol-order approximation argument used for the ΨDO branch, producing E_Ψ. Smoothing and out-of-window terms are absorbed into E_{res}, E_{tail}, and the frame/asymptotic/covering defects. The field-level statement is the same analysis–synthesis triangle inequality as before. ■

Theorem 25 is the correct home for high-frequency stress tests. It shows that branch count is only one term in the budget: improvement also requires small per-branch transport and symbol errors, a router that separates canonical tubes, limited cross-talk, and phase/landing accuracy that tightens with frequency. A flat branched Helmholtz table therefore identifies a finite-budget failure of these hypotheses while leaving the single-graph wave theorem intact.

Finite packet Egorov diagnostic. The next microlocal calculus question is whether $\mathcal{M}^{\mathrm{core}}_{h,K,P}$ not only approximates $U_+(t)$ but also conjugates pseudodifferential probes by the learned canonical map. For a frozen theorem-grade linearized Core, let $A^{\mathrm{pkt}}_{j,h} = W_h \operatorname{Op}_h(a_j) W_h^*$ be a finite family of order-zero packet probes, let $A^{\widehat{\kappa}}_{j,h} = W_h \operatorname{Op}_h(a_j \circ \widehat{\kappa}_\theta) W_h^*$, and let $S_{\theta,h}$ be the retained diagonal or local symbol action of the learned symbol branch. Define

$$\rho^{\mathrm{pkt}}_{\mathrm{Eg},h}(j) = \frac{\big\|(\mathcal{M}^{\mathrm{core}}_{\theta,h})^* A^{\mathrm{pkt}}_{j,h}\mathcal{M}^{\mathrm{core}}_{\theta,h} - S^*_{\theta,h} A^{\widehat{\kappa}}_{j,h} S_{\theta,h}\big\|_{\ell^2\to\ell^2}}{\|A^{\mathrm{pkt}}_{j,h}\|_{\ell^2\to\ell^2} + \epsilon}.$$

For a genuinely nonlinear executable model, the corresponding diagnostic must be applied to the local linearization $D\mathcal{M}_\theta[u]$ rather than to $\mathcal{M}_\theta$ itself. Thus the planned Egorov metric is

$$\rho^{\mathrm{Jac}}_{\mathrm{Eg},h}(u,j) = \frac{\big\|D\mathcal{M}_\theta[u]^* A^{\mathrm{pkt}}_{j,h} D\mathcal{M}_\theta[u] - S_{\theta,h}[u]^* A^{\widehat{\kappa}[u]}_{j,h} S_{\theta,h}[u]\big\|_{\ell^2\to\ell^2}}{\|A^{\mathrm{pkt}}_{j,h}\|_{\ell^2\to\ell^2} + \epsilon}.$$

This diagnostic records the finite object for branch-coupled Egorov measurements. The main closure of this paper remains packet propagation and finite wavefront proxies.

Corollary 26 (Packet-threshold wavefront localization) *Fix a packet threshold* $\tau > 0$ *and define the discrete* h*-wavefront proxy*

$$\mathrm{WF}_h^\tau(a) = \{z_\eta : \ |a_\eta| \geq \tau\}.$$

Let $a = W_h u$ *satisfy* $\|a\|_{\ell^1} \leq A$ *and let* $S = \mathrm{WF}_h^\tau(a)$*. Under the hypotheses of Theorem 22, the coefficients of*

$$\mathcal{M}_{h,K,P}^{\mathrm{core}}(t)a$$

outside the raw phase-space neighborhood of $\kappa_t(S)$ *obey*

$$\sum_{z_\gamma \notin (\kappa_t(S))_{h^{1/2}(R_K + C\epsilon_{\kappa,h} + C\delta_h)}} \left|(\mathcal{M}_{h,K,P}^{\mathrm{core}}(t)a)_\gamma\right| \leq CAK^{1-N/q} CA\big(E_{\mathrm{frame}}(h) + E_{\mathrm{asymp}}(h,M)\big)$$
$$+ CAC_{\mathrm{cov}}\delta_h + AE_{\mathrm{net}} + C\tau,$$

where

$$\epsilon_{\kappa,h} = h^{-1/2} \sup_{z\in\Omega_T} d(\widehat{\kappa}_\theta(z), \kappa_t(z)), \qquad E_{\mathrm{net}} = K\big(h^{-1/2} P_{\mathrm{tr}}^{-r/m_\Omega} + \delta_h\big) + KP_{\mathrm{sym}}^{-r/(m_\Omega + q_{\mathrm{loc}})} + \epsilon_{\mathrm{res}}.$$

Proof Split the input coefficients into the threshold-active part supported on S and the remainder of ℓ^1 mass at most $C\tau$ after restricting to the finite active packet window. The active part is transported into the canonical tube around $\kappa_t(S)$ by the retained-stencil construction; the raw tube radius is enlarged by $h^{1/2}\epsilon_{\kappa,h}$ for learned canonical-map error, by $h^{1/2}\delta_h$ for packet covering defect, and by the retained shell radius $R_K h^{1/2}$. Contributions outside that enlarged tube are exactly the discarded off-graph shells, packetization error, and learned-budget error from Theorem 22. Summing these terms gives the displayed estimate. ■

Classically, $(x, \xi) \notin \mathrm{WF}_h(u)$ means that some semiclassical cutoff supported near (x, ξ) annihilates u up to $O(h^\infty)$. The artifact measures the finite proxy above after packet analysis through packet-wavefront localization and transport-proxy tests. Corollary 26 is therefore the finite wavefront statement used by the branch-identifiability experiments: corrupting the transport branch should enlarge packet-wavefront localization error before it necessarily changes relative ℓ^2.

6.8 Restricted Hörmander-style calculus backbone

The previous subsection treats the hyperbolic case, where the dominant component is a canonical-graph FIO. The same packet calculus also describes elliptic and parabolic regimes, but the geometry changes. Elliptic parametrices are pseudodifferential operators with identity canonical relation, while parabolic updates are dissipative pseudodifferential semigroups. Thus the broader MiNO identity is not "packets always move." It is that the operator is represented by its packet matrix and that the microlocal structure of that matrix is exposed to the architecture.

Theorem 27 (Restricted FIO/ΨDO packet calculus) *Let T_h be an operator on a compact phase-space window in $\mathbb{R}^d$ admitting the microlocal decomposition*

$$T_h = \sum_{\ell=1}^{L} F_{\ell,h} + A_h + R_h,$$

where each $F_{\ell,h}$ is a semiclassical canonical-graph FIO satisfying the off-graph decay and shell-counting hypotheses of Theorem 17, $A_h = \mathrm{Op}_h(a)$ is a pseudodifferential operator with $a \in S^m_{1,0}$, and R_h is smoothing of order R. Let W_h be the same packet analysis map used above. For stencil budgets K_{fio} and K_{id}, there is a packet operator $\mathcal{M}^{\mathrm{calc}}_{h,K}$ with

$$\mathcal{M}^{\mathrm{calc}}_{h,K} = \sum_{\ell=1}^{L} \mathcal{M}^{\mathrm{fio}}_{\ell,h,K_{\mathrm{fio}}} + \mathcal{M}^{\Psi}_{h,K_{\mathrm{id}}} + \mathcal{M}^{\mathrm{sm}}_{h},$$

where the FIO components use canonical tubes, the ΨDO component uses the identity-canonical tube, and the smoothing component is an explicitly bounded residual, such that for every $N > q+1$ and every $M, R \geq 1$,

$$\begin{aligned}\|W_h T_h W_h^* - \mathcal{M}^{\mathrm{calc}}_{h,K}\|_{\ell^2\to\ell^2} &\leq C_{\mathrm{fio}} L K_{\mathrm{fio}}^{1-N/q} + C_\Psi K_{\mathrm{id}}^{1-N/q} \\ &\quad + C_R h^R + E_{\mathrm{frame}}(h) + E_{\mathrm{asymp}}(h,M) + C_{\mathrm{cov}}\delta_h. \qquad (10)\end{aligned}$$

Proof [Proof sketch] Apply Theorem 17 separately to each canonical-graph branch $F_{\ell,h}$. The sum over finitely many branches contributes the factor L. For $A_h = \mathrm{Op}_h(a)$, use the standard packet almost-diagonalization theorem for pseudodifferential operators: the packet matrix of A_h decays rapidly away from the identity relation $z_\gamma = z_\eta$, with the usual symbol-order envelope inherited from $a \in S^m_{1,0}$. The same shell-counting and Schur argument as in Theorem 17 gives the identity-tube truncation term $C_\Psi K_{\mathrm{id}}^{1-N/q}$. Finally, smoothing of order R contributes $C_R h^R$ on the compact packet window, while finite packetization is recorded by E_{frame}, E_{asymp}, and the normalized covering defect. Summing the branchwise estimates gives (10). ■

Theorem 27 is the restricted Hörmander-style calculus used by this paper. It is weaker than full Hörmander calculus because it assumes a finite branch decomposition, compact phase window, Euclidean packets, and classical packet almost-diagonalization input. It is stronger than the earlier wave-only bridge because it identifies the three regimes that the model must expose: nontrivial canonical transport, identity-canonical symbol action, and smoothing/dissipative residual structure.

Corollary 28 (Identity-canonical symbol-order approximation) *Let $A_h = \mathrm{Op}_h(a)$ with $a \in S^m_{1,0}$ on the compact packet window, and let $\mathcal{M}^{\Psi}_{h,K,P}$ be a* MiNO *identity-canonical branch whose order-aware symbol parameterization approximates a in the finitely many symbol seminorms required by Calderón–Vaillancourt with error $\epsilon_{s,P}$. Then, for every Sobolev index s in a fixed compact range,*

$$\begin{aligned}\|W_h^* \mathcal{M}^{\Psi}_{h,K,P} W_h - A_h\|_{H^s\to H^{s-m}} &\leq C_s\big(\epsilon_{s,P} + K^{1-N/q} + E_{\mathrm{frame}}(h) + E_{\mathrm{asymp}}(h,M) \\ &\quad + C_{\mathrm{cov}}\delta_h + E_{\mathrm{proj}}\big).\end{aligned}$$

Proof Classical pseudodifferential calculus gives $A_h : H^s \to H^{s-m}$ with constants controlled by finitely many $S^m_{1,0}$ seminorms of a. Packet almost-diagonalization reduces the operator to an identity-canonical packet matrix, with tail $K^{1-N/q}$ and packetization defects recorded by E_{frame}, E_{asymp}, and the normalized covering term. Replacing the retained symbol samples by the learned order-aware branch costs $\epsilon_{s,P}$ in the same finite seminorm envelope. The analysis/synthesis projection contributes E_{proj}. Combining these estimates gives the displayed Sobolev operator bound. ■

Corollary 28 states the mathematical target for an order-aware or derivative-regularized symbol branch. The implementation exposes symbol order and records a finite symbol-order proxy; full Hörmander-class control would require additional derivative seminorms.

Corollary 29 (Calculus-backbone MiNO approximation) *Assume the hypotheses of Theorem 27. Let a* MiNO *block contain matching branch types: bounded canonical FIO branches, an identity-canonical* Ψ*DO symbol branch, a dissipative symbol branch when the target family has parabolic smoothing, and an explicit residual path. If the learned branch budgets are*

$$\varepsilon_{\text{fio}}, \qquad \varepsilon_{\Psi}, \qquad \varepsilon_{\text{damp}}, \qquad \varepsilon_{\text{res}},$$

then the packet-space error is bounded by the right-hand side of (10) *plus the corresponding finite learned-budget term*

$$C_{\text{learn}}\big(\varepsilon_{\text{fio}} + \varepsilon_{\Psi} + \varepsilon_{\text{damp}} + \varepsilon_{\text{res}}\big).$$

Proof Combine the finite branchwise propagation theorem with Theorem 27. The finite part is exactly the machine-checked statement in `UnifiedCalculus.lean`: FIO-branch error, identity-ΨDO branch error, and smoothing residual error add linearly in packet ℓ^1. The continuous theorem supplies the classical truncation and discretization defects. ■

6.9 Admissible microlocal extension axes

The calculus backbone also clarifies which architectural extensions are mathematically admissible. An extension is admissible when it has (i) a corresponding microlocal object, (ii) an executable model hook, and (iii) a defect term that can be added to the same packet-budget ledger. This keeps each extension visible to the theorem instead of treating it as an unconstrained residual module.

Lemma 30 (Extension-budget bookkeeping) *Assume the hypotheses of Corollary 29. Suppose an architectural extension in Table 2 remains inside the restricted packet calculus and has a verified finite defect* E_{axis} *in the appropriate packet operator norm. Then the same field-level and packet-level conclusions hold with the right-hand side enlarged by* $C_{\text{axis}}E_{\text{axis}}$*. For a collection of such extensions,*

$$\|\mathcal{M}^{\text{ext}}_{\theta,h} - T_h\| \le \|\mathcal{M}^{\text{base}}_{\theta,h} - T_h\| + C_{\text{frame}}E_{\text{frame}} + C_{\text{branch}}E_{\text{branch}} + C_{S^m}E_{S^m} + C_{\text{rad}}E_{\text{rad}} + C_{\text{td}}E_{\text{td}}.$$

Extensions enter the theorem-facing MiNO*-Core claim once their finite defect term is available; otherwise they remain empirical hooks or residual paths.*

Extension axis	Microlocal object	Executable hook	Budget term
Anisotropic packets / curvelet limit	Directional packets with parabolic or elongated phase-space tiling	Current `anisotropic_gabor` and `directional_gabor` probes; full curvelet/shearlet tokenizer deferred	$E_{\mathrm{frame}}(\mathfrak{C}_{\mathrm{frame}}, \delta_h, \theta_{\mathrm{dir}})$
Finite multibranch canonical relation	Finite sum of canonical-graph FIOs on a compact window	`num_canonical_branches`, metadata or metadata-frequency softmax router, branchwise carrier/landing	$E_{\mathrm{route}} = E_{\mathrm{prior}} + E_{\mathrm{learned}}$, plus $E_{\mathrm{cross}} + \sum_\ell (E_{\mathrm{tr},\ell} + E_{\mathrm{sym},\ell} + E_{\mathrm{phase},\ell})$
Stronger symbol-class enforcement	Finite seminorm envelope for $S^m_{1,0}$ and Calderón–Vaillancourt bounds	`spectral_order`, identity-ΨDO branch, finite symbol-seminorm proxy	$E_{S^m} = E_{\mathrm{order}} + E_{\mathrm{seminorm}}$
Outgoing resolvent/radiation symbol	Local near-characteristic signed resolvent symbol for outgoing high-frequency resolvents	`helmholtz_resolvent` complex-pair multiplier, auto-calibrated characteristic shell, outgoing gate, optional explicit real/imaginary output losses	$E_{\mathrm{rad}} = E_{\mathrm{shell}} + E_{\mathrm{out}} + E_{\mathbb{C}} + E_{\mathrm{cap}} + E_{\mathrm{cplx}}$
Nonlinear transport–diffusion coupling	Linearized quasilinear step: advective FIO composed with dissipative ΨDO	Transport branch plus dissipative-ΨDO branch and residual pressure/closure path	$E_{\mathrm{td}} = E_{\mathrm{adv}} + E_{\mathrm{damp}} + E_{\mathrm{split}} + E_{\mathrm{closure}}$

Table 2: Microlocally admissible extension axes. The first four are partially instantiated in the artifact; the last row records the calculus reading of nonlinear or quasilinear transport–diffusion updates and is reserved for a dedicated empirical campaign.

Proof Each listed extension changes the retained packet operator by an explicitly named perturbation: frame discretization and anisotropic covering change the analysis/synthesis constants; branch routing and cross-talk perturb the finite FIO sum; symbol enforcement perturbs the finite S^m seminorm envelope; the outgoing resolvent proxy perturbs the near-characteristic shell, the signed radiation multiplier, and its finite complex-pair realization; and transport–diffusion splitting perturbs the product of advective and dissipative packet laws. The finite bridge is additive in these perturbations, so the result follows by the same analysis–synthesis triangle inequality used in Corollary 29. ■

6.10 Machine-checked bridge

The machine-checked component begins after the classical estimates have organized the packet matrix into canonical shells. The bridge has a concrete Lean decomposition:

- `CanonicalTube.lean` defines the finite shell and canonical-tube objects;
- `ShellCount.lean` isolates shell-cardinality and counting interfaces;
- `PacketTail.lean` proves the geometric shell-tail estimate;
- `AbstractSparsity.lean` packages off-graph decay plus shell counting into an abstract truncation bound;

- `RestrictedBridge.lean` packages the restricted non-stationary-phase tail bound with the cross-resolution transfer wrapper;
- `StencilPropagation.lean` proves the finite bounded-K propagation theorem used by the continuous-to-discrete stencil corollary;
- `StencilWaveTransfer.lean` composes the bounded-stencil coarse theorem with the scale-indexed transfer wrapper;
- `WaveTransfer.lean` connects the truncation layer back to the explicit cross-resolution theorem;
- `UnifiedCalculus.lean` proves the finite branch-composition spine for FIO, identity-ΨDO, and smoothing residual contributions.

In other words, the oscillatory-integral input remains classical, but the shell-counting, truncation, and transfer bridge is now modularized inside the theorem prover.

6.11 Why the finite core remains $K = 1$

The continuous theory now says that bounded canonical stencils are natural. The Lean-facing finite landing lemma in Section 5 nevertheless stays at $K = 1$. This is deliberate.

The $K = 1$ theorem is the cleanest machine-checkable algebraic core. It isolates transport mismatch, symbol mismatch, and residual error in the smallest exact class. The bounded-K continuous corollary above should therefore be read as an *upgrade path*: short-time wave and abstract FIO sparsity explain why the true operator often lives in a slightly richer class than the exact Lean theorem, while the Lean theorem identifies the primitive that remains visible inside that richer class.

The paper establishes the primitive, exhibits the canonical wave/FIO route to bounded-stencil refinement, and connects the classical continuous analysis to the machine-checked finite bridge.

7 PDE Specializations and Semi-Discrete Transplantation

Section 6 gives the continuous microlocal reason that a propagated packet law should exist. This section turns that continuous picture into PDE-specialized modeling statements and then lands back in an explicit packet-basis matrix class. The key idea is simple: once a *packet-basis discretization* of a PDE solver is close to a sparse canonical packet matrix, Theorem 6 gives the operator-level ℓ^2 envelope while Theorem 8 supplies the machine-checked finite landing algebra.

7.1 Packet-basis solver matrices

Fix a discretized PDE solver matrix K_h acting on a grid of size m. Let $U_h \in \mathbb{C}^{m\times n}$ be a packet synthesis matrix and let $V_h \in \mathbb{C}^{n\times m}$ be the corresponding analysis matrix. The packet-basis realization of the solver is

$$\mathcal{K}_h := V_h K_h U_h \in \mathbb{C}^{n\times n}.$$

The continuous microlocal reduction theorem says that, for oscillatory PDEs with hyperbolic transport structure, $\mathcal{K}_h$ should be close to a sparse transported-symbol matrix: most of the energy of an input packet should move to a small set of output packets centered near the propagated phase-space location, with amplitude determined by a local symbol. The finite class introduced below is the implementation-side landing spot for that continuous statement.

Definition 31 (Transported-symbol matrix class) *We say that a packet-basis matrix $\mathcal{K}_h$ belongs to the class $\mathfrak{T}(\varepsilon_{\mathrm{tr}}, \varepsilon_{\mathrm{sym}}, \varepsilon_{\mathrm{res}})$ if there exist a reference transport map $\pi^\star$, reference symbol values $s_i^\star$, an approximate transport map $\hat{\pi}$, approximate symbol values $\hat{s}_i$, and a residual operator R_h such that*

$$\mathcal{K}_h a = \left(s_i^\star a_{\pi^\star(i)}\right)_{i=1}^n + \left((\hat{s}_i - s_i^\star) a_{\hat{\pi}(i)}\right)_{i=1}^n + R_h a,$$

and the coefficient envelopes satisfy Assumption 7 on the coefficient vectors under consideration.

This definition is simply the theorem written in matrix language. It is precisely the right finite landing spot for PDE transplantation.

Proposition 32 (Semi-discrete transplantation principle) *Let K_h be a discretized PDE solver, let U_h, V_h define a packet basis, and let $\mathcal{K}_h = V_h K_h U_h$. If $\mathcal{K}_h$ belongs to the class $\mathfrak{T}(\varepsilon_{\mathrm{tr}}, \varepsilon_{\mathrm{sym}}, \varepsilon_{\mathrm{res}})$ for a family of coefficient vectors with envelopes B_a and B_s, then the induced packet-space approximation error of an approximate* MiNO *layer is bounded exactly as in Theorem 8.*

Proof By Definition 31, the action of $\mathcal{K}_h$ on the coefficient vectors under consideration admits the decomposition required by Theorem 8, with the same transport, symbol, and residual budgets. Therefore the theorem applies directly to the packet-space realization. ■

Proposition 32 is intentionally taut and finite. Its value is that it separates what must be established for each PDE family from what is already proved universally. Each PDE-specific section below can therefore focus on why a packet-basis solver should lie near the transported-symbol class.

7.2 Variable-coefficient Helmholtz

Consider the variable-coefficient Helmholtz problem

$$-\nabla \cdot (c(x)\nabla u) - \omega^2 n(x) u = f.$$

At high frequency, local oscillation and phase drift dominate. In packet language, a localized oscillatory packet does not remain stationary in physical space or frequency space. It moves according to the local Hamiltonian geometry induced by the principal symbol. The packet-basis matrix $\mathcal{K}_h$ is therefore expected to be dominated by a transport map in (x, ξ) together with a local amplitude factor and a lower-order leakage term.

For the purposes of this paper, the rigorous statement we need is the following finite modeling hypothesis.

Assumption 33 (Helmholtz packet hypothesis) *For a chosen packet basis and a resolved frequency range, the packet-basis inverse Helmholtz solver $\mathcal{K}_h$ lies in $\mathfrak{T}(\varepsilon_{\mathrm{tr}}, \varepsilon_{\mathrm{sym}}, \varepsilon_{\mathrm{res}})$ with budgets that improve as the packet frame is refined and the high-frequency solver is better localized.*

Assumption 33 is the discrete expression of the continuous microlocal intuition. Given this assumption, Proposition 32 immediately supplies the MiNO approximation guarantee. In other words, the theoretical work left to the PDE specialist is to justify the packet-basis transported-symbol model for the discretization at hand, not to re-prove the finite error law.

Local outgoing Helmholtz as a branch-factored resolvent instance. High-k Helmholtz is not a pure single-step wave propagator. Even in a local window, an outgoing high-frequency Green operator combines transported packet geometry with finite landing error, directional frame resolution, and a near-characteristic resolvent symbol. A finite sum of canonical branches is one admissible part of this structure, but branch multiplicity alone need not be the active bottleneck at a fixed finite budget. Local outgoing Helmholtz therefore fits the general branch-factored oscillatory envelope of Theorem 25, with additional radiation and landing budgets. The statement below treats the local parametrix regime and keeps global resolvent theory outside the theorem claim. Assume a no-trapping local outgoing parametrix admits a finite packet-window decomposition

$$K_h^{\mathrm{out}} \approx \sum_{\ell=1}^{L} K_{h,\ell}^{\mathrm{FIO}} + S_h,$$

where each $K_{h,\ell}^{\mathrm{FIO}}$ has a canonical graph κ_ℓ, transported symbol s_ℓ, and retained packet tube, and where S_h is smoothing or outside the retained frequency window. This is the local finite-branch regime tested in the high-k Helmholtz campaign; caustics, Maslov phases, glancing rays, trapping, and global radiation-condition analysis are not claimed.

Corollary 34 (Local finite-branch Helmholtz packet bound) *Suppose the local outgoing Helmholtz packet realization is within a finite L-branch FIO window as above. Suppose branch ℓ has transport, symbol, and optional phase defects $E_{\mathrm{tr},\ell}$, $E_{\mathrm{sym},\ell}$, and $E_{\mathrm{phase},\ell}$, the router has mismatch E_{route}, and retained branch tubes have cross-talk budget E_{cross}. Then Theorem 25 gives*

$$\begin{aligned} \|\mathcal{M}_{\theta,h}^{(L)} - K_h^{\mathrm{out}}\|_{\ell^2\to\ell^2} \le & \sum_{\ell=1}^{L} \big(E_{\mathrm{tr},\ell} + E_{\mathrm{sym},\ell} + E_{\mathrm{phase},\ell}\big) \\ & + \sum_{\ell=1}^{L} \big(E_{\mathrm{tsyn},\ell} + E_{\mathrm{car},\ell} + E_{\mathrm{land},\ell}\big) + E_{\mathrm{route}} + E_{\mathrm{cross}} \\ & + E_{\mathrm{res}} + E_{\mathrm{tail}} + E_{\mathrm{frame}} + E_{\mathrm{asymp}} + C_{\mathrm{cov}}\delta_h. \end{aligned}$$

At field level, the same projection constants as in Theorem 36 are added.

Proof This is Theorem 25 with $A_h = 0$ on the outgoing high-frequency window and with the smoothing or out-of-window part absorbed into E_{res}, E_{tail}, and the frame/asymptotic/covering defects. ■

Corollary 34 gives the precise hypothesis under which a branched carrier-bound MiNO can reduce the high-k gap: the target must have a small finite number of locally separable outgoing canonical branches, the learned router must select them with limited cross-talk, and the phase/landing error must be controlled tightly enough for the frequency range. The executed high-k probes indicate that, at the current local budget, branch multiplicity is not the identified active term; the stronger supported terms are the anisotropic frame, carrier-bound finite landing, and outgoing resolvent-symbol budgets introduced below. The branch diagnostic in Appendix D is interpreted as a finite-budget falsification of the branch-only hypothesis, not as evidence against the broader branch-factored microlocal structure.

Outgoing resolvent symbol layer. The previous corollary explains the multibranch canonical part, but high-k Helmholtz also contains a symbol difficulty that is absent from the short-time wave theorem. In semiclassical notation the local outgoing inverse is governed by a principal factor of the form

$$(p(x,\xi)+i0)^{-1}, \qquad p(x,\xi)=c(x)|\xi|^2-n(x),$$

after microlocal cutoffs and lower-order corrections. Thus an executable symbol branch that only exposes a smooth radial S^m order can miss the near-characteristic shell and outgoing attenuation structure of the resolvent. We isolate this as a separate finite budget rather than folding it into an anonymous residual.

Corollary 35 (Near-characteristic resolvent-symbol budget) *Assume the local outgoing Helmholtz window of Corollary 34. Suppose, in addition, that after branch localization the remaining identity-canonical factor has a cut-off resolvent symbol*

$$a_{\mathrm{rad},h}(z)=\chi_{\mathrm{out}}(z)\chi_{\mathrm{char}}(z)\,(p(z)+i0)^{-1}b_h(z),$$

with b_h satisfying the finite symbol-seminorm envelope required by Corollary 28. Let the learned radiation-aware multiplier have shell-distance error E_{shell}, outgoing-envelope error E_{out}, finite complex-pair realization error $E_{\mathbb{C}}$ for the signed resolvent multiplier, and bounded-cap error E_{cap} on the retained packet window. Then the local finite-branch Helmholtz estimate holds with the additional term

$$E_{\mathrm{rad}}:=E_{\mathrm{shell}}+E_{\mathrm{out}}+E_{\mathbb{C}}+E_{\mathrm{cap}},$$

so that the right-hand side of Corollary 34 is enlarged by $C_{\mathrm{rad}}E_{\mathrm{rad}}$.

Proof Apply the packet almost-diagonalization step used in Corollary 28 to the cut-off identity-canonical resolvent factor. The cutoff removes the global singularities and keeps the statement local in the retained packet window. Replacing the ideal near-characteristic multiplier by the executable layer changes the retained packet matrix by the displayed defects: shell localization, outgoing-envelope approximation, the finite complex-pair realization error $E_{\mathbb{C}}$ for multiplication by the signed real/imaginary resolvent factor, and the bounded cap used to keep the finite neural multiplier stable. The packet-matrix envelope is additive in retained symbol perturbations, so this produces the extra $C_{\mathrm{rad}} E_{\mathrm{rad}}$ term. ■

The artifact implements this layer as `helmholtz_resolvent`: a multiplier-only symbol parameterization with radial frequency, unit direction, an auto-calibrated characteristic shell, signed real/imaginary resolvent weights, and an outgoing gate. The code lifts packet features to a bias-free latent complex pair, applies the exact real formula for multiplication by $a + ib$, and projects back to the packet feature space. When a dataset provides genuine real/imaginary target channels, the runner can also add explicit complex-pair relative and phase/coherence losses; for scalar real-valued Helmholtz rows these losses are inactive, so no artificial quadrature channel is introduced. The layer is an executable hook for the near-characteristic symbol term isolated above. The direct `spectral_order` versus `helmholtz_resolvent` plausibility row in Table 4 shows that, at the current finite budget, the resolvent hook changes phase-sensitive behavior more clearly than relative-ℓ^2 error. It is therefore reported as a budget-local radiation/symbol diagnostic, not as an independent solved high-k mechanism. If high-k error remains large after enabling the hook, the remaining budgets are E_{frame}, E_{route}, E_{phase}, or the learned finite-budget E_{rad}.

Frequency-scaling benchmark contract. A high-k Helmholtz result is only a flagship result if it controls degradation as k increases, not merely one fixed cache row. The packet budget above gives the required scaling language. Let

$$\mathcal{E}_{\mathrm{Hk}}(k) = E_{\mathrm{frame}}(k)+E_{\mathrm{route}}(k)+E_{\mathrm{cross}}(k)+E_{\mathrm{phase}}(k)+E_{\mathrm{land}}(k)+E_{\mathrm{rad}}(k)+E_{\mathrm{res}}(k)+E_{\mathrm{tail}}(k).$$

Here E_{frame} includes the directional or anisotropic packet covering defect, E_{phase} includes phase/coherence error, E_{rad} is the near-characteristic resolvent-symbol defect from Corollary 35, and E_{res} includes the measured PDE and boundary/radiation residuals of the learned field. In a resolved no-trapping local window,

$$\|\mathcal{M}^{(L)}_{\theta,h}(k) - K^{\mathrm{out}}_{h,k}\| \le C(k)\, \mathcal{E}_{\mathrm{Hk}}(k),$$

with $C(k)$ inherited from the local resolvent stability constant on that window. The manuscript therefore uses "high-frequency robustness" only as a scaling claim: it requires reporting kL, points per wavelength, reference-solver residual, phase error, radiation/PDE residual, and error degradation across increasing k. A pollution-style learned-representation claim would additionally require showing that the learned degrees of freedom or packet budget grow no faster than the natural high-frequency resolution scale for the tested accuracy.

7.3 High-frequency wave propagation

For the scalar wave equation

$$\partial_t^2 u - c(x)^2 \Delta u = 0,$$

the one-step or short-time propagator is even more naturally interpreted in packet coordinates. Packets move along bicharacteristics and their amplitudes are corrected by lower-order transport factors. This is the cleanest setting for the microlocal viewpoint, because propagation itself is the fundamental phenomenon. In the notation of Section 6, the propagator is the direct model case for Theorem 19.

More precisely, the continuous spine splits the wave story into two steps. Theorem 15 supplies the PDE theorem: outside the canonical tube traced by the half-wave flow, packet matrix entries decay rapidly by non-stationary phase. Theorem 17 then converts that decay into an explicit bounded-stencil truncation rate. Theorem 19 is the wave/FIO consequence that lands directly in the theorem-grade MiNO class.

Transported synthesis and carrier binding. The theorem-grade model assumes that moved packet metadata is used not only to choose a message-passing stencil, but also to synthesize packets at their transported centers. Otherwise a finite implementation can satisfy the internal graph description while folding all coefficients back onto the original analysis grid, making the transport branch geometrically inert at the field level. The carrier-bound MiNO implementation therefore uses two transported field paths: a transported synthesis of the learned packet coefficients and a transported high-frequency input carrier built from the original analysis coefficients. The controls in Section 9 show that these paths are behaviorally separable, so the executable theorem tracks them separately. We write $W^*_{h,\widehat{\kappa}}$ for ideal transported synthesis using the learned packet centers, $\widetilde{W}^*_{h,\widehat{\kappa},\mathrm{coef}}$ for the executable transported synthesis of learned coefficients, and $\widetilde{W}^*_{h,\widehat{\kappa},\mathrm{car}}$ for the full executable carrier-bound synthesis including the high-frequency input carrier. Define

$$E_{\mathrm{tsyn}}(u,t) := \big\| W^*_{h,\widehat{\kappa}} \mathcal{M}^{\mathrm{core}}_{h,K,P}(t) W_h u - \widetilde{W}^*_{h,\widehat{\kappa},\mathrm{coef}} \mathcal{M}^{\mathrm{core}}_{h,K,P}(t) W_h u \big\|_{L^2},$$

and

$$E_{\mathrm{car}}(u,t) := \big\| \widetilde{W}^*_{h,\widehat{\kappa},\mathrm{coef}} \mathcal{M}^{\mathrm{core}}_{h,K,P}(t) W_h u - \widetilde{W}^*_{h,\widehat{\kappa},\mathrm{car}} \mathcal{M}^{\mathrm{core}}_{h,K,P}(t) W_h u \big\|_{L^2}.$$

The first term is the transported-coefficient synthesis defect. The second term records the high-frequency carrier channel; it is not part of the abstract FIO packet matrix, but is the executable field-level mechanism that makes transported metadata visible after synthesis. In the code path used for the controlled mechanism rows, both transported paths are implemented as packet-energy-weighted field warps from initial packet centers to final metadata, while the skip and refinement paths are low-pass restricted.

The code also exposes a learned transport-conditioned landing decoder D_θ that may be inserted after the two transported paths. The correct theoretical object is not an unrestricted post-hoc solver, but an admissible finite landing map. We call D_θ admissible when it is local with receptive radius r_{land}, bias-free, takes only the transported coefficient synthesis, transported input carrier, and low-pass lifted field as inputs, and is gated by transported-path energy so that

$$D_\theta(0,0,\ell) = 0 \qquad \text{for every low-pass field } \ell.$$

Let $\widetilde{W}^*_{h,\widehat{\kappa},\theta}$ denote the resulting learned executable landing map and define

$$E_{\mathrm{land},\theta}(u,t) := \big\| W^*_{h,\widehat{\kappa}} \mathcal{M}^{\mathrm{core}}_{h,K,P}(t) W_h u - \widetilde{W}^*_{h,\widehat{\kappa},\theta} \mathcal{M}^{\mathrm{core}}_{h,K,P}(t) W_h u \big\|_{L^2}.$$

Thus the nonlearned carrier-bound realization is controlled by $E_{\mathrm{tsyn}} + E_{\mathrm{car}}$, while the learned-landing realization is controlled by $E_{\mathrm{land},\theta}$. If the ideal finite landing correction belongs to a compact C^r local landing class on the retained packet window and the decoder has parameter budget P_{land}, standard local-network approximation gives the bookkeeping rate

$$E_{\text{land},\theta}(u,t) \le C_{\text{land}} P_{\text{land}}^{-r/m_{\text{land}}} \mathcal{E}_{\text{act}}(u) + E_{\text{gate}}(u,t),$$

where $\mathcal{E}_{\text{act}}(u)$ is the active transported packet energy and E_{gate} records any loss induced by the transported-energy gate. This is a finite executable landing budget, not a new microlocal propagation theorem. The ideal theorem corresponds to vanishing executable landing defect.

Theorem 36 (Microlocal Propagation Theorem for MiNO-Core) *Fix a no-caustic time window $|t| \le T$ for the half-wave propagator $U_+(t)$ generated by $p(x,\xi) = c(x)|\xi|$, and let*

$$K_h(t) = W_h U_+(t) W_h^*$$

be its packet realization on a packet frame of scale h and normalized covering defect δ_h (raw covering radius $h^{1/2}\delta_h$). Assume the compact packet window Ω_T swept out by the flow satisfies the smoothness hypotheses of Theorem 22: the canonical map κ_t is C^r with norm at most A_r, and the retained symbol is a local packet kernel

$$s_t(z,\nu;h), \qquad \nu = h^{-1/2}(z_\gamma - \kappa_t(z_\eta)),$$

with C^r norm at most A_r on a compact source–tube coordinate window of dimension $m_\Omega + q_{\text{loc}}$. Let $\mathcal{M}^{\text{core}}_{h,K,P}(t)$ be a theorem-grade MiNO*-Core block with stencil budget K, metadata-branch parameter budget P_{tr}, symbol-branch parameter budget P_{sym}, and explicit residual budget ε_{res}. Then for every $N > q+1$ and every retained asymptotic order $M \ge 1$,*

$$\begin{aligned}\|\mathcal{M}^{\text{core}}_{h,K,P}(t) - K_h(t)\|_{\ell^2\to\ell^2} \le{}& C_{\text{tr}} K\big(h^{-1/2} P_{\text{tr}}^{-r/m_\Omega} + \delta_h\big)\\ &+ C_{\text{sym}} K P_{\text{sym}}^{-r/(m_\Omega+q_{\text{loc}})} + C_{\text{res}}\varepsilon_{\text{res}}\\ &+ C_{\text{tail}} K^{1-N/q} + C_{\text{frame}} E_{\text{frame}}(h) + C_{\text{asymp}} E_{\text{asymp}}(h,M) + C_{\text{cov}}\delta_h,\end{aligned} \tag{11}$$

where $m_\Omega = \dim(\Omega_T)$ and the constants depend only on the wave-family window, packet frame, shell-counting geometry, compact-window C^r bounds, local tube dimension, and branch approximation class. The normalized covering defect δ_h is not automatically small: it tends to zero only for oversampled packet clouds, while for critically sampled executable frames it is a fixed packetization defect. Denote the right-hand side of (11) *by $E^{(2)}_{h,K,P}$; when no confusion is possible we write $E_{h,K,P}$ for this ℓ^2 packet-matrix envelope. Equivalently, for every packet vector a,*

$$\|\mathcal{M}^{\text{core}}_{h,K,P}(t)a - K_h(t)a\|_{\ell^2} \le E^{(2)}_{h,K,P}\|a\|_{\ell^2}.$$

If the synthesis map W_h^* has upper frame bound B_W, let $\mathcal{P}_h := W_h^* W_h$ and define the input and output projection defects

$$E^{\text{in}}_{\text{proj}}(u,t) := \|U_+(t)(I-\mathcal{P}_h)u\|_{L^2}, \qquad E^{\text{out}}_{\text{proj}}(u,t) := \|(I-\mathcal{P}_h)U_+(t)\mathcal{P}_h u\|_{L^2}.$$

Then the same theorem gives the direct field-level operator estimate for the nonlearned carrier-bound realization

$$\begin{aligned}
&\|\widetilde{W}^*_{h,\widehat{\kappa},\mathrm{car}}\mathcal{M}^{\mathrm{core}}_{h,K,P}(t)W_h u - U_+(t)u\|_{L^2} \\
&\quad\le B_W E^{(2)}_{h,K,P}\|W_h u\|_{\ell^2} + E^{\mathrm{in}}_{\mathrm{proj}}(u,t) + E^{\mathrm{out}}_{\mathrm{proj}}(u,t) \\
&\quad\quad + E_{\mathrm{tsyn}}(u,t) + E_{\mathrm{car}}(u,t).
\end{aligned}$$

For the learned admissible landing realization, the last two executable terms are replaced by $E_{\mathrm{land},\theta}(u,t)$*:*

$$\begin{aligned}
&\|\widetilde{W}^*_{h,\widehat{\kappa},\theta}\mathcal{M}^{\mathrm{core}}_{h,K,P}(t)W_h u - U_+(t)u\|_{L^2} \\
&\quad\le B_W E^{(2)}_{h,K,P}\|W_h u\|_{\ell^2} + E^{\mathrm{in}}_{\mathrm{proj}}(u,t) + E^{\mathrm{out}}_{\mathrm{proj}}(u,t) + E_{\mathrm{land},\theta}(u,t).
\end{aligned}$$

Thus the canonical reading of the theorem is a function-space approximation statement in the packet energy norm; the older coefficientwise ℓ^1 *estimate is the finite landing layer checked by the proof assistant.*

Proof This is Theorem 22 specialized to the PDE transplantation notation of the present section. Theorems 15, 17, and 19 reduce the half-wave propagator to a bounded-stencil packet law with tail, frame, asymptotic, and covering defects. The compact-window neural approximation input approximates κ_t and the local packet kernel $s_t(z,\nu;h)$ by the metadata and symbol branches at rates $P_{\mathrm{tr}}^{-r/m_\Omega}$ and $P_{\mathrm{sym}}^{-r/(m_\Omega+q_{\mathrm{loc}})}$, with the geometric transport error measured in packet-radius units and hence multiplied by $h^{-1/2}$. The Schur packet-matrix transfer in Theorem 6 converts these entrywise kernel and symbol errors into the $\ell^2 \to \ell^2$ learned-budget part of (11). Finally,

$$W_h^* K_h(t) W_h u = \mathcal{P}_h U_+(t)\mathcal{P}_h u,$$

so the difference between the packet-space realization and the true field propagator is exactly controlled by the two projection defects displayed in the theorem. Replacing the ideal transported synthesis $W^*_{h,\widehat{\kappa}}$ by the executable coefficient synthesis adds $E_{\mathrm{tsyn}}(u,t)$, and replacing the coefficient-only synthesis by the carrier-bound executable synthesis adds $E_{\mathrm{car}}(u,t)$, both by the triangle inequality. If the learned admissible landing map is used, the same triangle inequality is applied once, directly between $W^*_{h,\widehat{\kappa}}$ and $\widetilde{W}^*_{h,\widehat{\kappa},\theta}$, producing $E_{\mathrm{land},\theta}$. ■

Theorem 36 is the manuscript's Microlocal Propagation Theorem. The name is reserved for this theorem, not for the finite triangle-inequality lemma in Section 5, because this statement contains the actual propagation content: the PDE propagator has a canonical graph, the packet matrix decays away from the graph, bounded canonical stencils capture the significant entries, and MiNO-Core approximates the retained transported-symbol law with explicit network rates. It separates the budgets that the rest of the manuscript tracks empirically: network-realized transport, network-realized symbol, residual, packet-tail truncation, discretization/covering defect, transported-synthesis defect, carrier-channel discrepancy, and optional learned finite-landing discrepancy. The first two scale with branch capacity, not merely with assumed learned budgets; the tail and discretization terms are the price of replacing the exact packet matrix by a bounded canonical stencil on a finite frame; the final executable terms record field-synthesis effects that are not visible in a purely packet-matrix statement.

This is the main mathematical spine of the paper. Each architectural branch corresponds to one term in an explicit approximation law, and the restricted semiclassical packet-calculus chain has a computational landing layer:

$$\begin{aligned}\text{half-wave parametrix} &\Rightarrow \text{off-canonical-tube decay} \Rightarrow \text{packet sparsity}\\ &\Rightarrow \text{transported-symbol reduction} \Rightarrow \text{network-realizable}\\ &\Rightarrow \text{MiNO-Core approximation.}\end{aligned}$$

The theorem organizes the classical short-time FIO packet calculus into an approximation statement whose final finite bridge is machine-checkable and whose hypotheses map directly onto executable architecture knobs.

7.4 Transport certification and comparison boundary

The main theorem is an upper-bound theorem for a packet architecture. It does not by itself imply that MiNO must dominate every multiscale baseline. To make the logical status precise, we record two finite consequences that are used in the empirical interpretation. The first makes the theorem non-vacuous when a controlled benchmark supplies a flow target; the second states the limited separation that the current theory actually supports.

Proposition 37 (Flow certificate to packet transport budget) *Let Γ_h be the retained packet grid in the compact window and let $\hat\kappa_h$ be the learned metadata map. Suppose that, on the active packet set $A_h \subset \Gamma_h$,*

$$\epsilon_{\kappa,h} := \sup_{\eta\in A_h} d_h\big(\hat\kappa_h(z_\eta), \kappa_t(z_\eta)\big)$$

is finite in the packet-radius metric $d_h = h^{-1/2}d$. Assume that the retained packet overlap kernel is Lipschitz on the canonical tube:

$$\sum_{\gamma\in\Pi_K(\eta)} \big|G_h(z_\gamma,\hat z) - G_h(z_\gamma,z')\big| \le L_{\mathrm{ker}} d_h(\hat z, z')$$

for all $\hat z, z'$ in the enlarged tube, uniformly in η. Then the transport-kernel mismatch entering the finite transported-symbol bridge satisfies

$$\epsilon_{\mathrm{tr},h} \le L_{\mathrm{ker}}\big(\epsilon_{\kappa,h} + C\delta_h\big),$$

and the transport contribution in Theorem 36 may be bounded by

$$C_{\mathrm{tr}}K\, L_{\mathrm{ker}}\big(\epsilon_{\kappa,h} + C\delta_h\big).$$

Proof For each active source packet, compare the ideal retained kernel centered at $\kappa_t(z_\eta)$ with the learned retained kernel centered at $\hat\kappa_h(z_\eta)$. The covering defect moves the nearest executable packet center by at most $C\delta_h$ in packet-radius units. The Lipschitz kernel hypothesis gives the displayed row/column kernel mismatch. The finite bounded-stencil perturbation theorem then multiplies this transport mismatch by the retained degree K and the same transport constant as in Theorem 36. ■

Proposition 37 is the formal reading of the artifact's `test_canonical_flow_error` metric. A small measured canonical-flow proxy does not prove a full classical wavefront theorem, but it does certify the particular geometric hypothesis that the finite transport budget needs on controlled rows.

Proposition 38 (Conditional packet-transport advantage) *Fix a normalized input packet ϕ^h_η and suppose the target propagator satisfies*

$$U_+(t)\phi^h_\eta = s_\eta \phi^h_{\kappa_t(z_\eta)} + r_\eta, \qquad |s_\eta| \geq s_0, \qquad \|r_\eta\|_{L^2} \leq \rho.$$

Let $\mathcal{A}_R$ be any model class whose output on ϕ^h_η lies in the closed span of packets with centers inside the identity tube

$$\{z_\gamma : \ d_h(z_\gamma, z_\eta) \leq R\}.$$

Assume the transported packet is almost orthogonal to that identity tube:

$$\sup_{\substack{v \in \mathcal{A}_R(\phi^h_\eta) \\ \|v\|_{L^2}=1}} \left|\langle \phi^h_{\kappa_t(z_\eta)}, v\rangle\right| \leq \mu \qquad \textit{with } 0 \leq \mu < 1.$$

Then every identity-local model $A \in \mathcal{A}_R$ obeys

$$\|A\phi^h_\eta - U_+(t)\phi^h_\eta\|_{L^2} \geq s_0\sqrt{1-\mu^2} - \rho.$$

In contrast, a transported-synthesis packet model with

$$d_h(\hat{\kappa}_h(z_\eta), \kappa_t(z_\eta)) \leq \epsilon_{\kappa,h}, \qquad |\hat{s}_\eta - s_\eta| \leq \epsilon_s,$$

and Lipschitz packet synthesis on the tube has error

$$\|\hat{s}_\eta \phi^h_{\hat{\kappa}_h(z_\eta)} - U_+(t)\phi^h_\eta\|_{L^2} \leq C_{\mathrm{pkt}} s_0 \epsilon_{\kappa,h} + \epsilon_s + \rho.$$

Proof Let V_R be the identity-tube output subspace. The assumption says that the distance from $\phi^h_{\kappa_t(z_\eta)}$ to V_R is at least $\sqrt{1-\mu^2}$. Therefore every $v \in V_R$ has

$$\|v - s_\eta \phi^h_{\kappa_t(z_\eta)}\|_{L^2} \geq |s_\eta|\sqrt{1-\mu^2}.$$

Adding the residual r_η by the reverse triangle inequality gives the lower bound. The transported upper bound follows by adding and subtracting $s_\eta \phi^h_{\hat{\kappa}_h(z_\eta)}$ and using Lipschitz dependence of the packet atom on its center in packet-radius units. ■

The next proposition records the representation separation supplied by the packet geometry. It compares carrier-bound packet transport with primitives whose output is restricted to an identity tube plus a fixed finite Fourier-limited residual span. The result is a finite-subspace statement about representation geometry.

Proposition 39 (Primitive-level representation separation for packet transport) *Let the hypotheses and notation of Proposition 38 hold. Let V_R be the identity-tube packet subspace. Let $\mathcal{G}_Q$ be a fixed Q-dimensional Fourier-limited residual subspace. This subspace represents the part of a global spectral primitive that lacks an explicit packet-dependent landing map. Define*

$$\mathcal{B}_{R,Q}(\phi_\eta^h) \subset V_R + \mathcal{G}_Q.$$

Write $V_{R,Q} := V_R + \mathcal{G}_Q$. Assume the transported packet is separated from this combined primitive span:

$$\sup_{\substack{v \in V_{R,Q} \\ \|v\|_{L^2}=1}} \left|\langle \phi^h_{\kappa_t(z_\eta)}, v\rangle\right| \le \mu_{R,Q}, \qquad 0 \le \mu_{R,Q} < 1.$$

Then every $B \in \mathcal{B}_{R,Q}$ satisfies

$$\|B\phi_\eta^h - U_+(t)\phi_\eta^h\|_{L^2} \ge s_0\sqrt{1-\mu_{R,Q}^2} - \rho.$$

Meanwhile the transported-synthesis packet model has the same upper bound as in Proposition 38,

$$\|\hat{s}_\eta \phi^h_{\hat{\kappa}_h(z_\eta)} - U_+(t)\phi_\eta^h\|_{L^2} \le C_{\mathrm{pkt}} s_0 \epsilon_{\kappa,h} + \epsilon_s + \rho.$$

Proof The proof is the same projection argument as Proposition 38, with V_R replaced by the closed combined span $V_R + \mathcal{G}_Q$. The separation assumption lower bounds the distance from the transported packet to that combined span by $\sqrt{1-\mu_{R,Q}^2}$. The residual r_η is then handled by the reverse triangle inequality, and the transported-packet upper bound is unchanged. ■

Proposition 39 is a finite-subspace separation. When the correct high-frequency packet exits the identity tube and remains outside the fixed Fourier-limited residual span, an explicit packet-transport landing map avoids an error floor. Full UNO, FNO, WNO, or other nonlinear architectures may enlarge the effective span or learn an implicit transported-symbol law through pointwise mixing, multiscale decoding, or refinement. The proposition therefore supports a precise mechanism-efficiency statement: MiNO exposes the sparse canonical packet-matrix law directly, while non-transport primitives must represent the same law indirectly. Appendix C strengthens this single-packet statement into a many-packet rank-growth theorem. There, a fixed-rank or fixed-bandwidth output primitive that must land N_h almost-orthogonal transported packets incurs a rank lower bound proportional to N_h, while a sparse canonical packet matrix keeps bounded local stencil degree. This is the paper's strongest representation-complexity claim; it is still a restricted primitive separation rather than a universal lower bound against fully nonlinear neural operators.

Corollary 40 (Hamiltonian-branch specialization) *Assume the hypotheses of Theorem 36, and suppose the half-wave canonical flow is generated on Ω_T by a Hamiltonian $H \in C^{r+2}(\Omega_T)$ as in Proposition 21. If the metadata branch is the Hamiltonian-vector-field* *branch with scalar Hamiltonian budget P_H and Euler step Δ, then the transport contribution in* (11) *may be replaced by*

$$C_{\mathrm{ham}} K\Big(h^{-1/2}(P_H^{-(r-2)/m_\Omega} + \Delta) + \delta_h\Big),$$

with the same packet-tail, symbol, residual, and projection terms as in Theorem 36. The branch also has canonical-defect proxy

$$\sup_z \big\| D\Psi_{P_H,\Delta}(z)^\top \Omega_0 D\Psi_{P_H,\Delta}(z) - \Omega_0 \big\| \le C_T \Delta^2 \|H_{P_H}\|_{C^2}^2.$$

If the branch is the separable `hamiltonian_verlet` *branch, the same statement holds with the discretization term Δ replaced by Δ^2, and the exact-map canonical-defect proxy is zero.*

Proof Proposition 21 gives the geometric canonical-map error

$$\epsilon_\kappa \le C_T(P_H^{-(r-2)/m_\Omega} + \Delta).$$

The packet-kernel bridge in Theorem 22 converts geometric error into packet-radius units, producing the factor $h^{-1/2}$, and then adds the covering term δ_h. The remaining terms are unchanged. The canonical-defect estimate is exactly the second conclusion of Proposition 21. For `hamiltonian_verlet`, the separable Störmer–Verlet part of Proposition 21 gives the improved Δ^2 consistency term and exact symplecticity of the metadata map. ■

Corollary 40 is the theorem-level reason for exposing Hamiltonian transport in the branch-identifiability campaign. The old `hamiltonian_euler` path is a cheap scalar-generator diagnostic. The `hamiltonian_verlet` path is a structured separable/splitting-biased parameterization: it enforces symplecticity for separable Hamiltonian metadata updates and makes the canonical-defect metric meaningful. The main Microlocal Propagation Theorem requires a learned near-canonical map with the displayed compact-window error budget; it does not require Verlet, nor does it assert that separable Verlet is a general representation theorem for the half-wave Hamiltonian. Since $p(x,\xi) = c(x)|\xi|$ is generally nonseparable, the executable Verlet branch should be read as one theorem-clean way to parameterize a portion of the transport budget.

Corollary 41 (Active-support field-level microlocal propagation bound) *Assume Theorem 36. Also assume that the synthesis map $W_h^* : \ell^2(\Gamma_h) \to L^2(\mathbb{R}^d)$ has upper frame bound B_W. Let*

$$\begin{aligned} E_{h,K,P} = {} & C_{\mathrm{tr}} K\big(h^{-1/2} P_{\mathrm{tr}}^{-r/m_\Omega} + \delta_h\big) + C_{\mathrm{sym}} K P_{\mathrm{sym}}^{-r/(m_\Omega + q_{\mathrm{loc}})} \\ & + C_{\mathrm{res}}\varepsilon_{\mathrm{res}} + C_{\mathrm{tail}} K^{1-N/q} + C_{\mathrm{frame}} E_{\mathrm{frame}}(h) + C_{\mathrm{asymp}} E_{\mathrm{asymp}}(h, M) + C_{\mathrm{cov}}\delta_h. \end{aligned}$$

Let $\mathcal{P}_h = W_h^* W_h$ and define $E_{\mathrm{proj}}^{\mathrm{in}}(u,t)$, $E_{\mathrm{proj}}^{\mathrm{out}}(u,t)$, $E_{\mathrm{tsyn}}(u,t)$, $E_{\mathrm{car}}(u,t)$, and $E_{\mathrm{land},\theta}(u,t)$ as in Theorem 36. Then for every u with packet coefficients $a = W_h u$,

$$\begin{aligned} & \|\widetilde{W}_{h,\widehat{\kappa},\mathrm{car}}^* \mathcal{M}_{h,K,P}^{\mathrm{core}}(t) W_h u - U_+(t) u\|_{L^2} \\ & \quad \le B_W E_{h,K,P} \|W_h u\|_{\ell^2} + E_{\mathrm{proj}}^{\mathrm{in}}(u,t) + E_{\mathrm{proj}}^{\mathrm{out}}(u,t) \\ & \qquad + E_{\mathrm{tsyn}}(u,t) + E_{\mathrm{car}}(u,t). \end{aligned}$$

If the packet error is supported on at most s active packets and obeys a pointwise envelope $E_\infty(a)$, then the sharper active-support estimate

$$\begin{aligned}\|\widetilde{W}^*_{h,\widehat{\kappa},\mathrm{car}}\mathcal{M}^{\mathrm{core}}_{h,K,P}(t)W_h u - U_+(t)u\|_{L^2} \\ \le B_W\sqrt{s}\,E_\infty(a) + E^{\mathrm{in}}_{\mathrm{proj}}(u,t) + E^{\mathrm{out}}_{\mathrm{proj}}(u,t) \\ + E_{\mathrm{tsyn}}(u,t) + E_{\mathrm{car}}(u,t)\end{aligned}$$

also holds. For the learned admissible landing map, replace $E_{\mathrm{tsyn}} + E_{\mathrm{car}}$ by $E_{\mathrm{land},\theta}$ in either displayed estimate.

Proof The first estimate follows from Theorem 36. For the active-support estimate, apply the synthesis bound and the norm chain

$$\|W_h^* e\|_{L^2} \le B_W \|e\|_{\ell^2},$$

with $e = (\mathcal{M}^{\mathrm{core}}_{h,K,P}(t) - K_h(t))W_h u$, and use $\|e\|_{\ell^2} \le \sqrt{s}\|e\|_{\ell^\infty}$. The two projection defects, transported-synthesis defect, carrier-channel term, and optional learned-landing term are then added as in the theorem. ∎

Corollary 42 (Rate balancing for field-level convergence) *Let $h \downarrow 0$ and allow the stencil and branch budgets to depend on h. Under the hypotheses of Theorem 36, suppose*

$$K(h) \to \infty, \qquad K(h)\delta_h \to 0, \qquad K(h)h^{-1/2}P_{\mathrm{tr}}(h)^{-r/m_\Omega} \to 0,$$

$$K(h)P_{\mathrm{sym}}(h)^{-r/(m_\Omega+q_{\mathrm{loc}})} \to 0, \qquad \varepsilon_{\mathrm{res}}(h) \to 0, \qquad K(h)^{1-N/q} \to 0,$$

and $E_{\mathrm{frame}}(h) + E_{\mathrm{asymp}}(h,M) + \delta_h \to 0$. If, for the input family under consideration, $E^{\mathrm{in}}_{\mathrm{proj}}(u_h,t) + E^{\mathrm{out}}_{\mathrm{proj}}(u_h,t) \to 0$ and $\|W_h u_h\|_{\ell^2}$ remains bounded, then

$$\|W_h^*\mathcal{M}^{\mathrm{core}}_{h,K(h),P(h)}(t)W_h u_h - U_+(t)u_h\|_{L^2} \to 0.$$

Proof Under the displayed scaling assumptions every term in $E_{h,K,P}$ tends to zero. The field-level estimate in Theorem 36 then gives convergence after adding the vanishing projection defects. ∎

The discrete transplantation principle applies exactly as above. One picks a time-step operator $K_{h,\Delta t}$, analyzes it in a packet basis, and checks whether the realized matrix lies near the transported-symbol class. In this setting transport error is genuinely a wavefront error: it measures how much the learned packet motion departs from the discrete propagation law.

This is also the PDE family in which Theorem 11 is easiest to interpret. For nested packet systems at scales h and $2h$, the reference transfer defect

$$\|P_{2h\to h}\mathcal{T}^\star_{2h}R_{h\to 2h}a_h - \mathcal{T}^\star_h a_h\|_{\ell^1}$$

measures how much the packetized short-time propagator changes when the phase-space covering is refined. Standard microlocal almost-diagonalization heuristics suggest that this defect should decay with packet radius and time step away from caustics and branch splitting. The theorem therefore turns wave-resolution transfer into a quantitative two-term problem: one term for coarse learned error and one term for the intrinsic packetization defect of the propagator itself. The new bounded-stencil viewpoint sharpens this interpretation further: the reference defect is exactly the error made when the fine packet matrix is compressed into a coarse canonical tube and then lifted back. The Lean file `WaveTransfer.lean` records this finite bridge in the same language as the scale-indexed theorem.

Corollary 43 (Local-window Helmholtz parametrix corollary) *Suppose a variable-coefficient Helmholtz resolvent is microlocally represented on a local frequency window by a finite superposition of short-time canonical-graph pieces with the same packet geometry assumptions as Theorem 36. Then the packet realization of the windowed resolvent satisfies the same* MiNO-*Core approximation law, up to constants that additionally depend on the window partition and the local parametrix amplitudes.*

Proof On each local window, the Helmholtz parametrix is treated as a finite sum of canonical-graph pieces. Apply Theorem 36 to each piece and sum the resulting constants across the partition of unity. The bounded number of window pieces is absorbed into the aggregate constants. ■

7.5 Darcy and elliptic smoothing

The Darcy map is not high-frequency transport dominated. It is primarily elliptic and smoothing. This does not make it non-microlocal. It puts it in the pseudodifferential regime of the calculus.

For a uniformly elliptic operator such as

$$Lu = -\nabla \cdot (A(x)\nabla u),$$

the parametrix is microlocally a pseudodifferential operator of negative order. Its principal inverse symbol has the form

$$q_{-2}(x, \xi) \simeq (\xi^\top A(x)\xi)^{-1}$$

on the elliptic cone. The associated canonical relation is the identity relation, not a nontrivial flow. Therefore, for a Darcy-type elliptic map, the packet-basis matrix is expected to be close to an identity-canonical local symbol action with smoothing residuals:

$$(\mathcal{K}_h a)_i \approx m_h(x_i, \xi_i)a_i + \rho_i(a).$$

In the language of Proposition 4, the transport budget is near zero because $\kappa = \mathrm{id}$, while the symbol and smoothing budgets carry the approximation burden. This is the elliptic specialization of the same microlocal packet calculus.

7.6 One-step linearized Navier–Stokes

For a one-step linearized Navier–Stokes update around a local background velocity field, one can decompose the step into an advection-diffusion structure. The advection part is transport dominated, while the diffusion part acts like a dissipative symbol. In packet coordinates this is a natural mixed regime:

- the advection component pushes packets in position and frequency;
- the diffusion component damps higher frequencies through a dissipative symbol, locally of the form $\exp[-\Delta t\, q(x, \xi)]$;
- nonlinear closure and truncation are absorbed into the residual term.

This is precisely the kind of setting where the packet-budget spine is useful. It says that the model should be judged separately on transport, symbol, and residual quality, rather than through a single opaque operator error.

7.7 Role of the PDE Connection

The PDE layer has a specific role. It connects classical microlocal structure to the finite packet theorem by proving:

1. a finite landing lemma for transported-symbol packet operators;
2. an operator-level packet-matrix transfer from packet-basis discretizations to the theorem;
3. an explicit-rate short-time wave-family approximation theorem for MiNO-Core, plus a local-window Helmholtz corollary;
4. explicit PDE-specific explanations of how the same transplantation should be instantiated in other families.

The continuous reduction theorem is built from standard microlocal ingredients; the finite landing algebra is machine-checked; and the ℓ^2 packet-matrix transfer is the interface where the two meet. This is the level needed for the present architecture paper: it justifies the sparse canonical packet-matrix primitive, identifies the relevant error budgets, and separates short-time wave propagation from the larger calculus program around Egorov consistency, caustics, and operator microscopy.

8 Benchmark Contract

The experiments follow the same decomposition as the theorem: transport, symbol, residual, packetization, and refinement are measured separately whenever the data support it. The benchmark uses a unified scenario registry over synthetic surrogates, cached TCNO PDE families, imported mature reference rows, and optional sequence-valued rollout paths. One-step approximation rows are kept separate from nonlinear rollout diagnostics.

Model split. All theorem-facing claims attach to MiNO-Core: analytic packet tokenizer, bounded canonical stencil, local symbol branch, transported synthesis, transported high-frequency carrier, and explicit residual. MiNO-Plus keeps the same packet backbone and adds confidence-gated transport and a low-pass post-synthesis local refinement head. Reported gains from MiNO-Plus are empirical enlargements of the theorem-facing Core, while the carrier-bound mechanism test still uses the same transported packet primitive.

Staged runs. Stage 0 is a two-epoch smoke check. Stage 1 is a one-seed breadth pass over the synthetic and cached PDE slate. Stage 2 repeats the most important wave, diffusion, and Helmholtz rows over three seeds. Stage 3 is the higher-budget oscillatory stress path. Runs are written as raw JSON plus aggregate CSV rows, and the empirical-closure runner uses stable row identifiers so interrupted campaigns resume with `--skip-existing`.

Objectives. The breadth and shortlist matrices use field loss,

$$\mathcal{L}_{\text{field}} = \|u_{\text{pred}} - u_{\text{target}}\|_2^2.$$

The theorem-aligned mechanism tests add packet proxies,

$$\mathcal{L}_{\text{field}} + \lambda_{\text{tr}}\mathcal{L}_{\text{tr}} + \lambda_{\text{sym}}\mathcal{L}_{\text{sym}} + \lambda_{\text{can}}\mathcal{L}_{\text{can}} + \lambda_{\text{ord}}\mathcal{L}_{\text{ord}} + \lambda_{\text{pkt}}\mathcal{L}_{\text{pkt}} + \lambda_{\text{hf}}\mathcal{L}_{\text{hf}} + \lambda_{\text{flow}}\mathcal{L}_{\text{flow}},$$

where the terms measure transported packet geometry, packet-amplitude mismatch, canonical consistency, symbol-order behavior, direct packet-space reconstruction, high-frequency core reconstruction, and controlled-flow metadata displacement when a known packet drift is available. The carrier-bound profile trains the core first, freezes local refinement during warmup, releases the refinement head only as a low-pass residual, and routes high-frequency reconstruction through transported packet metadata.

The executed carrier-bound weights are fixed in the runner rather than tuned per scenario:

$$\begin{aligned}
\lambda_{\text{tr}} &= 0.20, \quad \lambda_{\text{sym}} = 0.05, \quad \lambda_{\text{can}} = 0.02, \\
\lambda_{\text{ord}} &= 0.01, \quad \lambda_{\text{pkt}} = 0.05, \quad \lambda_{\text{hf}} = 1.00, \\
\lambda_{\text{flow}} &= 2.00.
\end{aligned}$$

The mechanism profile uses `hamiltonian_verlet` transport, `spectral_order` symbol modulation, Gaussian/Gabor packets by default, sparse top-K aggregation after neighbor selection, transported synthesis scale one, transported input-carrier scale one, token-level refinement scale zero, and skip/refinement low-pass cutoff 0.16.

In the theorem-facing `spectral_order` mode, the executable symbol action is multiplier-only: additive local shifts are disabled and belong instead to the residual or finite-landing terms. The symbol-order proxy used in $\mathcal{L}_{\text{ord}}$ is computed from the symbol output without detaching the packet values, so it is a training penalty as well as a logged diagnostic. The generic `mlp` symbol branch may still use an affine scale–shift action in legacy empirical rows, but those shifts are not used to justify the finite S^m-proxy statement.

Learning certificate. The approximation theorem is an existence statement over the MiNO hypothesis class; it does not claim that stochastic gradient descent reaches the global optimum. The connection to supervised training is the standard excess-risk certificate below. Let $\mathcal{D}$ be a distribution over input–target pairs (u, Tu) and let

$$\mathcal{R}_{\text{field}}(\theta) := \mathbb{E}_{(u,Tu)\sim\mathcal{D}}\|M_\theta(u) - Tu\|_{L^2}^2.$$

Let the theorem-aligned training objective be

$$\mathcal{J}(\theta) := \mathcal{R}_{\text{field}}(\theta) + \mathcal{P}_{\text{proxy}}(\theta), \qquad \mathcal{P}_{\text{proxy}}(\theta) \geq 0,$$

where $\mathcal{P}_{\text{proxy}}$ collects the nonnegative transport, symbol, canonical, packet, high-frequency, flow, and optional Egorov penalties used by the selected profile.

Proposition 44 (Realizability-to-learning certificate) *Assume the hypotheses of Theorem 36 on the support of $\mathcal{D}$. Suppose there exists a theorem-grade parameter $\theta^\star$ such that*

$$\|M_{\theta^\star}(u) - Tu\|_{L^2} \le e_{h,K,P}(u) \quad \textit{for } \mathcal{D}\textit{-almost every } u,$$

and

$$\mathcal{P}_{\text{proxy}}(\theta^\star) \le \Pi_{\text{proxy}}.$$

Let $\widehat{\mathcal{J}}_n$ be the empirical version of $\mathcal{J}$ on n training pairs. If the trained parameter $\hat{\theta}$ satisfies the empirical optimization gap

$$\widehat{\mathcal{J}}_n(\hat{\theta}) \le \inf_{\theta \in \mathcal{H}_{\text{MINO}}} \widehat{\mathcal{J}}_n(\theta) + \eta_{\text{opt}},$$

and the profile class has uniform generalization gap

$$\sup_{\theta \in \mathcal{H}_{\text{MINO}}} \left| \mathcal{J}(\theta) - \widehat{\mathcal{J}}_n(\theta) \right| \le \eta_{\text{gen}},$$

then

$$\mathcal{R}_{\text{field}}(\hat{\theta}) \le \mathbb{E}_{\mathcal{D}}\, e_{h,K,P}(u)^2 + \Pi_{\text{proxy}} + \eta_{\text{opt}} + 2\eta_{\text{gen}}.$$

Proof Since $\mathcal{P}_{\text{proxy}} \ge 0$, $\mathcal{R}_{\text{field}}(\hat{\theta}) \le \mathcal{J}(\hat{\theta})$. Uniform generalization gives

$$\mathcal{J}(\hat{\theta}) \le \widehat{\mathcal{J}}_n(\hat{\theta}) + \eta_{\text{gen}} \le \widehat{\mathcal{J}}_n(\theta^\star) + \eta_{\text{opt}} + \eta_{\text{gen}} \le \mathcal{J}(\theta^\star) + \eta_{\text{opt}} + 2\eta_{\text{gen}}.$$

The realizability hypothesis and proxy bound give

$$\mathcal{J}(\theta^\star) \le \mathbb{E}_{\mathcal{D}} e_{h,K,P}(u)^2 + \Pi_{\text{proxy}},$$

which proves the claim. ■

Proposition 44 is intentionally modest. It says that the microlocal approximation theorem becomes a supervised learning guarantee once the optimizer, sample size, and proxy weights deliver small η_{opt}, η_{gen}, and Π_{proxy}. The experiments therefore test not only expressivity but also whether the registered training protocol makes the theorem-grade mechanism reachable in finite budget.

Mechanism-identifiability protocol. The completed `branch_id_v3` campaign is the main branch attribution test. It uses `hamiltonian_verlet` metadata transport, `gabor_gaussian` tokenization, sparse top-K aggregation after dense neighbor search, proxy losses, core-first warmup, low-pass residual refinement, transported synthesis, transported high-frequency input carrier, and same-refinement FNO/UNO/WNO controls. Its ablations include full, no transport, identity transport, randomized metadata, no symbol, oracle transport where a known flow is available, and same-refinement baselines.

The ablation semantics are fixed in the runner. `no_transport` replaces the propagation block by a zero message and leaves metadata at the analysis centers, so the carrier path is still present but cannot move high-frequency content. `identity_transport` keeps the propagation/message block active while setting the learned displacement scale to zero.

`randomized_metadata` permutes the packet metadata across tokens before propagation, preserving the metadata multiset while breaking coefficient–metadata association. Same-refinement baselines attach the identical low-pass image-space refinement head, cutoff, route initialization, and training loss budget to the baseline prediction; they do not receive MiNO's packet tokenizer, transported synthesis, transported high-frequency carrier, or token-level packet refinement.

The interpretation rule is fixed: canonical transport is empirically credited only when transport corruption or metadata randomization degrades field or packet diagnostics, and when the same local refinement head does not by itself explain the microlocal branch effect. The carrier-bound rows pass this rule on controlled wave probes: removing transport and randomizing metadata consistently worsens field error, while the same-refinement rows remain the absolute-accuracy controls.

The controlled-flow rows also use the known packet drift through $\mathcal{L}_{\text{flow}}$. This theorem-aligned supervision term tests whether a carrier-bound architecture can use a canonical metadata channel when the target flow is available. The executable control campaign adds two fairness gates: a flow-off row, which removes $\mathcal{L}_{\text{flow}}$ at fixed architecture, and a carrier-off transport row, which removes both transport and transported carrier paths. These rows separate supervision dependence, carrier placement, and transport learning.

Egorov pilot diagnostics. Egorov consistency is evaluated as a diagnostic rather than as a default training loss. Choose a fixed probe family $\mathcal{A}_{\text{probe}} = \{A^{\text{pkt}}_{j,h}\}_{j=1}^{J}$ of order-zero localization multipliers on the packet window. For theorem-grade linear Core rows we evaluate $\rho^{\text{pkt}}_{\text{Eg},h}(j)$ from Section 6; for nonlinear MiNO-Plus rows we evaluate the Jacobian version $\rho^{\text{Jac}}_{\text{Eg},h}(u,j)$. The executable runner also exposes two cheaper controlled-flow probes: a self-intertwining probe comparing $A_{\text{out}}M(u)$ with $M(A_{\text{in}}u)$, and a target-calibrated probe comparing $M(A_{\text{in}}u)$ with $A_{\text{out}}v$ for the supervised target v. The target-calibrated probe is the identifying one: it can detect a model that is self-consistent but transports to the wrong location. If used as a training term, the full residual enters a theorem-aligned optional profile,

$$\mathcal{L}_{\text{field}} + \lambda_{\text{Eg}} \frac{1}{J} \sum_{j=1}^{J} \rho^{\text{Jac}}_{\text{Eg},h}(u,j)^2,$$

with `egorov_loss_weight` reported separately. The relevant ablations are `egorov_on`, `egorov_off`, `no_transport`, and `no_symbol`. The diagnostic target is separation in the calculus metric: transport corruption should increase the pullback part of the residual, while symbol removal should increase the left/right symbol-coupling part.

As a minimal sanity check, we ran the finite-difference proxy on the bicharacteristic control row with three seeds, six epochs, and sample caps of 8/4/4. The self-Jacobian proxy is not transport-identifying: it is 0.075 for `full`, 0.067 for `no_symbol`, and 0.036 for `no_transport`, showing that a model can remain self-consistent under an incorrect transport. The target-calibrated finite-difference Jacobian proxy is more informative: it is 0.402 for `full`, 0.437 for `no_symbol`, and 0.848 for `no_transport`, matching the field-error ordering 0.426, 0.465, and 0.879. The target-calibrated proxy is therefore a useful calculus diagnostic; the main empirical verdict remains based on packet propagation and carrier-bound branch identifiability.

Table 3: Executed regime matrix for the MINO artifact.

Regime	Scenarios	Status	Reading
Smooth elliptic/parabolic	Poisson, diffusion, Darcy rows	smoke/breadth plus diffusion shortlist	strong on diffusion; mixed on Poisson/Darcy
Oscillatory Helmholtz	cached Helmholtz families	breadth plus selected shortlist	MINO-Plus improves over Core; references still lead hard rows
Short-time wave	`wave_synth`	breadth, shortlist, local wrapper comparison	positive MINO-Plus signal
Carrier-bound mechanism	five controlled wave rows	completed `branch_id_v3` + controls	transport ablations are field-causal
Rollout stress	Navier–Stokes rollout rows	executable path	outside the one-step theorem claim

Cross-resolution protocol. The repository also exposes a `cross_resolution_wave` grid,

$$64 \to 64, \qquad 64 \to 128, \qquad 96 \to 192, \qquad 128 \to 256,$$

with fixed/scaled packet budgets and fixed/scaled stencil budgets. It is retained as the executable counterpart of Theorem 11, but it is not used as completed evidence in the current empirical verdict.

9 Empirical Evidence

The empirical section evaluates the carrier-bound MINO architecture defined in Sections 4 and 7. The mechanism question is precise: if high-frequency packet content is forced through transported packet metadata, do transport corruption and metadata randomization degrade the learned operator? On controlled wave probes the answer is yes. The section therefore treats branch identifiability as the primary evidence, with absolute field error reported as context.

9.1 Regime Matrix

9.2 Context Shortlist

The stage-2 shortlist repeats the broad one-step rows over three seeds. Representative relative ℓ^2 results are:

- `wave_synth`: MINO-Plus 0.157 ± 0.003 versus MINO-Core 0.561 ± 0.023;
- `diffusion_neumann_control`: MINO-Core 0.00423 ± 0.00008 and MINO-Plus 0.00436 ± 0.00008, compared with WNO-style 0.00481 ± 0.00067;

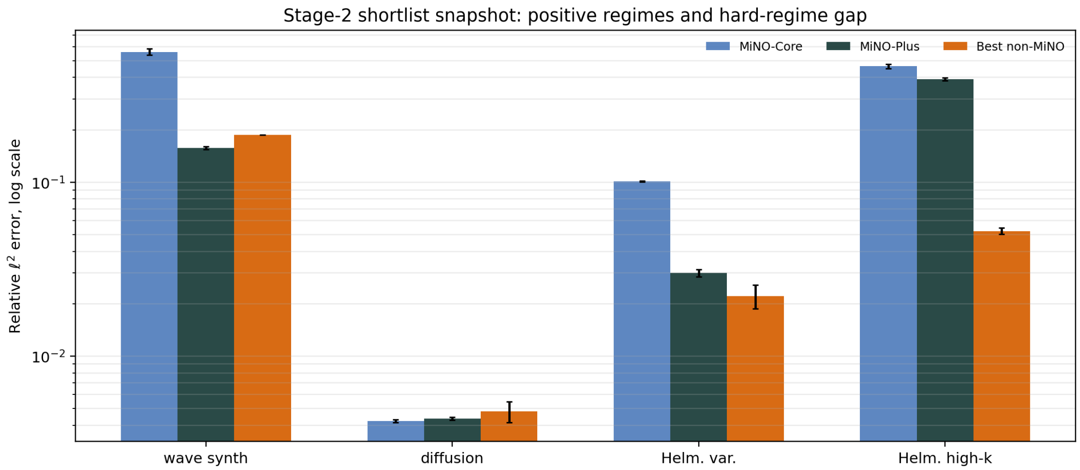

Figure 5: Stage-2 shortlist snapshot. The plot puts the positive rows and the high-k Helmholtz motivation on the same log-scale axis. MiNO-Plus improves over MiNO-Core on the wave and Helmholtz rows and is competitive on diffusion; the high-k row motivates the branch/resolvent extension rather than serving as the main mechanism claim.

- `helmholtz_dirichlet_control`: MiNO-Plus 0.0626 ± 0.0032 versus MiNO-Core 0.153 ± 0.0004, with the strongest imported reference at about 0.019;
- `helmholtz_variable_control`: MiNO-Plus 0.0300 ± 0.0015 versus MiNO-Core 0.101 ± 0.001, close to imported references at about 0.022–0.024;
- `helmholtz_highk_positive`: MiNO-Plus 0.391 ± 0.009 versus MiNO-Core 0.464 ± 0.013, behind UNO at about 0.052.

These rows establish that the packetized model is useful beyond a toy setting. They also motivate the stronger high-k Helmholtz layer developed in Section 7: local outgoing resolvents require branch structure, anisotropic packet resolution, and a near-characteristic radiation symbol beyond the base wave-facing profile.

9.3 Branched High-k Helmholtz Probe

The high-k Helmholtz row is the regime in which a single canonical graph is least plausible. We therefore add a targeted finite-branch probe for Helmholtz. The model uses $L = 3$ independently parameterized canonical branches, token-wise metadata-softmax routing, branch entropy/diversity regularization, branch-summed transported synthesis and carrier paths, and the same admissible learned finite-landing layer used in the carrier-bound wave controls. The default MiNO model remains the $L = 1$ carrier-bound architecture; the branched profile is used for the `helmholtz_branched_highk` campaign.

The executed rows are sample-capped and answer a narrow question: whether branch multiplicity alone is the active missing budget before adding the stronger outgoing resolvent-symbol layer.

The ablations are designed to separate finite-branch structure from ordinary transport use. `single_branch` forces the router onto branch 1 at fixed architecture, `no_branch_routing` replaces the learned router by uniform routing, `no_transport` removes branch transport, `no_transport_no_carrier` additionally removes transported synthesis/carrier/landing, and `no_symbol` removes symbol modulation. The criterion is conservative: a useful branch result should reduce the high-k gap relative to the single-branch row, and otherwise the row is treated as a guide for the next microlocal budget rather than as a central claim.

The sample-capped branch-only result serves as a budget-local branch probe. Across the three Helmholtz rows, the full $L = 3$ branched model is close to the forced single-branch model, the uniform-routing ablation, and the no-symbol ablation. This points to a sharper implementation requirement already predicted by the calculus ledger: branch multiplicity alone is too weak unless the frame, router, phase/landing, and outgoing resolvent-symbol budgets are all controlled. The full table, routing plot, and field panels are kept in Appendix D; the main text uses the row to motivate the `helmholtz_resolvent` extension.

The follow-up `helmholtz_highk_careful` profile is therefore theory-aligned differently: it keeps the anisotropic packet frame and carrier-bound landing path, but replaces the generic `spectral_order` multiplier by the executable `helmholtz_resolvent` symbol layer from Corollary 35. The layer uses an auto-calibrated shell and finite complex-pair multiplication for the signed outgoing resolvent factor, so the comparison is no longer merely an envelope-only symbol ablation. Table 4 reports the completed local mid-scale row and two matched profile checks. Increasing the finite packet/resolvent capacity from the minimal smoke profile reduces the high-k relative ℓ^2 error from roughly 0.71 to 0.50. The ablations also delimit the mechanism: forcing a single branch leaves the result unchanged, and removing the transport message alone is nearly flat, while removing both transport and the transported high-frequency carrier degrades strongly. The `no_symbol` row is close in relative ℓ^2 and can be slightly better at some finite budgets, while its phase-sensitive diagnostics are worse; the matched `spectral_order` row has essentially the same relative ℓ^2 but worse phase than `helmholtz_resolvent`. The safe reading is therefore that the outgoing symbol layer is active and phase-sensitive, but is not yet a standalone relative-ℓ^2 improvement. The runner now separates this budget into `source_only_symbol`, `no_edge_symbol`, and `no_resolvent_phase` rows, with a small symbol-identity regularizer available when the local-kernel correction should remain a diagnostic rather than a default field-error path. The matched `gabor_gaussian` row has lower phase error but substantially worse relative ℓ^2, so the high-k field-error gain is attributed primarily to anisotropic packet resolution plus carrier-bound landing rather than to the isotropic Gabor frame. Thus the current high-k evidence supports anisotropic carrier-bound reconstruction more strongly than multibranch canonical transport, with the resolvent/local-kernel symbol remaining a phase/radiation budget under refinement.

Flagship high-frequency contract. The mid-scale row is not the full high-k claim. The registered flagship protocol is stricter: train on moderate frequencies, test on larger

Table 4: Mid-scale `helmholtz_highk_careful` probe on `helmholtz_highk_positive`. Entries are three-seed means with standard deviations. The ablation block uses anisotropic packets, the `helmholtz_resolvent` complex-pair symbol layer, two canonical branches, width 48, six modes, a transported landing decoder with eight channels, and 16 training epochs. The matched-profile block changes one calculus hook at a time. Lower is better.

Ablation	Rel. ℓ^2	Phase error	Imaginary resolvent energy
`full`	0.503 ± 0.008	1.511 ± 0.028	0.0349
`single_branch`	$\mathbf{0.502} \pm 0.005$	1.512 ± 0.008	0.0322
`no_transport`	0.506 ± 0.011	$\mathbf{1.509} \pm 0.006$	0.0229
`no_transport_no_carrier`	0.842 ± 0.002	1.499 ± 0.015	0.0424
`no_symbol`	0.509 ± 0.012	1.602 ± 0.022	0.0000

Matched full-model profile	Rel. ℓ^2	Phase error
`anisotropic_gabor` + `helmholtz_resolvent`	0.503 ± 0.008	1.511 ± 0.028
`anisotropic_gabor` + `spectral_order`	$\mathbf{0.502} \pm 0.007$	1.609 ± 0.037
`gabor_gaussian` + `helmholtz_resolvent`	0.584 ± 0.017	$\mathbf{1.486} \pm 0.014$

k, report nondimensional frequency kL, wavelengths across the domain, points per wavelength, reference-solver residual, complex relative ℓ^2 when real/imaginary channels are present, amplitude and phase error, PDE residual, boundary/radiation residual, receiver or far-field trace error when available, runtime, memory, and error degradation as k increases. The runner now logs three finite proxies for this contract in addition to relative ℓ^2: `test_complex_relative_l2_proxy`, which uses genuine real/imaginary channel pairs when the dataset supplies them; `test_amplitude_error_proxy`, which measures envelope mismatch; and `test_boundary_trace_error_proxy`, which records a receiver-style boundary trace mismatch. The baseline set is also tiered. General neural operators (FNO, UNO/U-FNO, WNO, CNO/CANO-style attention when available) test whether the packet matrix improves over generic operator primitives. Helmholtz-specialized solvers, including neural multigrid/Fourier and probabilistic high-frequency Helmholtz references, are required before claiming competitiveness in the high-k solver literature. The runner exposes this as `helmholtz_highk_flagship`; the campaign uses anisotropic packets, carrier-bound finite landing, `helmholtz_resolvent`, OOD high-k rows, and explicit complex-pair losses only when the dataset provides real/imaginary channels. This protocol is the path from the current mechanism evidence to a high-k flagship benchmark.

9.4 Carrier-Bound Branch Identifiability

The final MiNO mechanism test is the carrier-bound `branch_id_v3` profile. It keeps packet metadata batch-specific, uses Hamiltonian Störmer–Verlet metadata transport, synthesizes learned coefficients at transported packet centers, and adds a transported high-frequency input carrier from the original packet coefficients. Skip and refinement paths are low-pass restricted, token-level refinement is disabled, and controlled-flow rows use the known packet drift as a metadata-flow target. This makes transport a field-level mechanism: disabling

Table 5: Carrier-bound `branch_id_v3` mechanism test. Entries are mean relative ℓ^2 over three seeds; lower is better. Scenario aliases match the five controlled wave rows listed in the text.

Scenario	MINO	No transport	Random metadata	No symbol	Best same-refine
bichar.	0.380	0.714	0.912	0.395	0.056 (UNO)
chirp	0.316	0.839	1.048	0.335	0.070 (UNO)
two-packet	0.204	0.777	1.037	0.236	0.022 (UNO)
var-speed	0.239	0.456	0.890	0.240	0.074 (UNO)
wave synth	0.295	0.537	0.900	0.298	0.154 (UNO)

Table 6: Registered `branch_id_v3_controls` field-error checks. Entries are mean relative ℓ^2 over three seeds; lower is better.

Scenario	Full	No flow	No tr.	Identity	No tr./car.	Rand. meta	No sym.
bichar.	0.378	0.379	0.715	0.714	0.815	0.912	0.392
chirp	0.315	0.319	0.840	0.806	0.826	1.044	0.334
two-packet	0.205	0.185	0.779	0.749	0.801	1.038	0.234
var-speed	0.238	0.238	0.455	0.456	0.753	0.892	0.240
wave synth	0.296	0.296	0.534	0.535	0.774	0.905	0.298

transport leaves the high-frequency carrier at the analysis centers rather than the learned transported centers.

For fairness, the ablations are defined at the executable level. `no_transport` zeroes the propagation message and keeps identity metadata; it does not randomize packet centers. `identity_transport` keeps the propagation/message path but freezes displacement. `randomized_metadata` permutes metadata across packets, preserving the metadata multiset while breaking the packet–metadata association. The same-refinement baseline column uses the same low-pass image-space refinement mechanism and training loss budget as MINO-Plus, but without MiNO's packet tokenizer, transported synthesis, transported carrier, or token-level packet refinement. The registered `branch_id_v3_controls` campaign adds two fairness controls: `full_no_flow_supervision`, which removes metadata-flow supervision at fixed architecture, and `no_transport_no_carrier`, which removes both transport and the transported high-frequency carrier. These controls separate flow-supervision dependence, carrier binding, and learned transport corruption.

The controls table below is generated from the completed `branch_id_v3_controls` summary. It is the fairness counterpart to Table 5: `full_no_flow_supervision` tests whether the effect is merely flow-supervision driven, and `no_transport_no_carrier` tests whether the degradation is explained by the transported carrier path alone. Across all five rows, removing flow supervision changes relative ℓ^2 by at most 0.020, while removing transport increases relative ℓ^2 by 0.217–0.574. Removing both transport and the high-frequency carrier remains worse than the full model in every row, and is worse than `no_transport` in four of the five rows.

The ablation pattern is coherent. Removing transport increases field error in every row; identity transport behaves like the transport-off control; randomizing metadata increases the error further; removing only the symbol branch has a smaller effect on these transport-

Table 7: Theorem-to-metric map for the carrier-bound mechanism evidence. The table records what each theorem budget predicts experimentally and which registered row probes it.

Theorem term	Artifact metric	Registered probe	Status
Transport budget E_{tr}	relative ℓ^2, canonical-flow proxy, WF-transport proxy	`no_transport`, `identity_transport`, `randomized_metadata`	tested
Transported synthesis E_{tsyn}	field error under transported synthesis corruption	`identity_transport` vs full	tested
Carrier channel E_{car}	extra degradation after removing transported carrier	`no_transport_no_carrier` vs `no_transport`	tested
Symbol budget E_{sym}	relative ℓ^2, symbol-order proxy	`no_symbol`	tested
Packet tail E_{tail}	packet-WF localization and future K sweep	fixed-K controls; budget sweep deferred	partial
Frame/asymptotic/covering $E_{\mathrm{frame}}, E_{\mathrm{asymp}}, \delta_h$	tokenizer reconstruction, frame diagnostics, resolution sweep	cross-resolution campaign	deferred

dominant wave probes. The same-refinement baselines are included as absolute-accuracy controls. Tables 5–6 therefore establish the main empirical point: the final MiNO architecture uses packet transport as a causal field-level branch under the registered carrier-bound protocol.

This table also fixes the interpretation of the mechanism claim. Field degradation under transport corruption is a causal-use test. On controlled rows we additionally log canonical-flow and packet-wavefront proxies; budget-scaling in K, transport capacity, symbol capacity, and packet resolution is the natural quantitative follow-up.

Executable landing diagnostic. The theory idealizes a packet synthesis operator with controlled frame constants, while the executable model must land transported packet content on a finite pixel grid. We therefore ran a synthesis-mode probe on the two cleanest wavefront scenarios. Two analytic replacements for the packet-energy warp, moved Gabor-atom splatting and moved local-patch folding, were substantially worse. A transport-conditioned finite landing decoder was the only tested landing map that reduced the UNO gap in the single-seed diagnostic, reaching relative ℓ^2 0.056 on `wave_chirp_propagation` and 0.034 on `wave_two_packet_no_interaction`. The narrow theoretical reading is that the learned decoder reduces the executable landing terms E_{tsyn} and E_{car}; it does not change the canonical propagation theorem and it is not a new microlocal calculus result.

We then promoted the learned finite-landing row to a three-seed `branch_id_v3_controls` follow-up. Table 9 and Figure 7 show the outcome. The learned landing layer sharply improves absolute field error relative to the analytic finite-landing controls, but it does not erase the transport mechanism: `no_transport` remains worse than `full` in every row, and removing both transport and the high-frequency carrier remains much worse. The effect is smaller than in the non-decoder carrier-bound table, so the decoder is treated as a finite implementation layer that reduces landing error while preserving a residual transport penalty. The `no_symbol` rows remain flat or slightly better in several scenarios; these learned-landing controls therefore strengthen finite-landing evidence, not symbol-branch attribution.

Table 8: Single-seed executable landing probe. This table is diagnostic, not a replacement for the registered three-seed mechanism tables. It tests whether the remaining UNO gap is partly a finite landing problem rather than a canonical-flow problem.

Scenario	Landing map	Full	No transport	UNO+ref
chirp	warp	0.316	0.839	0.070
	atom splat	0.801	0.841	0.048
	patch fold	0.758	0.662	0.047
	learned landing	0.056	0.093	0.048
two-packet	warp	0.204	0.777	0.022
	atom splat	0.680	0.836	0.021
	patch fold	0.764	0.743	0.022
	learned landing	0.034	0.053	0.022

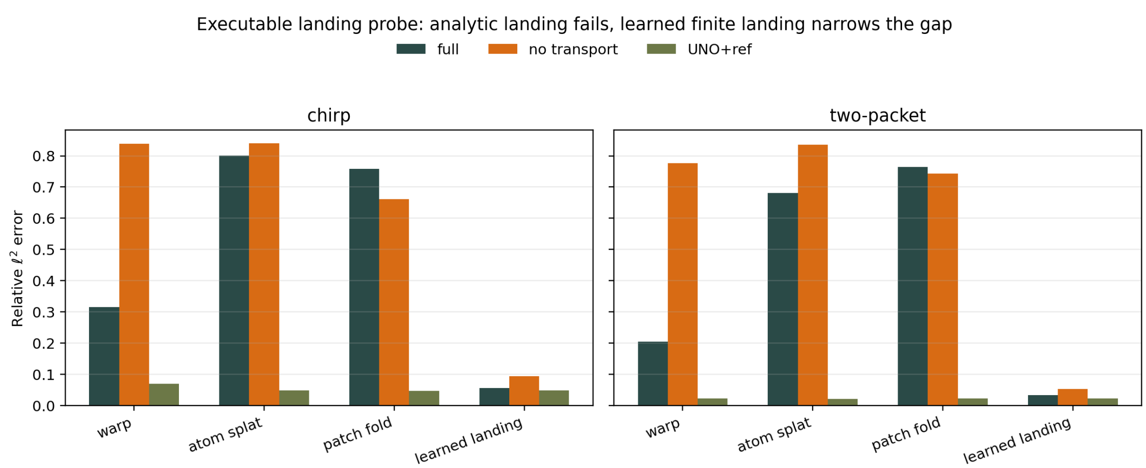

Figure 6: Single-seed executable landing probe. The instability of analytic atom-splat and moved-patch folding indicates that the UNO gap is partly an executable finite-landing problem. The learned landing decoder narrows this gap while preserving a transport ablation penalty, so it is treated as a finite landing layer for E_{tsyn} and E_{car}.

9.5 Claim Matrix

The low-budget local-window Helmholtz and synthetic Navier–Stokes panels are kept as appendix diagnostics (Figures 14–15). They delimit extension regimes and are not part of the main positive evidence.

9.6 Interpretation

The empirical evidence aligns with two mechanism statements. On controlled wave probes, the carrier-bound model makes transported metadata field-causal: transport corruption de-

Table 9: Three-seed learned finite-landing controls. Entries are mean relative ℓ^2; lower is better. The transport-conditioned decoder reduces absolute field error relative to the analytic finite-landing maps, while transport removal and carrier removal remain visible.

Scenario	Full	No flow	No tr.	No tr./car.	Rand. meta	No sym.
bichar.	0.060	0.059	0.090	0.814	0.722	0.050
chirp	0.057	0.067	0.104	0.862	0.845	0.053
two-packet	0.036	0.051	0.050	0.828	0.888	0.031
var-speed	0.072	0.086	0.144	0.754	0.752	0.101
wave synth	0.179	0.180	0.219	0.771	0.779	0.178

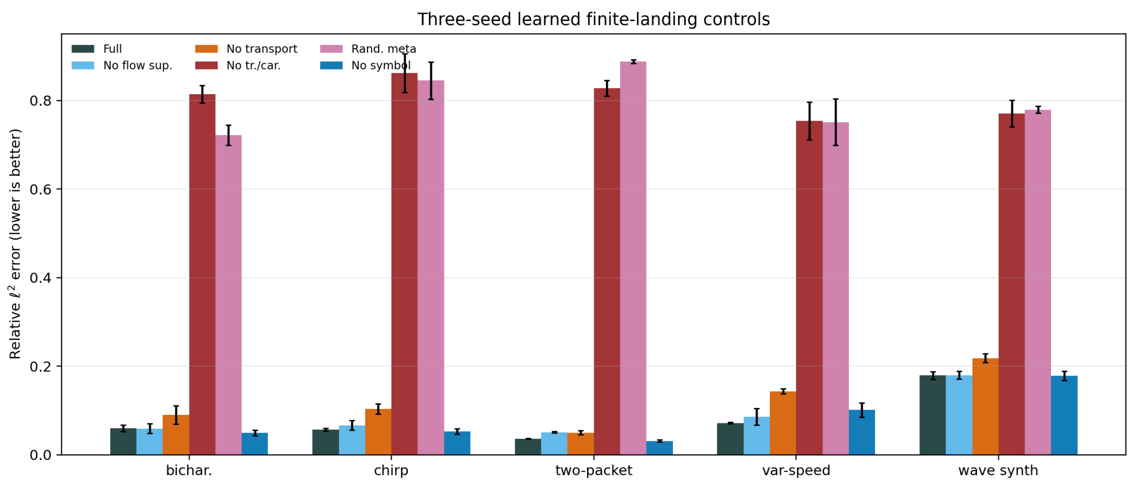

Figure 7: Three-seed learned finite-landing controls. The transport-conditioned decoder reduces absolute field error on controlled wave probes, while `no_transport`, `no_transport_no_carrier`, and metadata randomization remain degraded. The result supports a finite landing correction that preserves transport dependence.

grades field error after low-frequency residual paths are restricted. On the high-k Helmholtz row, the active mechanism is different: the robust field-error signal is carried by anisotropic packet resolution, the transported high-frequency carrier, and finite landing. The high-k mid-scale ablation does not identify multibranch canonical transport, since `single_branch` and `no_transport` are essentially flat against `full`. The matched profile checks separate the calculus terms: replacing the resolvent/local-kernel symbol changes phase-sensitive behavior more than relative ℓ^2, while replacing anisotropic packets by isotropic Gabor packets worsens field error.

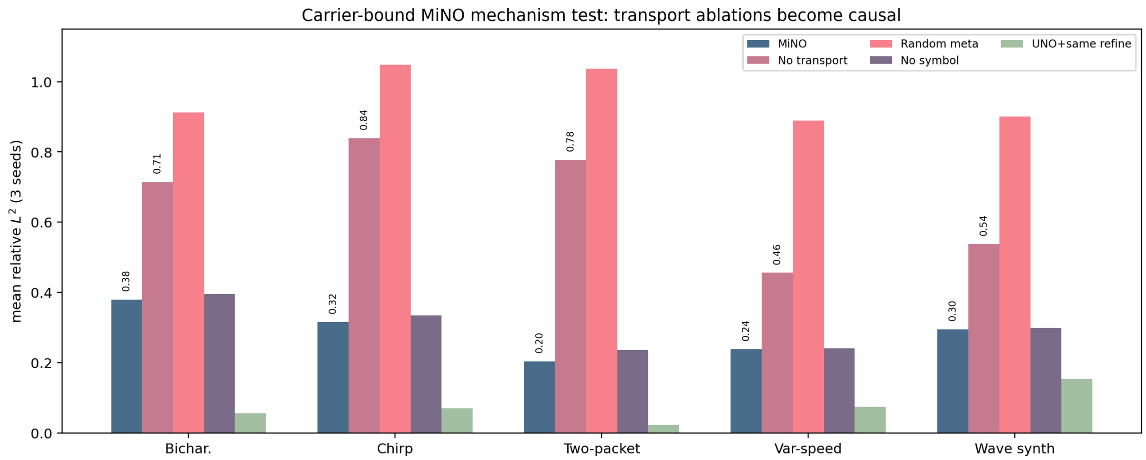

Figure 8: Carrier-bound branch-identifiability results. Transport removal and metadata randomization produce a consistent degradation pattern across controlled wave probes, while same-refinement baselines remain a separate absolute-accuracy control.

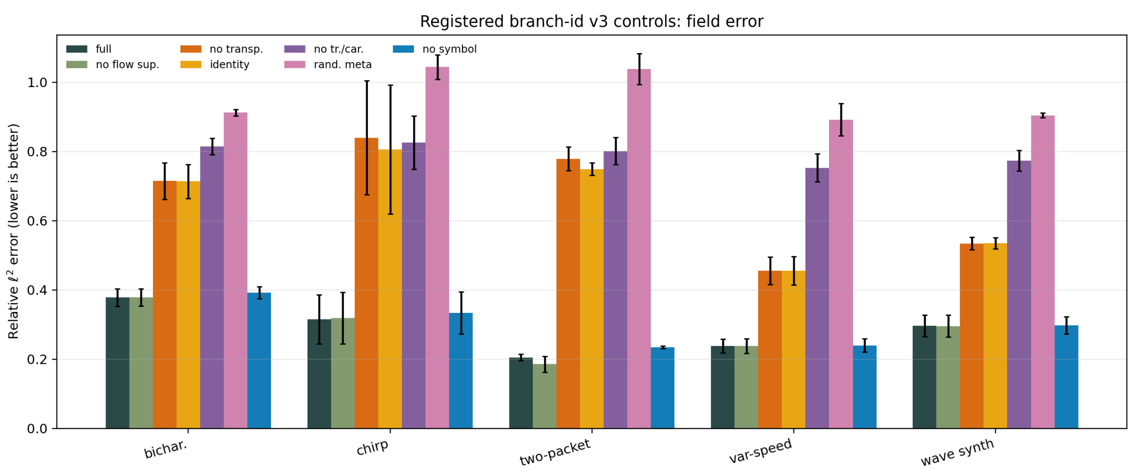

Figure 9: Registered `branch_id_v3_controls` field-error checks. The key comparisons are `full_no_flow_supervision` versus `full`, and `no_transport_no_carrier` versus `no_transport`. Flow supervision is not needed for the reported field-error effect, while carrier removal separates the high-frequency carrier contribution from the transport-corruption contribution.

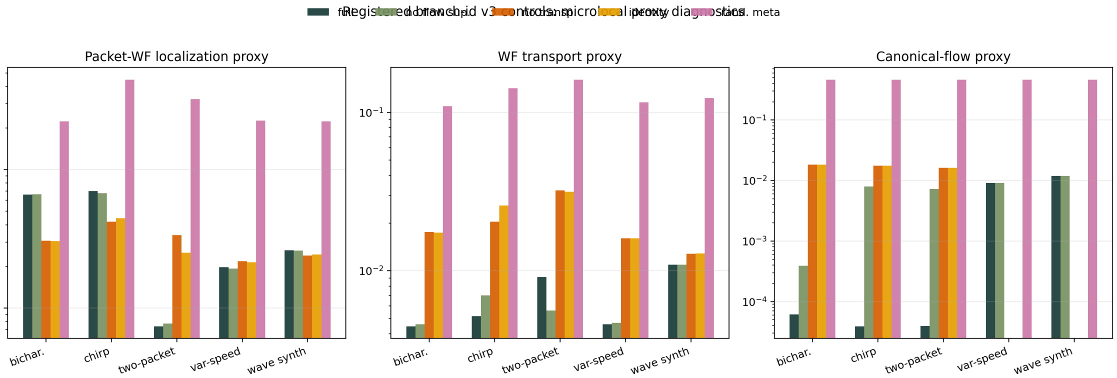

Figure 10: Registered **branch_id_v3_controls** microlocal-proxy checks. The plotted diagnostics correspond to Corollary 26: packet-wavefront localization error, wavefront-transport proxy error, and canonical-flow proxy error. These are mechanism diagnostics rather than aggregate SOTA metrics.

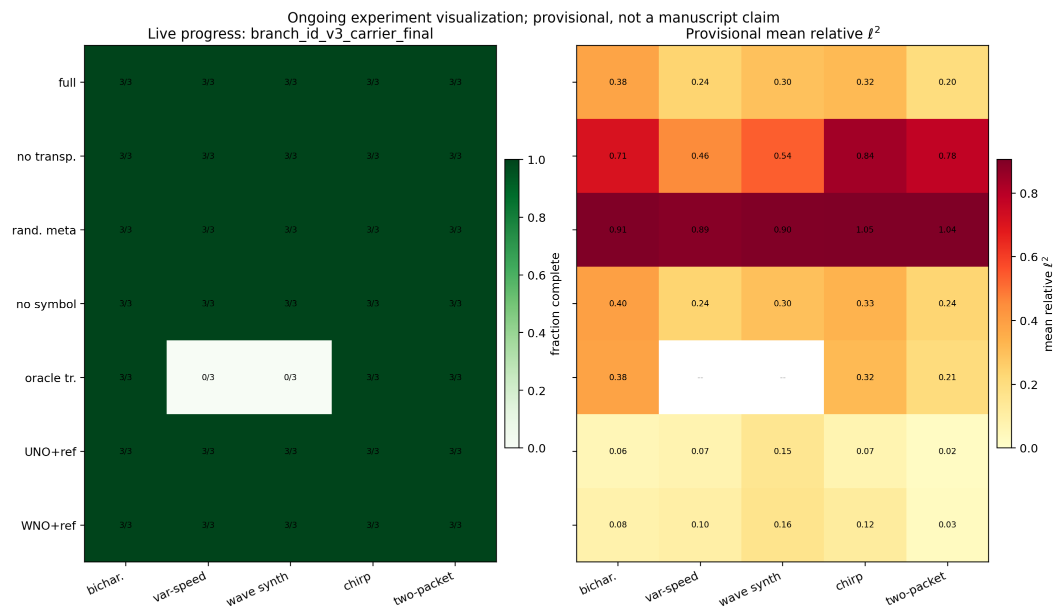

Figure 11: Completed **branch_id_v3** campaign matrix generated by the resumable runner. Each cell records the completed seed count and the mean relative ℓ^2 used to build Table 5.

10 Artifact and Discussion

10.1 Artifact

The executable artifact separates the theorem-facing model, empirical enlargement, benchmark runner, imported references, and Lean finite bridge: the core paths are

Table 10: Claim-to-regime matrix for the carrier-bound MINO manuscript.

Claim	Evidence	Verdict
MINO-Core is a theorem-facing packet architecture	ℓ^2 packet-matrix theorem plus finite Lean landing layer and executable Core model	supported
Carrier-bound MINO makes transport field-causal	branch-ID ablations and controls	supported on controlled wave probes
Learned finite landing reduces executable synthesis error	learned-landing controls	supported as finite implementation evidence; transport penalty remains
Symbol branch is the dominant wave mechanism	`no_symbol` rows	not the dominant effect on these transport-dominant probes
Identity/dissipative packet regimes are useful	diffusion rows near strongest imported references	supported as aggregate performance; symbol-branch attribution still requires calculus-regime ablations
High-k Helmholtz is a stress diagnostic for anisotropic carrier-bound reconstruction	Table 4 and Corollary 35	local mid-scale improvement; anisotropic packets and carrier-bound landing drive the field-error gain, the resolvent/local-kernel symbol is phase-sensitive but not field-error-positive at every budget, and branch multiplicity is not identified in this row
Local-window Helmholtz/Navier probes delimit extension regimes	local-window Helmholtz and synthetic Navier calculus probes	scoping diagnostics for branch/resolvent and transport–diffusion profiles

`mino/models/`, `mino/data/`, `scripts/`, `lean/MiNO/Microlocal/`, and `manuscript/jmlr/`. The manuscript-source bundle and executable artifact may be distributed separately; an artifact-inclusive bundle contains these paths together with raw JSON/CSV summaries, seed manifests, and exact runner commands. The Lean component verifies the finite packet landing layer once the classical microlocal estimates have supplied a packet matrix with tails and budgets. The Python artifact writes raw JSON, aggregate CSV, and manifest files so that benchmark rows can be audited and resumed.

10.2 Closed Results

The finite landing spine is closed for the stated packet model: transported-symbol perturbation, bounded-stencil stability, shell/tube truncation, cross-resolution transfer, and the finite landing layer are checked or replayed in the Lean companion. The operator-level theorem spine is the ℓ^2 sparse packet-matrix transfer coupled to the continuous no-caustic FIO reduction: short-time half-wave propagation gives a canonical graph, non-stationary phase gives off-tube decay, and packet sparsity converts that decay into a bounded-stencil approximation. Together these pieces justify carrier-bound MiNO-Core as a theorem-guided architecture for short-time no-caustic wave propagation.

At the executable level, the theorem-facing `spectral_order` branch is aligned with the finite symbol statement: it is a pure multiplier on packet coefficients, while additive feature shifts are assigned to the empirical branch or to residual/landing terms. For high-k Helmholtz, the artifact exposes `helmholtz_resolvent`, a near-characteristic radiation proxy corresponding to Corollary 35. This branch uses an auto-calibrated shell, signed real/imaginary resolvent weights, an outgoing gate, and a bias-free latent complex-pair action for the signed multiplier. Repository tests check zero-input equivariance of the theorem-facing symbol output, exact finite real multiplication by complex pairs, and differentiability of the symbol-order proxy with respect to packet values.

10.3 Mechanism Evidence

The carrier-bound `branch_id_v3` and `branch_id_v3_controls` campaigns supply the central mechanism evidence. They keep final metadata batch-specific, apply transported synthesis by a packet-energy-weighted warp, transport a high-frequency input carrier from the original packet coefficients, low-pass restrict skip and refinement paths, and disable token-level refinement in the binding profile. Removing metadata-flow supervision leaves field error essentially unchanged; removing transport consistently worsens field error; randomizing metadata worsens it further. This is the empirical alignment point between the sparse packet-matrix theorem and the carrier-bound executable model.

10.4 Evidence Boundary

The completed evidence has a clear center: the sparse canonical packet matrix is a field-level mechanism on controlled short-time wave probes. The theory identifies the packet operator class, the architecture implements a carrier-bound realization, and the benchmark corrupts the corresponding branch. The conclusion is primitive-level: when high-frequency content must be landed after nontrivial canonical motion, explicit packet transport is a directly testable representation of the law. Same-refinement UNO/WNO rows show that strong multiscale baselines can learn useful transported behavior implicitly; MiNO's contribution is to expose and test that law directly.

The high-k Helmholtz and Navier–Stokes rows serve as calculus stress paths. For Helmholtz, the local outgoing resolvent requires anisotropic packet resolution, branch separation, phase/landing accuracy, and a near-characteristic radiation symbol. The `helmholtz_branched_highk` probe shows that branch multiplicity alone is not the active budget at the current scale; the `helmholtz_highk_careful` profile points instead to anisotropic carrier-bound resolvent reconstruction. For Navier–Stokes-style dynamics, the relevant extension is an advective FIO branch coupled to a dissipative ΨDO branch and pressure/closure residuals. These rows motivate the extension ledger in Section 6.9.

High-k Helmholtz is the most direct future flagship benchmark for the microlocal primitive, because it stresses phase coherence, outgoing radiation, and frequency-scaling rather than generic smooth interpolation. A solver-level claim requires the stronger contract in Section 9.3: increasing-k rows, phase and radiation residuals, reference-solver accuracy, and comparisons against both general neural operators and specialized Helmholtz solvers.

Navier–Stokes is a secondary nonlinear stress path; its microlocal hook is transport–diffusion splitting, not a standalone turbulence theory.

Three implementation axes are separated from the main wave theorem. The separable Störmer–Verlet metadata branch gives the cleanest symplectic metadata statement; the main theorem itself only requires a learned near-canonical map with the displayed compact-window error budget. The local symbol branch is secondary on transport-dominant wave probes and is naturally tested in identity-canonical elliptic, dissipative/parabolic, and outgoing-resolvent regimes. The learned landing decoder is treated as an admissible finite implementation layer for $E_{\mathrm{land},\theta}$, E_{tsyn}, and E_{car}; transport corruption remains visible after the decoder is enabled.

The extension ledger records how additional microlocal structure enters the same packet budget. Anisotropic packets, finite multibranch canonical relations, stronger S^m symbol control, outgoing resolvent-symbol enforcement, and nonlinear transport–diffusion coupling each has a packet-calculus object and an executable hook. Once the hook has a verified finite defect term, it enters the packet-budget theorem additively.

Two foundation directions are separated from the present paper. An Egorov-style neural calculus would test whether learned operators conjugate pseudodifferential probes by the learned canonical map; the finite objects are $\rho^{\mathrm{pkt}}_{\mathrm{Eg},h}$ for a frozen linear Core and $\rho^{\mathrm{Jac}}_{\mathrm{Eg},h}(u,j)$ for the nonlinear executable model. An operator-microscopy program would recover and compile a microlocal fingerprint, in the no-caustic graph case $\mathfrak{F}(T) = (\kappa, S, a, R)$, consisting of canonical map, action/phase primitive, amplitude, and residual. Multi-chart caustics and Maslov transition data require a separate Lagrangian atlas.

10.5 Conclusion

MiNO advances a testable microlocal design principle: neural operators for oscillatory PDEs should expose phase-space packet structure rather than rely only on point-space correlation or global Fourier filtering. The mathematical contribution is a sparse canonical packet-matrix primitive with a finite machine-checkable landing layer and a short-time wave/FIO approximation theorem for carrier-bound MiNO-Core. The empirical contribution is a mechanism-identifiability result: in the carrier-bound architecture, high-frequency reconstruction is tied to transported packet metadata, and transport corruption consistently degrades controlled wave rows.

The resulting claim is specific. The theorem identifies the operator class, the architecture implements one carrier-bound realization of the transported packet primitive in field synthesis, and the benchmark contract tests whether the trained model uses that primitive. The high-frequency Helmholtz and Navier–Stokes rows define separate calculus hooks—branch/resolvent structure for outgoing oscillatory inverses and transport–diffusion splitting for nonlinear dynamics—rather than broadening the main mechanism claim.

Appendix A. Supplementary Proofs and Derivations

A.1 Proof anatomy of the finite transported-symbol perturbation bound

The proof of Theorem 8 is short enough to appear in the main text, and its structure reveals what stronger continuous theorems preserve.

Step 1: add and subtract the transported coefficient. The only useful algebraic move is to add and subtract the mixed term $s_i^\star a_{\hat{\pi}(i)}$. This is what isolates transport error from symbol error.

Step 2: factor the three mechanisms. After that addition and subtraction, the difference splits into:

1. reference symbol times transport mismatch,
2. symbol mismatch times approximate transported coefficient,
3. residual.

This is the whole reason for using a transported-symbol architecture rather than an undifferentiated interaction block.

Step 3: apply envelopes. The theorem then requires only the envelope hypotheses on transported coefficients and reference symbol size.

Step 4: sum. The ℓ^1 bound is simply the sum of the pointwise bound over all packets.

The bounded-stencil wave/FIO section preserves the same separation: even when several neighboring packets are retained, the dominant error breaks into transport, symbol, and residual components, summed over a small canonical stencil instead of a single destination.

A.2 Proof of Proposition 32

The semi-discrete transplantation principle is immediate, but we record its logic in a more explicit way.

Let K_h be a discretized PDE solver and $\mathcal{K}_h = V_h K_h U_h$ its packet-basis realization. Suppose $\mathcal{K}_h$ belongs to the transported-symbol class $\mathfrak{T}(\varepsilon_{\mathrm{tr}}, \varepsilon_{\mathrm{sym}}, \varepsilon_{\mathrm{res}})$. By definition this means there exists a reference transported-symbol action together with approximate transport, approximate symbol, and residual terms satisfying Assumption 7 for the coefficient vectors of interest. The packet-space map induced by the approximate MiNO layer therefore satisfies the exact hypotheses of Theorem 8. The coefficient-level error bound follows immediately. No PDE-specific argument is used at this stage; the PDE work is entirely in verifying membership of the packet-basis realization in the transported-symbol class.

A.3 Proof anatomy of Theorem 11

The cross-resolution transfer theorem is finite and introduces a scale-indexed structure beyond the basic propagation theorem. Its proof has two steps.

Step 1: split transferred error into model and reference parts. Write

$$P_{c\to f}\widehat{\mathcal{T}}_c R_{f\to c} a_f - \mathcal{T}_f^\star a_f = \big(P_{c\to f}\widehat{\mathcal{T}}_c R_{f\to c} a_f - P_{c\to f}\mathcal{T}_c^\star R_{f\to c} a_f\big) + \big(P_{c\to f}\mathcal{T}_c^\star R_{f\to c} a_f - \mathcal{T}_f^\star a_f\big).$$

The first bracket is the lifted coarse model error. The second bracket is the reference transfer defect.

Step 2: apply prolongation stability and the two rate assumptions. The prolongation stability constant κ turns the coarse model error into a fine-resolution bound, while the second bracket is already controlled by the packetization defect rate. Summing the two contributions yields the final estimate

$$\|P_{c\to f}\widehat{\mathcal{T}}_c R_{f\to c}a_f - \mathcal{T}_f^\star a_f\|_{\ell^1} \le \kappa C_c h^\beta + C_t h^\alpha.$$

The reason this theorem matters is not its algebraic complexity. It matters because it separates two effects that standard neural-operator reporting often conflates: coarse learned error and intrinsic packet transfer defect across resolutions.

A.4 Why the global-scalar obstruction matters

Proposition 5 is simple and isolates a useful obstruction.

Suppose a method corrects all packets using the same scalar gain. In a completely smooth and translation-invariant regime, such a simplification may be acceptable because neighboring packets can indeed share almost the same local symbol. The obstruction appears when local wavevector variation matters. Then two packets with the same incoming coefficient but different outgoing amplitude demands cannot both be matched. The proposition makes this impossibility precise in the simplest possible finite setting. It is the toy version of the microlocal claim that one should not compress a genuinely local transported-symbol law into a single global scalar.

A.5 A longer discussion of the identity-canonical reduction

Proposition 4 says that MiNO specializes to identity transport when the PDE regime has no leading nontrivial canonical motion. Many operator-learning benchmarks mix smooth and oscillatory families, and the reduction proposition explains how the architecture covers the identity-canonical pseudodifferential regime, with the usual spectral multiplier appearing as a further translation-invariant limit:

1. if the transport map is the identity,
2. if the symbol is diagonal in packet coordinates,
3. and if the packet frame degenerates to the global Fourier frame,

then the model recovers a classical spectral residual law. Thus the microlocal design refines spectral neural operators by placing them in the correct subregime. Elliptic and parabolic operators stress the identity-canonical and dissipative-symbol branches instead of the nontrivial transport branch.

A.6 Proof sketch for a weighted packet norm variant

The finite landing lemma is written in ℓ^1 because that is the exact norm formalized in Lean and because it makes the coefficient bookkeeping transparent. The main theorem in the body uses the ℓ^2 packet-matrix Schur transfer. For applications that need importance weights, the same finite landing proof has the weighted variant

$$\|a\|_{w,1} = \sum_{i=1}^{n} w_i |a_i|, \qquad w_i \geq 0.$$

The same proof gives

$$\|\widehat{\mathcal{T}}a - \mathcal{T}^\star a\|_{w,1} \leq \Big(\sum_{i=1}^{n} w_i\Big)\big(B_s \varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{sym}} B_a + \varepsilon_{\mathrm{res}}\big),$$

because each pointwise inequality is simply multiplied by w_i before summation. This weighted variant is useful when packet importance depends strongly on scale or on physically important spatial regions.

A.7 How the theorem extends to bounded canonical stencils

The exact theorem of Section 5 moves each packet to one destination. The continuous section shows that the natural refinement is a bounded canonical stencil

$$(\mathcal{T}^\star a)_i = \sum_{j \in \mathcal{N}(i)} s^\star_{ij}\, a_j.$$

The proof pattern would be the same:

1. compare the approximate stencil weights to the reference stencil weights,
2. compare the approximate neighbor set to the reference neighbor set,
3. absorb the difference between them into a residual.

The bookkeeping becomes heavier, but the conceptual error split stays intact. This matters because the revised continuous theorem now predicts exactly this bounded-K regime for short-time wave and canonical-graph FIO packet matrices.

A.8 PDE-wise derivation notes

Helmholtz. The high-frequency inverse Helmholtz map is the most demanding oscillatory test of the transported-symbol law. In a resolved packet basis, outgoing energy should concentrate near branchwise propagated phase-space locations, but the inverse operator also carries a near-characteristic resolvent symbol and radiation condition. The theorem therefore reads as a direct estimate of how much the learned packet transport, branch router, radiation-aware symbol, and residual deviate from the resolved solver's packet realization. This is why high-k Helmholtz naturally motivates the branch/resolvent extension rather than a single-graph wave-style profile.

Wave propagation. For the wave equation, transport error is physically interpretable as a wavefront drift. The theorem's first term is thus not only mathematically meaningful but diagnostically meaningful.

Darcy. For elliptic smoothing, the packet dynamics should be nearly static and the symbol should be smoothing. In this regime the theorem explains why the transport term should be small and why the low-frequency branch can dominate.

Navier–Stokes. For a one-step linearization, the packet transport term corresponds to advection while the symbol term corresponds to dissipative or pressure-related modulation. The theorem therefore matches the advection–diffusion decomposition at the semi-discrete level.

A.9 Role of the finite landing theorem in the JMLR manuscript

The finite transported-symbol theorem is the right *machine-checked landing lemma* for this paper, not the flagship operator theorem:

- it isolates the transport, symbol, and residual budgets used by the architecture;
- it is explicit enough to be machine-checked;
- it is flexible enough to support bounded-stencil and cross-resolution finite bridges;
- it is narrow enough to interface cleanly with classical continuous inputs.

The operator-level spine is instead the ℓ^2 Schur packet-matrix transfer in Theorem 6 and its continuous wave realization in Theorem 36. The finite lemma supplies the formal algebraic core that those operator-level statements use after the microlocal packet matrix has been reduced to retained stencils and explicit budgets.

Appendix B. Continuous Microlocal Derivation

This appendix records the continuous microlocal layer in more detail than the main text. The main body uses a dual spine: a short-time half-wave off-graph decay theorem and an abstract canonical-graph packet sparsity theorem. This appendix gives the lemma chain behind those statements.

For the single flagship manuscript, the FoCM-style proof map is:

$$\begin{gathered}\text{short-time parametrix} \Rightarrow \text{restricted non-stationary phase}\\ \Rightarrow \text{packet almost-diagonalization} \Rightarrow \text{canonical-tube sparsity}\\ \Rightarrow \text{transported-symbol reduction} \Rightarrow \text{network-realizable MiNO-Core theorem.}\end{gathered}$$

The finite landing layers of this map are machine-checked. The continuous arrows are classical manuscript mathematics restricted to Euclidean compact phase windows, Gaussian/Gabor packet frames, finite branch count, and a no-caustic short-time half-wave regime. This division gives continuous microlocal structure for the architecture and finite formalization for the computational landing layer.

B.1 Short-time half-wave parametrix

Let

$$p(x,\xi) = c(x)|\xi|, \qquad 0 < c_{\min} \le c(x) \le c_{\max} < \infty,$$

with $c \in C^\infty(\mathbb{R}^d)$. On a short time interval $|t| \le T$ chosen before caustic formation on the frequency band seen by the packet frame, the positive half-wave propagator $U_+(t)$ admits a standard oscillatory representation

$$U_+(t)u(x) \sim (2\pi h)^{-d} \iint e^{\frac{i}{h}(\Phi_t(x,\xi)-y\cdot\xi)} a_t(x,\xi;h)\, u(y)\, dy\, d\xi,$$

where:

1. Φ_t generates the canonical graph of the Hamiltonian flow κ_t associated with p;
2. a_t is a classical amplitude with uniform symbol bounds on compact time intervals;
3. the graph remains single-valued on the chosen time window.

This is the exact place where we restrict to short time and Euclidean local coordinates. The theorem does not try to address global caustic geometry.

B.2 Packet testing and localized phase

Let φ_η^h and φ_γ^h be analytic Gaussian/Gabor-type packets centered at $z_\eta = (x_\eta, \xi_\eta)$ and $z_\gamma = (x_\gamma, \xi_\gamma)$. Inserting the packets into the parametrix gives

$$M_{\gamma\eta}^h(t) := \langle U_+(t)\varphi_\eta^h, \varphi_\gamma^h\rangle = (2\pi h)^{-d} \iiint e^{\frac{i}{h}\Psi_{t,\gamma,\eta}(x,y,\xi)} b_{t,\gamma,\eta}(x,y,\xi;h)\, dx\, dy\, d\xi,$$

where the localized phase $\Psi_{t,\gamma,\eta}$ is the half-wave phase plus the packet phases, and $b_{t,\gamma,\eta}$ is a compactly localized amplitude coming from the packet windows and the FIO amplitude.

The continuous packet problem is therefore reduced to an oscillatory-integral question: how does the integral behave when z_γ is far from $\kappa_t(z_\eta)$?

B.3 Phase-gradient lower bound outside the canonical tube

The key microlocal lemma is a non-stationarity statement.

Lemma 45 (Phase-gradient lower bound) *Fix a tube radius parameter $\rho > 0$. There exist constants $c_\rho, C_\rho > 0$ such that if*

$$\operatorname{dist}\big(z_\gamma, \kappa_t(z_\eta)\big) \ge \rho h^{1/2},$$

then on the support of the localized amplitude one has

$$|\nabla_{x,y,\xi}\Psi_{t,\gamma,\eta}(x,y,\xi)| \ge c_\rho \operatorname{dist}\big(z_\gamma, \kappa_t(z_\eta)\big) - C_\rho h^{1/2}.$$

In particular, after enlarging ρ if necessary,

$$|\nabla_{x,y,\xi}\Psi_{t,\gamma,\eta}(x,y,\xi)| \gtrsim \operatorname{dist}\big(z_\gamma, \kappa_t(z_\eta)\big)$$

uniformly outside the canonical tube.

Proof [Proof idea] The packets localize the variables to neighborhoods of size $h^{1/2}$ around (x_γ, ξ_γ) and (x_η, ξ_η). On such neighborhoods, the stationary point equations for $\Psi_{t,\gamma,\eta}$ reduce to the canonical relation for the half-wave phase. If z_γ is outside the tube around $\kappa_t(z_\eta)$, no stationary point survives on the localized support. Smooth dependence of the parametrix on (x, ξ) and the no-caustic hypothesis then yield a quantitative lower bound. ■

Lemma 45 is the PDE-specific step that replaces the old abstract assumption. Once it is available, the remainder of the packet argument is a classical non-stationary phase calculation.

B.4 Restricted non-stationary phase bridge

For the packet theorem used here, the full continuous calculus is not needed at the point where packet almost-diagonalization begins. The exact bridge can be isolated as follows.

Proposition 46 (Restricted packet non-stationary phase theorem) *Consider a packet-localized oscillatory integral representation*

$$M^h_{\gamma\eta} = \int e^{\frac{i}{h}\Psi^h_{\gamma\eta}(z)} b^h_{\gamma\eta}(z)\, dz$$

on a compact phase-space window. After the packet rescaling $z = z^0_{\gamma\eta} + h^{1/2}\zeta$, write the integral as

$$M^h_{\gamma\eta} = \int_{\mathbb{R}^m} e^{i\Theta^h_{\gamma\eta}(\zeta)} a^h_{\gamma\eta}(\zeta)\, d\zeta,$$

where $a^h_{\gamma\eta}$ is compactly supported in a fixed compact set independent of h, γ, η. Set

$$D_{\gamma\eta} := 1 + h^{-1/2}\mathrm{dist}(z_\gamma, \kappa(z_\eta)).$$

Fix $N \geq 1$. Assume:

1. *the phase comes from a single canonical graph on the chosen short-time window;*
2. *the rescaled phase has no stationary point off the canonical tube and satisfies, for all multiindices $1 \leq |\alpha| \leq N+1$,*
$$\left|\partial^\alpha_\zeta \Theta^h_{\gamma\eta}(\zeta)\right| \leq C_\alpha D_{\gamma\eta}, \qquad \left|\nabla_\zeta \Theta^h_{\gamma\eta}(\zeta)\right| \geq c D_{\gamma\eta};$$
3. *the rescaled localized amplitude satisfies the uniform L^1 seminorm bound*
$$\sum_{|\alpha|\leq N} \|\partial^\alpha_\zeta a^h_{\gamma\eta}\|_{L^1} \leq B_N.$$

Then there is a constant C'_N depending only on N, c, the phase derivative constants C_α, the dimension, and the fixed support set such that

$$|M^h_{\gamma\eta}| \leq C'_N B_N \big(1 + h^{-1/2}\mathrm{dist}(z_\gamma, \kappa(z_\eta))\big)^{-N}.$$

Proof Let

$$v(\zeta) := \frac{\nabla_\zeta \Theta^h_{\gamma\eta}(\zeta)}{|\nabla_\zeta \Theta^h_{\gamma\eta}(\zeta)|^2}, \qquad L := \frac{1}{i} v(\zeta) \cdot \nabla_\zeta.$$

Then $Le^{i\Theta^h_{\gamma\eta}} = e^{i\Theta^h_{\gamma\eta}}$. Since the amplitude is compactly supported in a fixed compact set, repeated integration by parts gives

$$M^h_{\gamma\eta} = \int e^{i\Theta^h_{\gamma\eta}(\zeta)} (L^*)^N a^h_{\gamma\eta}(\zeta)\, d\zeta.$$

It remains to bound $(L^*)^N a^h_{\gamma\eta}$ in L^1. The assumptions imply a uniform coefficient estimate for the vector field v: for every multiindex $|\alpha| \le N$,

$$|\partial^\alpha_\zeta v(\zeta)| \le C_{\alpha,c}\, D^{-1}_{\gamma\eta}.$$

Indeed, v is a rational expression in the first derivatives of $\Theta^h_{\gamma\eta}$ with denominator $|\nabla\Theta^h_{\gamma\eta}|^2$; differentiating v produces sums of terms whose numerators contain derivatives of $\Theta^h_{\gamma\eta}$ bounded by $C_\alpha D_{\gamma\eta}$ and whose denominators contain at least one net power of $|\nabla\Theta^h_{\gamma\eta}|$. The lower bound $|\nabla\Theta^h_{\gamma\eta}| \ge cD_{\gamma\eta}$ therefore leaves one factor $D^{-1}_{\gamma\eta}$ in every coefficient derivative used below.

The formal adjoint has the form

$$L^* f = i^{-1}\big[-v \cdot \nabla f - (\div v) f\big],$$

up to the harmless sign convention for complex conjugation. Thus L^* is a first-order differential operator whose coefficients and coefficient derivatives up to order $N-1$ are $O(D^{-1}_{\gamma\eta})$. Expanding $(L^*)^N$ by induction yields

$$(L^*)^N a^h_{\gamma\eta} = \sum_{|\alpha| \le N} c_{\alpha,N,\gamma,\eta}(\zeta) \partial^\alpha_\zeta a^h_{\gamma\eta}(\zeta), \qquad \|c_{\alpha,N,\gamma,\eta}\|_{L^\infty} \le C''_N D^{-N}_{\gamma\eta}.$$

Consequently,

$$\|(L^*)^N a^h_{\gamma\eta}\|_{L^1} \le C''_N D^{-N}_{\gamma\eta} \sum_{|\alpha| \le N} \|\partial^\alpha_\zeta a^h_{\gamma\eta}\|_{L^1} \le C''_N B_N D^{-N}_{\gamma\eta}.$$

Taking absolute values in the integrated-by-parts representation proves the stated estimate. ■

B.5 Repeated integration by parts

Define the standard non-stationary phase operator

$$L := \frac{\nabla\Psi_{t,\gamma,\eta} \cdot \nabla_{x,y,\xi}}{i|\nabla\Psi_{t,\gamma,\eta}|^2}.$$

By construction,

$$Le^{\frac{i}{h}\Psi_{t,\gamma,\eta}} = h^{-1} e^{\frac{i}{h}\Psi_{t,\gamma,\eta}}.$$

Integrating by parts N times gives

$$M^h_{\gamma\eta}(t) = h^N(2\pi h)^{-d} \iiint e^{\frac{i}{h}\Psi_{t,\gamma,\eta}}(L^*)^N b_{t,\gamma,\eta}\, dx\, dy\, d\xi.$$

Lemma 45 shows that every application of L^* contributes one inverse power of the tube distance, while analyticity and bounded overlap of the packet windows keep differentiated amplitudes summable. This yields

$$|M^h_{\gamma\eta}(t)| \le C_{T,N}\Big(1 + h^{-1/2}\operatorname{dist}\big(z_\gamma, \kappa_t(z_\eta)\big)\Big)^{-N},$$

which is Theorem 15.

B.6 Shell decomposition for abstract FIO sparsity

Fix a source packet η and write

$$\mathcal{S}_\ell(\eta) = \Big\{\gamma \in \Gamma_h : 2^\ell h^{1/2} \le \operatorname{dist}\big(z_\gamma, \kappa(z_\eta)\big) < 2^{\ell+1}h^{1/2}\Big\}, \qquad \ell \ge 0.$$

Assume the canonical-graph packet matrix satisfies:

1. *off-graph decay:*
$$|F^h_{\gamma\eta}| \le C_N 2^{-N\ell} \qquad \text{for } \gamma \in \mathcal{S}_\ell(\eta);$$
2. *shell counting:*
$$\#\mathcal{S}_\ell(\eta) \le C_{\mathrm{cnt}} 2^{q\ell}.$$

Then the shell mass satisfies
$$\sum_{\gamma\in\mathcal{S}_\ell(\eta)} |F^h_{\gamma\eta}| \le C 2^{(q-N)\ell}.$$

Summing shells $\ell \ge L$ gives

$$\sum_{\ell\ge L}\sum_{\gamma\in\mathcal{S}_\ell(\eta)} |F^h_{\gamma\eta}| \le C' 2^{(q-N)L}.$$

The number of packets retained through shell $L-1$ is $O(2^{qL})$, so if K is the retained stencil size then $K \asymp 2^{qL}$ and therefore

$$2^{(q-N)L} \asymp K^{1-N/q}.$$

This is the entire proof of Theorem 17.

B.7 Restricted packet almost-diagonalization bridge

Once Proposition 46 and the shell-counting hypothesis are available, the remaining step toward a theorem-grade packet law is local. Near the canonical graph, stationary phase identifies the retained entries of the packet matrix with sampled amplitudes; away from the graph, the dyadic shell tail is summable. The combination gives a restricted packet almost-diagonalization theorem:

1. the far-graph packet mass contributes a $K^{1-N/q}$ tail;
2. the retained tube contributes a local sampled-symbol law plus frame, asymptotic, and covering-defect remainders;
3. together they imply the bounded-stencil transported-symbol decomposition of Theorem 19.

The key point is structural. The almost-diagonalization statement required by MiNO is not a vague appeal to the literature. It is the explicit sum of one restricted non-stationary-phase estimate and one local stationary-phase estimate.

B.8 Bounded-stencil transported-symbol reduction

The shell argument only tells us that far-graph entries are small. To get the actual MiNO reference law one still needs a near-graph approximation on the retained tube. This is a stationary-phase calculation rather than a non-stationary one.

Let $\Pi^{\star}_{h,K}(t,\gamma)$ denote the K source packets whose propagated centers lie nearest to z_γ. For $\eta \in \Pi^{\star}_{h,K}(t,\gamma)$, stationary phase around the localized canonical point gives

$$\langle U_+(t)\varphi^h_\eta, \varphi^h_\gamma\rangle = s^{\star}_{h,\gamma\eta}(t) + O(h^{1/2}(1+\delta_h)),$$

where $s^{\star}_{h,\gamma\eta}(t)$ is the sampled FIO amplitude. Hence

$$K_h(t) = \mathcal{T}^{\star(K)}_h(t) + R^{(K,M)}_h(t), \qquad (\mathcal{T}^{\star(K)}_h(t)a)_\gamma = \sum_{\eta\in\Pi^{\star}_{h,K}(t,\gamma)} s^{\star}_{h,\gamma\eta}(t)a_\eta,$$

with residual consisting of:

1. the discarded far-shell tail from Theorem 17;
2. the local stationary-phase remainder $O(h^M)$;
3. the packet covering mismatch $O(\delta_h)$.

This is exactly Theorem 19.

B.9 Identity-canonical pseudodifferential branch

The elliptic side of the restricted calculus uses the same packet argument with the canonical graph replaced by the identity relation. Let $A_h = \mathrm{Op}_h(a)$ with $a \in S^m_{1,0}$ on the compact phase window. Standard pseudodifferential packet almost-diagonalization gives, for every $N \geq 1$,

$$|\langle A_h\varphi^h_\eta, \varphi^h_\gamma\rangle| \leq C_N\langle\xi_\eta\rangle^m\Big(1 + h^{-1/2}\mathrm{dist}(z_\gamma, z_\eta)\Big)^{-N}.$$

Thus the relevant tube is not a transported tube around $\kappa(z_\eta)$ but the identity tube around z_η. The same row/column shell-counting proof and Schur transfer then yield

$$\|W_hA_hW_h^* - (W_hA_hW_h^*)^{(K_{\mathrm{id}})}\|_{\ell^2\to\ell^2} \leq C_\Psi K_{\mathrm{id}}^{1-N/q},$$

after absorbing the symbol-order envelope into the compact-window constant or into a weighted packet norm. This is the mathematical reason the elliptic branch of MiNO should be local-symbol dominated rather than transport dominated.

B.10 Dissipative pseudodifferential branch

For parabolic families, the microlocal law is not merely identity-canonical; it is also dissipative. In the model case where the generator has principal symbol $q(x,\xi)$ with $\operatorname{Re} q(x,\xi) \geq c|\xi|^m$ on the high-frequency window, the short-time semigroup has principal packet action

$$m_t(x,\xi) \approx \exp[-tq(x,\xi)].$$

The executable branch therefore parameterizes a multiplier of the form

$$m_\theta(x,\xi,\Delta t) = \exp[-\Delta t\ \mathrm{softplus}(q_\theta(x,\xi))],$$

which enforces $0 < m_\theta \leq 1$ before any learned residual correction. This is not a global Fourier fallback. It is the dissipative-symbol regime of the same Hörmander-style packet calculus.

B.11 Unified branch summation

The restricted calculus theorem in Section 6.8 is obtained by summing three packet laws:

1. FIO branches: off-canonical-tube decay plus shell counting;
2. ΨDO branch: off-identity-tube decay plus shell counting;
3. smoothing branch: compact-window residual controlled by smoothing order.

The finite counterpart is simple: once each branch has an ℓ^1 budget, the total packet error is bounded by their sum. This summation step is machine-checked in `UnifiedCalculus.lean`; the continuous FIO and ΨDO almost-diagonalization inputs are classical.

B.12 Lean-friendly lemma chain

The oscillatory integral proof is classical. The decomposition below gives each continuous layer a finite mirror.

Layer 1: canonical shells and tubes. `CanonicalTube.lean` defines the finite shell and tube objects and proves the basic inclusion lemmas used by any truncation argument.

Layer 2: shell counting interface. `ShellCount.lean` isolates shell-cardinality bounds from the continuous geometry that produced them. This is the place where Euclidean phase-space counting enters the finite theorem chain.

Layer 3: geometric shell tails. `PacketTail.lean` proves the finite geometric tail estimate from shell mass bounds.

Layer 4: abstract sparsity bridge. `AbstractSparsity.lean` packages shell counting plus entry decay into a canonical-tube truncation theorem.

Layer 5: restricted bridge wrapper. `RestrictedBridge.lean` packages the restricted off-graph tail with the finite transfer wrapper. This is the Lean-facing mirror of the restricted non-stationary-phase / almost-diagonalization bridge used in the manuscript.

Layer 6: wave-to-transfer wrapper. `WaveTransfer.lean` turns the truncation error into the same language as the explicit cross-resolution theorem.

The wave/FIO analysis is classical. Once it is reduced to shell counts and off-graph decay, the theorem prover handles the finite layer.

B.13 Classical continuous inputs

The classical inputs are:

1. The short-time half-wave parametrix and the identification of the canonical graph are classical.
2. The phase-gradient lower bound of Lemma 45 is classical.
3. The restricted repeated-integration-by-parts argument is proved at manuscript level in Proposition 46.
4. The near-graph stationary-phase identification of the retained amplitude samples is classical.

These inputs feed the finite MiNO theorem spine through explicit shell/tube estimates.

B.14 Scope of the continuous theorem

The theorem chain has the following scope.

Short time only. The half-wave theorem is stated only before caustics and branch crossing. Long-time propagation will generally require multibranch packet laws and more delicate geometry.

Euclidean local charts. The current proof uses $\mathbb{R}^d$ packet frames. Manifold generality would require additional chart and partition-of-unity bookkeeping.

Canonical graph regime. The abstract sparsity theorem assumes a single canonical graph. Multigraph or reflected propagators can still fit the framework, but only after splitting into branches.

Bounded-K rather than dense transport. The whole point of the dual spine is to show that bounded stencils are natural. If the true packet matrix is structurally dense, then the transported-symbol thesis is simply the wrong primitive for that operator family.

B.15 FoCM-strength extensions

The extension path is:

1. enlarge the finite theorem core from $K = 1$ to general bounded-K stencils;
2. formalize the shell-counting law for a concrete Euclidean packet frame;
3. replace the restricted bridge used here by a genuinely continuous Lean proof of Proposition 46;

4. extend the half-wave theorem to multibranch or reflected settings.

These steps target the remaining gap between the restricted manuscript calculus and a fully formal continuous development.

Appendix C. Extended Theorem Work for the Packet-Matrix Primitive

This appendix records the strongest theorem package that the present MINO-Direct manuscript can support without turning into the broader operator-microscopy program. The statements are deliberately phrased as packet-matrix theorems: the continuous microlocal input is isolated as an assumption, and the finite operator-learning consequences are proved at the level of packet matrices, fingerprints, and budgets.

C.1 Compiled packet-matrix approximation

The central object is not a pointwise kernel or a global Fourier multiplier, but a sparse packet matrix generated by a microlocal fingerprint. In the no-caustic graph case we write the fingerprint schematically as

$$\mathfrak{F}(T_h) = (\kappa_h, S_h, a_h, R_h),$$

where κ_h is the canonical packet map, S_h is a fixed action/phase convention, a_h is the transported amplitude or symbol, and R_h is a smoothing or unresolved residual. The executable MINO layer is a compiler for an approximate fingerprint.

Theorem 47 (Fingerprint-to-compiler stability) *Let $W_h : H^s \to \ell^2(\Omega_h)$ and $W_h^* : \ell^2(\Omega_h) \to H^{s-m}$ be analysis and synthesis maps with frame constant C_W on the compact packet window. Let $T_h : H^s \to H^{s-m}$ have packet realization $K_h = W_h T_h W_h^*$. Suppose that, on the active packet window,*

$$K_h = K_{\kappa,S,a} + R_{\mathrm{tail}} + R_{\mathrm{proj}}, \qquad \|R_{\mathrm{tail}} + R_{\mathrm{proj}}\|_{\ell^2\to\ell^2} \le E_{\mathrm{mic}},$$

where $K_{\kappa,S,a}$ is the sparse canonical packet matrix compiled from (κ, S, a). Suppose the executable packet matrix M_θ satisfies

$$\|M_\theta - K_{\kappa,S,a}\|_{\ell^2\to\ell^2} \le E_{\mathrm{tr}} + E_{\mathrm{sym}} + E_{\mathrm{phase}} + E_{\mathrm{res}} + E_{\mathrm{tsyn}} + E_{\mathrm{car}} + E_{\mathrm{land}}.$$

Then

$$\|W_h^* M_\theta W_h - T_h\|_{H^s\to H^{s-m}} \le C_W\Big(E_{\mathrm{mic}} + E_{\mathrm{tr}} + E_{\mathrm{sym}} + E_{\mathrm{phase}} + E_{\mathrm{res}} + E_{\mathrm{tsyn}} + E_{\mathrm{car}} + E_{\mathrm{land}}\Big).$$

Proof Insert the packet realization:

$$W_h^* M_\theta W_h - T_h = W_h^*(M_\theta - K_h)W_h + \big(W_h^* K_h W_h - T_h\big).$$

The second term is the analysis/synthesis projection defect and is included in E_{mic}. For the first term,

$$\|M_\theta - K_h\| \le \|M_\theta - K_{\kappa,S,a}\| + \|K_{\kappa,S,a} - K_h\|.$$

The first summand is controlled by the executable budgets and the second by E_{mic}. The frame bound for $W_h^*(\cdot)W_h$ contributes the constant C_W. ■

The theorem is intentionally conditional. Classical microlocal analysis supplies $K_h = K_{\kappa,S,a} + R_{\mathrm{tail}} + R_{\mathrm{proj}}$ for the short-time FIO regimes used in the paper. The neural model is responsible for making the executable budgets small.

C.2 Representation separation for sparse canonical packet matrices

The primitive-level claim is not that MiNO dominates every neural operator. It is that there are high-frequency canonical motions for which a sparse packet matrix represents the law directly, while identity-local or fixed Fourier-limited primitives must represent the same law indirectly.

Theorem 48 (Packet-matrix representation separation) *Let ψ_h be a unit packet and suppose that a target operator obeys*

$$T_h\psi_h = b_h\psi_h^\kappa + r_h, \qquad |b_h| \geq b_0 > 0, \qquad \|r_h\|_{L^2} \leq \rho.$$

Let $V_{R,Q}$ be the closed output subspace generated by an identity tube of radius R around ψ_h together with a fixed Fourier-limited residual span of dimension Q. Assume the transported packet is separated from this subspace:

$$\mathrm{dist}_{L^2}(\psi_h^\kappa, V_{R,Q}) \geq \gamma_{R,Q} > 0.$$

Then every model A whose output on ψ_h lies in $V_{R,Q}$ satisfies

$$\|A\psi_h - T_h\psi_h\|_{L^2} \geq b_0\gamma_{R,Q} - \rho.$$

In contrast, a carrier-bound packet-matrix compiler that maps ψ_h to $b_h\psi_h^\kappa$ up to landing error η_{land} satisfies

$$\|M_\theta\psi_h - T_h\psi_h\|_{L^2} \leq \eta_{\mathrm{land}} + \rho.$$

Proof For any $v \in V_{R,Q}$,

$$\|v - T_h\psi_h\|_{L^2} = \|v - b_h\psi_h^\kappa - r_h\|_{L^2} \geq \mathrm{dist}_{L^2}(b_h\psi_h^\kappa, V_{R,Q}) - \|r_h\|_{L^2}.$$

The distance term is $|b_h|\,\mathrm{dist}(\psi_h^\kappa, V_{R,Q})$ and is at least $b_0\gamma_{R,Q}$. This proves the lower bound. The upper bound for the packet compiler follows from the triangle inequality:

$$\|M_\theta\psi_h - T_h\psi_h\| \leq \|M_\theta\psi_h - b_h\psi_h^\kappa\| + \|r_h\| \leq \eta_{\mathrm{land}} + \rho.$$

■

This is the precise separation supported by the present manuscript. It does not lower-bound fully nonlinear UNO, FNO, WNO, or attention models; those architectures may enlarge the effective span and learn the transported law implicitly. The theorem separates the sparse canonical packet matrix from restricted identity-local or fixed spectral residual primitives.

Theorem 49 (Rank growth for many transported packets) *Let $\{\psi_{h,i}\}_{i=1}^{N_h}$ be unit input packets whose transported outputs*

$$e_{h,i} := \psi_{h,i}^{\kappa}$$

form a Bessel family with bound B_{out}:

$$\Big\|\sum_{i=1}^{N_h} c_i e_{h,i}\Big\|_{L^2}^2 \le B_{\mathrm{out}} \sum_{i=1}^{N_h} |c_i|^2.$$

Assume a target operator satisfies

$$T_h\psi_{h,i} = b_{h,i}e_{h,i} + r_{h,i}, \qquad |b_{h,i}| \ge b_0 > 0, \qquad \|r_{h,i}\|_{L^2} \le \rho,$$

for every active packet i. Let V_Q be any Q-dimensional output subspace, representing a fixed-rank spectral or identity-local residual primitive without an explicit packet-dependent landing map. If a model whose outputs all lie in V_Q achieves

$$\|A\psi_{h,i} - T_h\psi_{h,i}\|_{L^2} \le \epsilon \qquad \textit{for all } i,$$

then

$$Q \ge \frac{N_h}{B_{\mathrm{out}}}\left(1 - \frac{\epsilon+\rho}{b_0}\right)^2.$$

By contrast, a sparse canonical packet matrix with one retained transported entry per active input packet represents the leading term $b_{h,i}e_{h,i}$ with $O(N_h)$ retained nonzeros and $O(1)$ local stencil degree per packet.

Proof Let P_Q be the orthogonal projection onto V_Q. The assumed error bound and the reverse triangle inequality give

$$\mathrm{dist}(e_{h,i}, V_Q) \le \frac{\epsilon+\rho}{|b_{h,i}|} \le \frac{\epsilon+\rho}{b_0}.$$

Hence

$$\|P_Q e_{h,i}\|_{L^2}^2 \ge 1 - \left(\frac{\epsilon+\rho}{b_0}\right)^2.$$

Let $S = \sum_i e_{h,i} \otimes e_{h,i}^*$ be the finite frame operator associated with the transported outputs. The Bessel bound says $\|S\| \le B_{\mathrm{out}}$. Therefore

$$\sum_{i=1}^{N_h} \|P_Q e_{h,i}\|_{L^2}^2 = \mathrm{tr}(P_Q S) \le Q\|S\| \le Q B_{\mathrm{out}}.$$

Combining the lower and upper bounds and rearranging gives the rank lower bound. The packet-matrix upper bound is the retained-stencil construction: each input packet stores the transported output packet and its amplitude as one local canonical entry, so the row/column degree is bounded independently of the total number of active packets when the canonical graph has finite overlap. ∎

For a non-affine canonical shear or short-time wave flow on a fixed compact phase window, one can choose N_h active packets with pairwise separation on the packet scale; standard packet almost-orthogonality gives $B_{\mathrm{out}} = O(1)$ as $h \to 0$. The theorem then says that a fixed-rank or fixed-bandwidth residual primitive must grow its effective output rank with the number of active transported packets, while the sparse canonical packet matrix keeps bounded local degree. This is the representation-complexity separation that the present paper can support without claiming a lower bound against fully nonlinear FNO, UNO, CANO, or WNO architectures.

C.3 Learnability certificate for theorem-grade budgets

The approximation theorems become learning guarantees only when the training protocol reaches small empirical loss and generalizes. The next result states this link without making an optimization claim.

Theorem 50 (Finite-sample learning certificate) *Let $\mathcal{F}_P$ be a parameter-bounded class of executable packet-matrix operators and let*

$$\mathcal{L}(\theta) = \mathbb{E}_{(u,v)}\ell(M_\theta u, v), \qquad \widehat{\mathcal{L}}_N(\theta) = \frac{1}{N}\sum_{i=1}^{N}\ell(M_\theta u_i, v_i),$$

with $0 \le \ell \le B$. Let $\widehat{\theta}$ be an empirical optimizer up to η_{opt}:

$$\widehat{\mathcal{L}}_N(\widehat{\theta}) \le \inf_{\theta\in\mathcal{F}_P} \widehat{\mathcal{L}}_N(\theta) + \eta_{\mathrm{opt}}.$$

Then, with probability at least $1-\delta$,

$$\mathcal{L}(\widehat{\theta}) \le \inf_{\theta\in\mathcal{F}_P} \mathcal{L}(\theta) + 2\mathfrak{R}_N(\ell\circ\mathcal{F}_P) + B\sqrt{\frac{\log(2/\delta)}{2N}} + \eta_{\mathrm{opt}},$$

where $\mathfrak{R}_N(\ell\circ\mathcal{F}_P)$ is the empirical Rademacher complexity of the induced loss class. If a nonnegative proxy penalty $\Pi(\theta)$ is used with weight λ and the theorem-grade budgets are bounded by $B_{\mathrm{thm}}(\theta) \le c_0\mathcal{L}(\theta) + c_1\Pi(\theta)$, then the same inequality gives a high-probability bound on $B_{\mathrm{thm}}(\widehat{\theta})$ after adding $c_1\lambda^{-1}$ times the achieved regularized objective.

Proof Uniform convergence for bounded losses gives, with probability at least $1-\delta$,

$$\sup_{\theta\in\mathcal{F}_P} |\mathcal{L}(\theta) - \widehat{\mathcal{L}}_N(\theta)| \le \mathfrak{R}_N(\ell\circ\mathcal{F}_P) + B\sqrt{\frac{\log(2/\delta)}{2N}}/2,$$

after the standard symmetrization and Hoeffding concentration step. Applying this inequality to $\widehat{\theta}$ and to an exact population minimizer candidate and then using the η_{opt} empirical optimality gives the displayed bound, with the constants absorbed into the conventional $2\mathfrak{R}_N$ form. The proxy-budget consequence follows by the assumed deterministic certificate $B_{\mathrm{thm}} \le c_0\mathcal{L} + c_1\Pi$ and by the nonnegativity of the regularized objective terms. ∎

Thus the paper's learning statement is conditional but non-vacuous: once finite optimization, sample complexity, and proxy alignment are controlled, the microlocal approximation budgets become reachable supervised quantities.

C.4 Packet microscope identifiability

The operator-microscopy program requires more than field prediction. The basic identifiability statement is already available in a finite packet form. For packet centers $z, w \in \Omega_h$, define the microscope response

$$\mathcal{X}_h[T](w,z) = |\langle T_h \varphi_z, \varphi_w \rangle|.$$

Theorem 51 (Active canonical ridge recovery) *Let $\Lambda_{\mathrm{ell}} \subset \Omega_h \times \Omega_h$ be an elliptic, branch-separated active region of a canonical graph. Suppose there are constants $c_{\mathrm{ell}} > c_{\mathrm{off}} > 0$ and a raw packet radius $r_h = Ch^{1/2} + \delta_h$ such that:*

1. *if (w,z) is within distance r_h of Λ_{ell}, then $\mathcal{X}_h[T](w,z) \geq c_{\mathrm{ell}}$ for at least one packet w in the r_h-tube over each active z;*
2. *if (w,z) has distance greater than r_h from Λ_{ell}, then $\mathcal{X}_h[T](w,z) \leq c_{\mathrm{off}}$;*
3. *measurement noise is bounded by $\sigma < (c_{\mathrm{ell}} - c_{\mathrm{off}})/2$.*

For any threshold τ satisfying $c_{\mathrm{off}} + \sigma < \tau < c_{\mathrm{ell}} - \sigma$, the thresholded ridge of the noisy response lies inside the r_h-tube of Λ_{ell} and intersects that tube over every active source packet. Consequently the recovered active canonical relation has raw Hausdorff error at most r_h on the resolved active window.

Proof Outside the r_h-tube the noisy response is at most $c_{\mathrm{off}} + \sigma$, which is below τ; hence no thresholded point can lie outside the tube. Inside the active tube, for every active source packet the ellipticity assumption supplies an output packet whose noiseless response is at least c_{ell}, and the noisy response is at least $c_{\mathrm{ell}} - \sigma > \tau$. Thus the thresholded set is contained in the tube and is nonempty over each active source. These two inclusions are exactly the stated Hausdorff recovery bound on the active window. ■

The theorem identifies the active graph and, with complex responses rather than magnitudes only, the transported phase convention S_h up to the chosen packet-frame gauge. Multi-chart caustics and Maslov transition data are not identified by this single-chart statement.

C.5 Finite packet Egorov theorem

The full continuous Egorov theorem is not formalized here. What the present paper needs is the finite packet-matrix consequence: if the learned packet operator is close to a compiled canonical-symbol packet law, then conjugating packet observables is stable with the same budget.

Theorem 52 (Finite packet Egorov stability) *Let M be an executable packet matrix and let $M_0 = S_\theta K_{\widehat{\kappa}}$ be the compiled canonical-symbol packet law on a finite packet window. Suppose $\|M - M_0\|_{\ell^2 \to \ell^2} \leq \epsilon$. For every packet observable A,*

$$\|M^* A M - M_0^* A M_0\|_{\ell^2 \to \ell^2} \leq \|A\| \epsilon (2\|M_0\| + \epsilon).$$

If classical Egorov identifies $M_0^* A M_0$ with the packet realization of the pulled-back symbol observable up to residual $E_{\mathrm{Eg,cl}}$, then the executable Egorov residual is bounded by

$$E_{\mathrm{Eg,cl}} + \|A\|\epsilon(2\|M_0\| + \epsilon).$$

Proof Write $M = M_0 + \Delta$ with $\|\Delta\| \le \epsilon$. Expanding gives

$$M^* A M - M_0^* A M_0 = \Delta^* A M_0 + M_0^* A \Delta + \Delta^* A \Delta.$$

Taking operator norms and using submultiplicativity yields

$$\|\Delta^* A M_0\| + \|M_0^* A \Delta\| + \|\Delta^* A \Delta\| \le 2\epsilon \|A\| \|M_0\| + \epsilon^2 \|A\|.$$

The final statement adds the classical Egorov residual by the triangle inequality. ∎

This is why the Egorov diagnostic is a calculus metric rather than a new architecture theorem: it tests whether the learned packet law and the learned canonical-symbol law are coupled in the correct operator sense.

C.6 Composition closure in fingerprint norm

A field-level composition bound alone is mostly a triangle inequality. The meaningful closure statement is that the packet fingerprint remains sparse and composes according to the FIO law.

Theorem 53 (Packet-fingerprint composition closure) *Let K_1 and K_2 be exact sparse packet laws with fingerprints $(\kappa_1, S_1, a_1, R_1)$ and $(\kappa_2, S_2, a_2, R_2)$ on a compact packet window. Assume their classical FIO composition has fingerprint*

$$\kappa_{21} = \kappa_2 \circ \kappa_1, \qquad S_{21} = S_1 + S_2 \circ \kappa_1, \qquad a_{21} = a_2 \circ \kappa_1 \, a_1 + a_{\mathrm{sub}},$$

with packet cross-tail E_{cross}. Let executable packet matrices M_i satisfy $\|M_i - K_i\| \le \epsilon_i$ and $\|M_i\|, \|K_i\| \le L_i$. Then

$$\|M_2 M_1 - K_{21}\| \le L_2 \epsilon_1 + L_1 \epsilon_2 + \epsilon_1 \epsilon_2 + E_{\mathrm{cross}},$$

where K_{21} is the retained packet compiler for the composed fingerprint.

Proof Add and subtract $K_2 K_1$:

$$M_2 M_1 - K_{21} = (M_2 - K_2) K_1 + K_2 (M_1 - K_1) + (M_2 - K_2)(M_1 - K_1) + (K_2 K_1 - K_{21}).$$

The first three terms are bounded by $L_1 \epsilon_2$, $L_2 \epsilon_1$, and $\epsilon_1 \epsilon_2$ respectively. The last term is precisely the retained-stencil cross-tail from the classical packet composition law. Summing gives the result. ∎

The theorem is finite and does not claim long-time stability. It states the local algebra needed for rollout theory: canonical maps compose, action primitives add, amplitudes multiply through the usual symbol product, and the packet stencil remains controlled when E_{cross} is small.

C.7 Local outgoing Helmholtz packet budget

High-k Helmholtz requires more than a finite branch count. The outgoing resolvent also carries a near-characteristic symbol and radiation convention. The following theorem is the local statement justified by the current manuscript.

Theorem 54 (Local outgoing Helmholtz packet budget) *Consider a no-trapping local window for the outgoing Helmholtz resolvent at wavenumber $k = h^{-1}$. Assume the cut-off resolvent has a finite packet parametrix with L outgoing branches, anisotropic frame defect E_{frame}, branch-routing defect E_{route}, branch cross-talk E_{cross}, branch transport/symbol budgets $\{E_{\mathrm{tr},\ell}, E_{\mathrm{sym},\ell}\}_{\ell=1}^{L}$, phase/landing budget $E_{\mathrm{phase}} + E_{\mathrm{land}}$, and radiation-symbol budget E_{rad} for the local $(p+i0)^{-1}$ multiplier. Then the branch-factored carrier-bound packet compiler satisfies*

$$\|M^{(L)}_{\theta,h} - R^{\mathrm{out}}_h\|_{\ell^2\to\ell^2} \le C\,\mathcal{E}_{\mathrm{Helm}},$$

$$\mathcal{E}_{\mathrm{Helm}} = E_{\mathrm{frame}} + E_{\mathrm{route}} + E_{\mathrm{cross}} + \sum_{\ell=1}^{L}(E_{\mathrm{tr},\ell} + E_{\mathrm{sym},\ell}) + E_{\mathrm{phase}} + E_{\mathrm{land}} + E_{\mathrm{rad}} + E_{\mathrm{tail}}.$$

Proof The local outgoing parametrix decomposes the packet realization of the resolvent into a finite sum of outgoing branch packet laws, a radiation-symbol multiplier near the characteristic shell, and a remainder. Apply Theorem 47 to each branch, sum the branch-wise transport and symbol errors, add the router error for mass assigned to the wrong branch, and add the cross-talk error for overlapping retained tubes. The anisotropic frame defect, radiation-symbol defect, landing/phase defect, and packet tail are independent perturbations of the retained packet operator, so the final estimate follows by the triangle inequality and finite overlap of the branch tubes. ■

The negative branch-only Helmholtz diagnostic is therefore not a contradiction of the microlocal theory. It says that reducing only the branch-count budget does not control the full local resolvent packet budget; the radiation-symbol, frame, phase, and landing terms are load-bearing.

C.8 Formalization status for the extended theorem package

The Lean companion formalizes the finite algebraic parts of this appendix: thresholded ridge inclusion for packet microscope responses, additive fingerprint budgets, finite Egorov budget expansion, and finite composition budget accumulation. It does not formalize the classical inputs: stationary phase, the continuous Egorov theorem, the local outgoing resolvent parametrix, or the analytic construction of wavepacket frames. Those remain manuscript proofs using standard microlocal assumptions, while the finite landing algebra is machine-checked.

Table 11: Branched MiNO high-k Helmholtz diagnostic. Entries are mean relative ℓ^2 over available seeds under the local sample-capped protocol; lower is better. The table tests whether finite canonical-branch structure reduces the high-k gap at this budget, not a global Helmholtz resolvent theorem or full-budget benchmark.

Scenario	Full	Single br.	Uniform route	No tr.	No tr./car.	No sym.
Helm. high-k ctl.	0.548	0.548	0.548	0.559	0.554	0.548
Helm. high-k	0.705	0.705	0.705	0.706	0.722	0.705
Helm. var.+	0.735	0.735	0.735	0.735	0.734	0.735

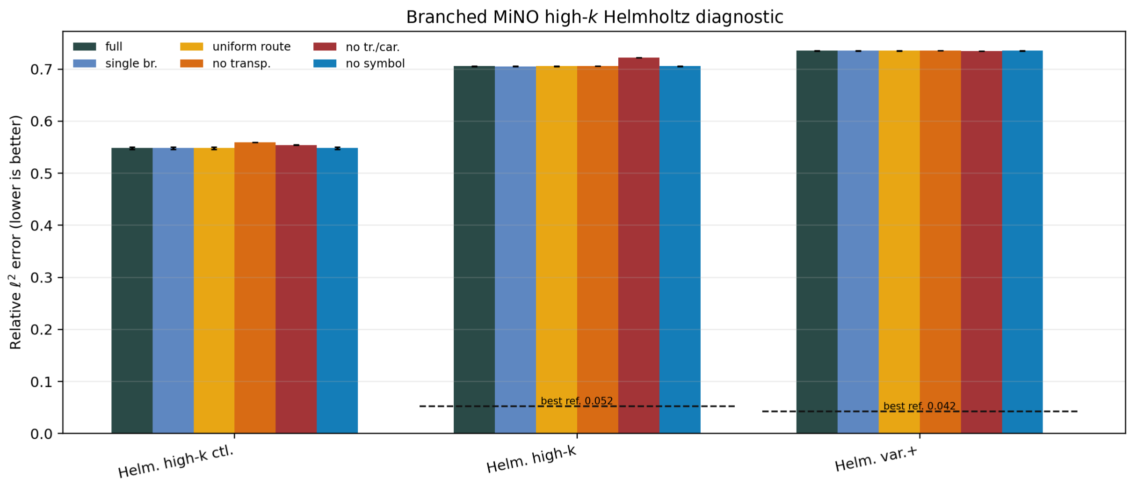

Figure 12: Branched MiNO high-k Helmholtz diagnostic. The figure compares the full branched carrier-bound model against single-branch, uniform-routing, transport-removal, carrier-removal, and symbol-removal ablations under the local sample-capped protocol. Dashed reference marks are imported baselines where available.

Appendix D. Additional Diagnostic Figures

D.1 Branched High-k Helmholtz Diagnostic

The branched Helmholtz campaign records the local branch probe used in the main text. It tests whether the high-k Helmholtz stress row is improved by a finite outgoing-branch implementation at the sample-capped budget used in the manuscript.

D.2 Low-Budget Calculus-Regime Diagnostics

The following panels are stress diagnostics for local-window Helmholtz and synthetic Navier–Stokes profiles. They are included to document boundary conditions for future calculus hooks, not as main success claims.

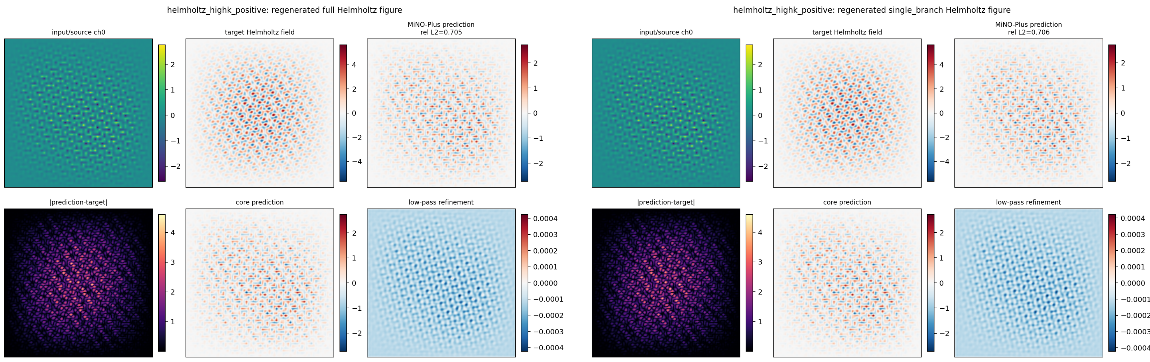

Figure 13: High-k Helmholtz prediction panels for the sample-capped branched diagnostic. Left: full $L = 3$ branched model. Right: forced single-branch control at the same budget and sample. The nearly identical relative errors show that branch multiplicity alone leaves the main high-k correction budget to the frame, router, phase/landing, and resolvent-symbol terms.

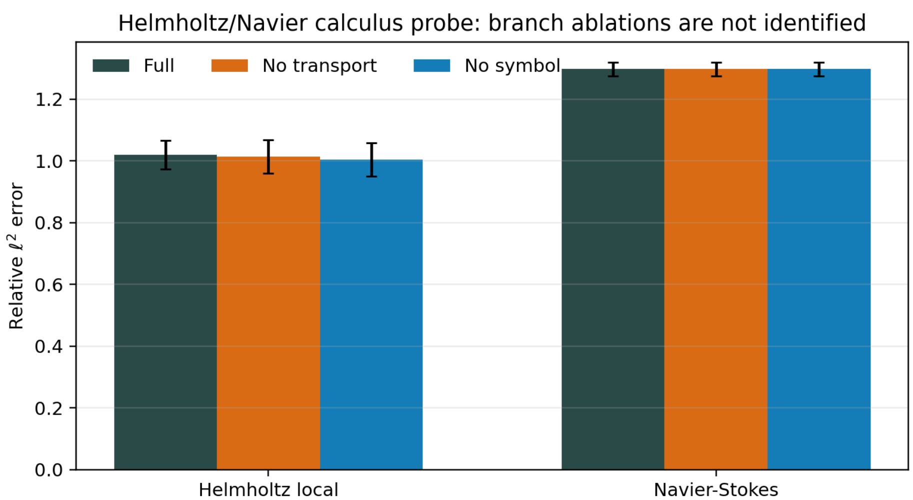

Figure 14: Low-budget calculus-regime probe on local-window Helmholtz and synthetic Navier–Stokes. Each row uses three seeds, twelve epochs, and restricted train/validation/test samples. The nearly flat ablation bars identify which branch effects are not active at this scale and motivate the dedicated branch/resolvent and transport–diffusion profiles.

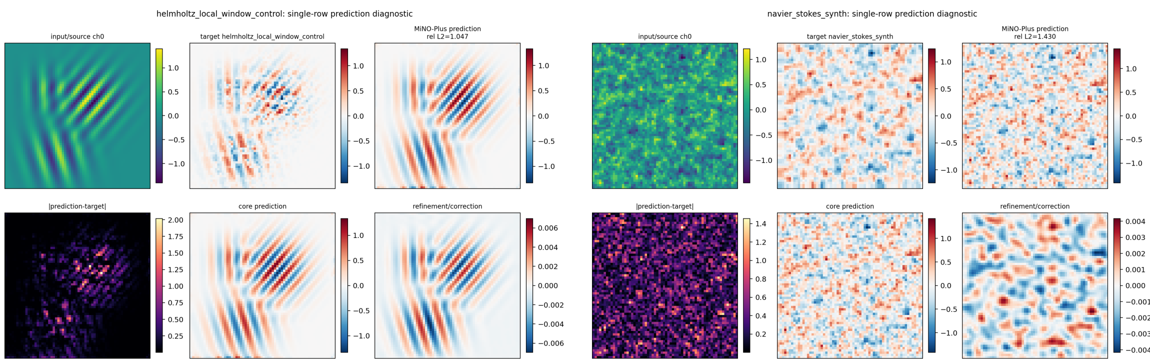

Figure 15: Single-row prediction diagnostics for the same low-budget Helmholtz/Navier probe. The panels visualize input, target, prediction, absolute error, core prediction, and refinement/correction for one held-out sample. These rows show the regimes that motivate the next calculus hooks.

Appendix E. Machine-Proof Companion

The Lean companion is intentionally finite. It checks the landing layer reached after the classical microlocal estimates have produced finite packet tails, stencils, and budgets. It does not formalize oscillatory integrals, stationary phase, or the full Hörmander calculus.

E.1 Lean Scope

The Lean files in `lean/MiNO/Microlocal/` formalize:

- finite packet data structures and transported-symbol actions;
- pointwise and ℓ^1 finite propagation bounds;
- bounded-stencil finite propagation;
- finite approximation and identity-canonical reduction lemmas;
- the global-scalar obstruction inequality;
- semi-discrete transplantation and cross-resolution transfer;
- canonical-tube, shell-counting, packet-tail, and abstract sparsity bridges;
- finite packet-matrix tail accumulation;
- restricted non-stationary-phase certificate handoff;
- finite FIO/ΨDO/smoothing branch composition;
- finite packet-microscope ridge inclusion, fingerprint-budget stability, finite Egorov budget expansion, composition-budget accumulation, and learning-budget transfer for the extended theorem package.

This is the machine-checked finite part of the proof spine used by Sections 5–7. The new ℓ^2 packet-matrix Schur transfer is an elementary manuscript-level operator-norm step built on these finite row/column and tail objects; it is not advertised as a continuous Lean formalization.

E.2 Proof-Status Inventory

Table 12: Proof-status inventory for the manuscript.

Claim layer	Status	Source
Finite transported-symbol perturbation	fully formalized	Lean + Section 5
Bounded-stencil finite propagation	fully formalized	Lean + Section 5
Schur packet-matrix transfer	manuscript proof from finite row/column bounds	Theorem 6
Cross-resolution transfer	fully formalized	Lean + Section 5
Canonical tube, shell count, packet tail	fully formalized finite bridge	Lean + Section 6
Packet-matrix tail accumulation	fully formalized finite bridge	Lean `PacketMatrix.lean`
Restricted non-stationary-phase handoff	formal finite wrapper with classical certificate input	Lean `NonstationaryPhase.lean`
Unified FIO/ΨDO/smoothing composition	formal finite wrapper with classical input	Lean `UnifiedCalculus.lean`
Packet-microscope ridge inclusion and finite fingerprint/Egorov/composition budgets	fully formalized finite algebra	Lean `Flagship.lean`
Representation separation, fingerprint compiler stability, local Helmholtz packet budget	manuscript proof with classical packet/FIO input	Appendix C
Short-time half-wave off-graph decay	classical manuscript proof	Section 6, Appendix B
Abstract canonical-graph packet sparsity	classical derivation plus finite bridge	Section 6
Continuous transported-symbol reduction	classical manuscript proof	Section 6, Appendix B
Hamiltonian metadata consistency	manuscript proof	Proposition 21
Microlocal Propagation Theorem for MiNO-Core	classical continuous input + standard NN approximation + finite Lean bridge	Theorem 36
PDE specialization statements	manuscript-level derivations and modeling assumptions	Section 7

E.3 Executable Certificates

Two lightweight certificate paths accompany the Lean project. The SymPy transport certificate expands the finite coefficient identity behind the transported-symbol bound. The interval-style script checks concrete budget instances and writes the result to `certificates/interval_check.json`. These certificates are sanity checks for the finite algebra; they are not substitutes for the continuous microlocal argument.

E.4 Next Formalization Targets

The next useful Lean targets are weighted packet norms, confidence-weighted stencils, stronger finite packet-matrix residual operators, finite-to-asymptotic bridge lemmas indexed by the packet scale h, and finite matrix-norm versions of the Egorov and composition lemmas in Appendix C. A more substantive proof-assistant target is a finite-frequency version of Calderón–Vaillancourt continuity for order-aware packet symbols, together with finite symplectic metadata maps and their canonical-defect estimates. These targets are closer to the actual microlocal content than the elementary finite triangle inequalities, while still avoiding the full analytic burden of continuous oscillatory integrals. Full continuous microlocal formalization would require substantial new libraries for oscillatory integrals, pseudodifferential symbol classes, Hamiltonian flows, Sobolev spaces, and wavefront sets.

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